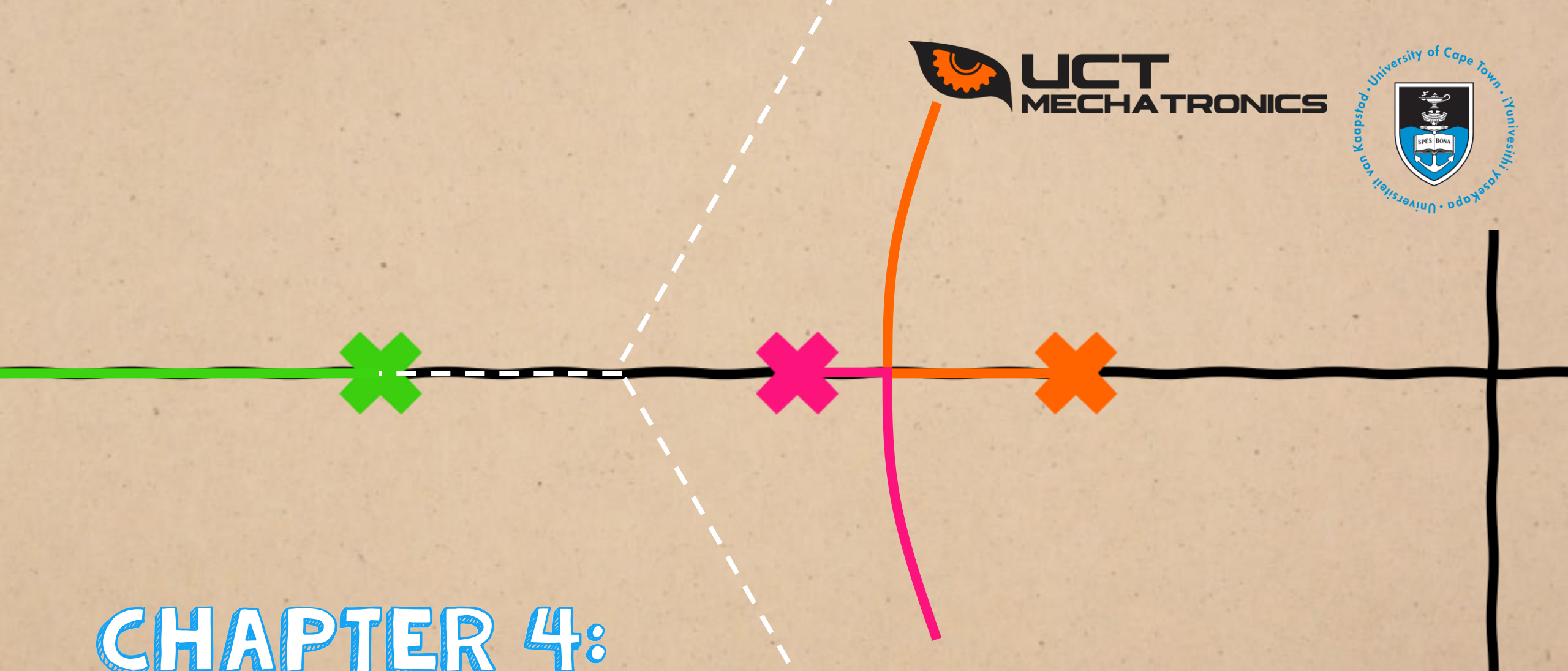
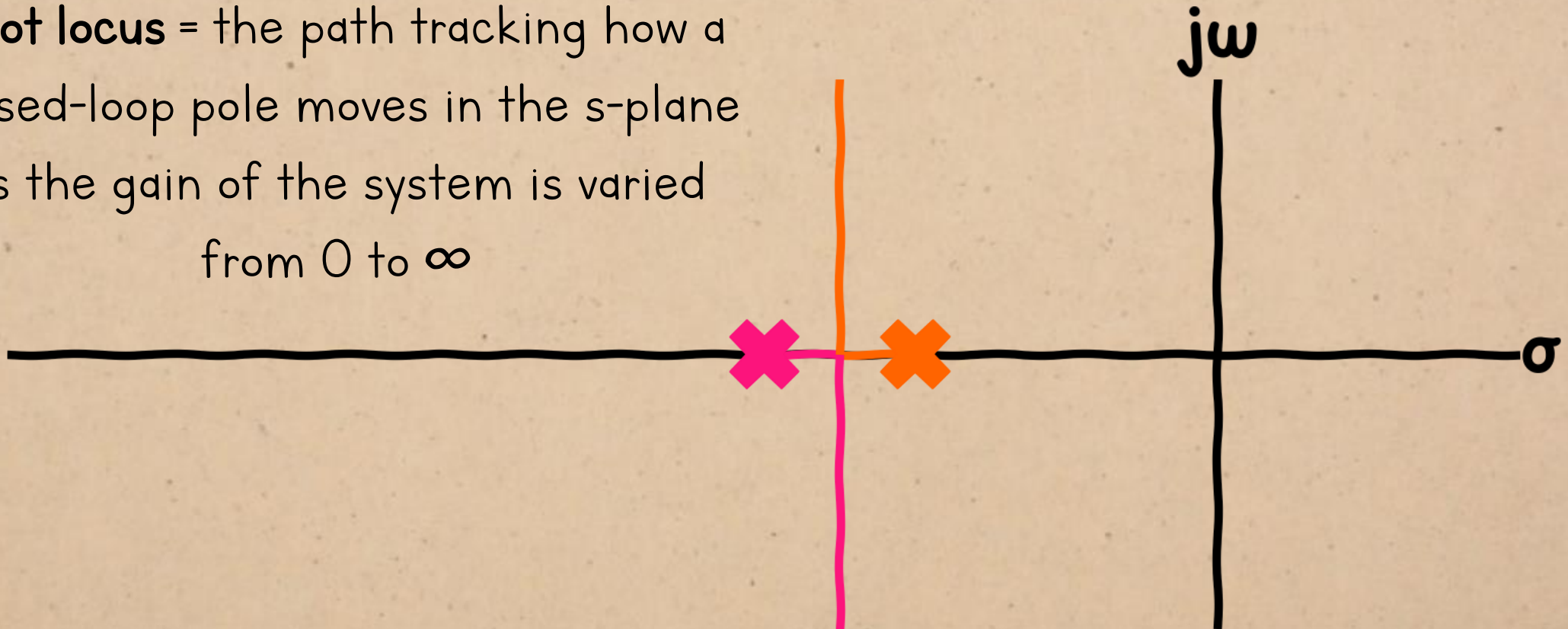




CHAPTER 4: ROOT LOCUS 2 SKETCHING RULES

LAST TIME:

Root locus = the path tracking how a closed-loop pole moves in the s-plane as the gain of the system is varied from 0 to ∞

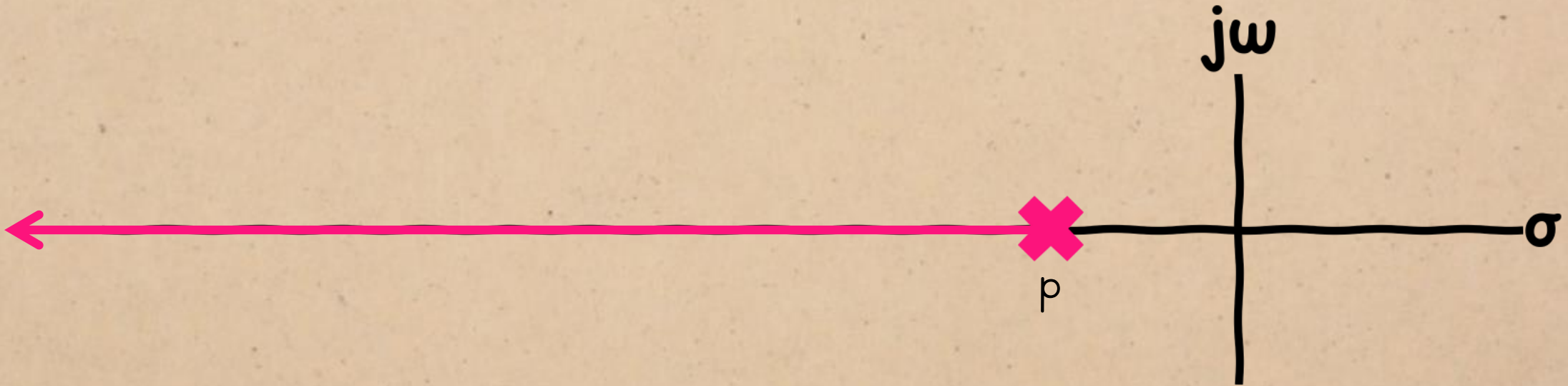


LAST TIME:



Poles start at their open-loop locations and move to the zeros...

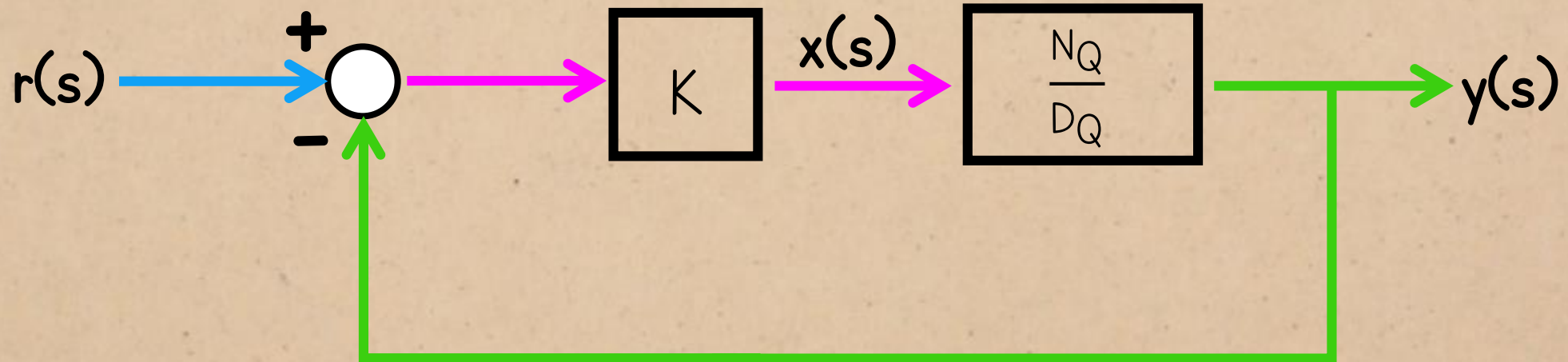
LAST TIME:



Poles start at their open-loop locations and move to the zeros, or to infinity, if there aren't enough zeros to pair with every pole.

LAST TIME:

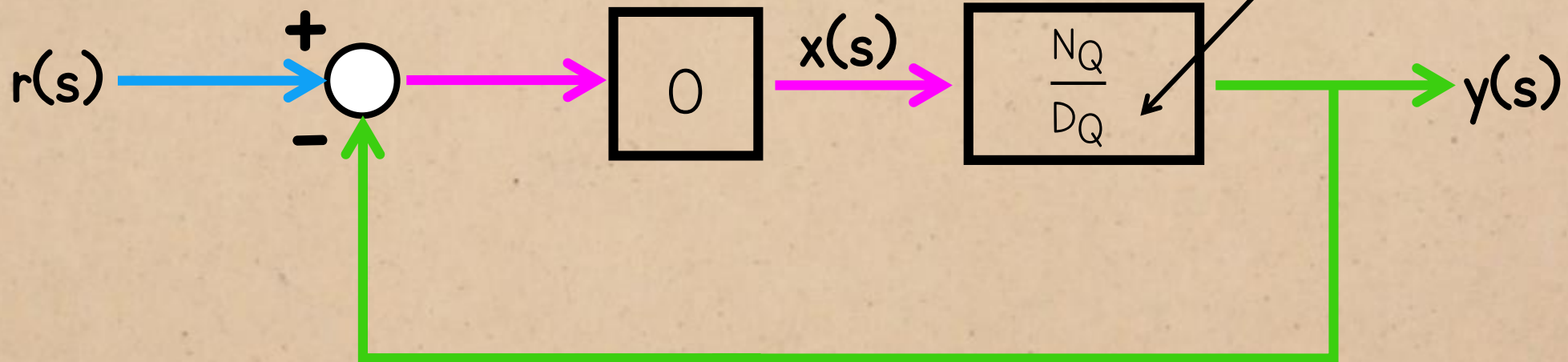
characteristic equation: $KN_Q + D_Q = 0$



LAST TIME:

When the gain = 0, the open- and closed-loop characteristic equations (and $\therefore$ poles) are the same.

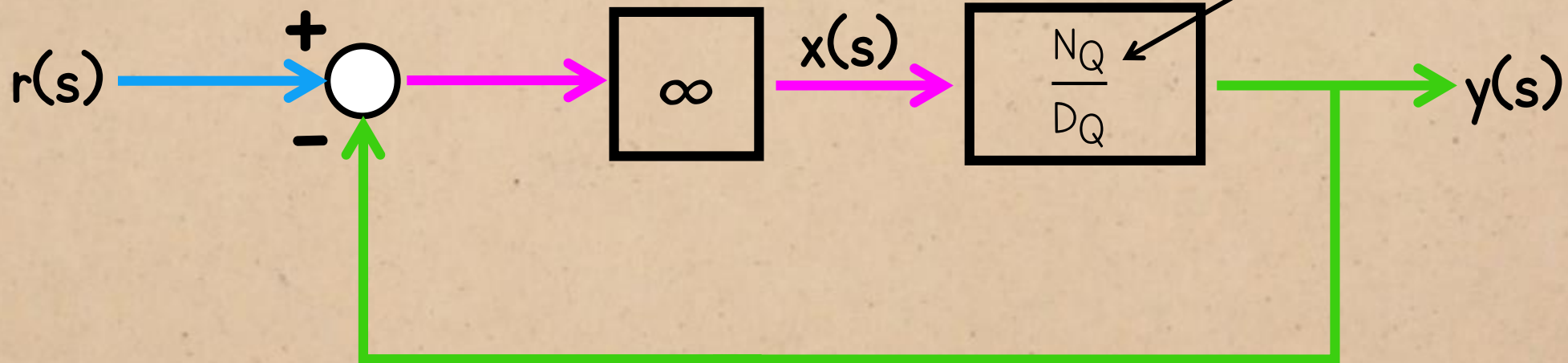
characteristic equation: $D_Q = 0$



LAST TIME:

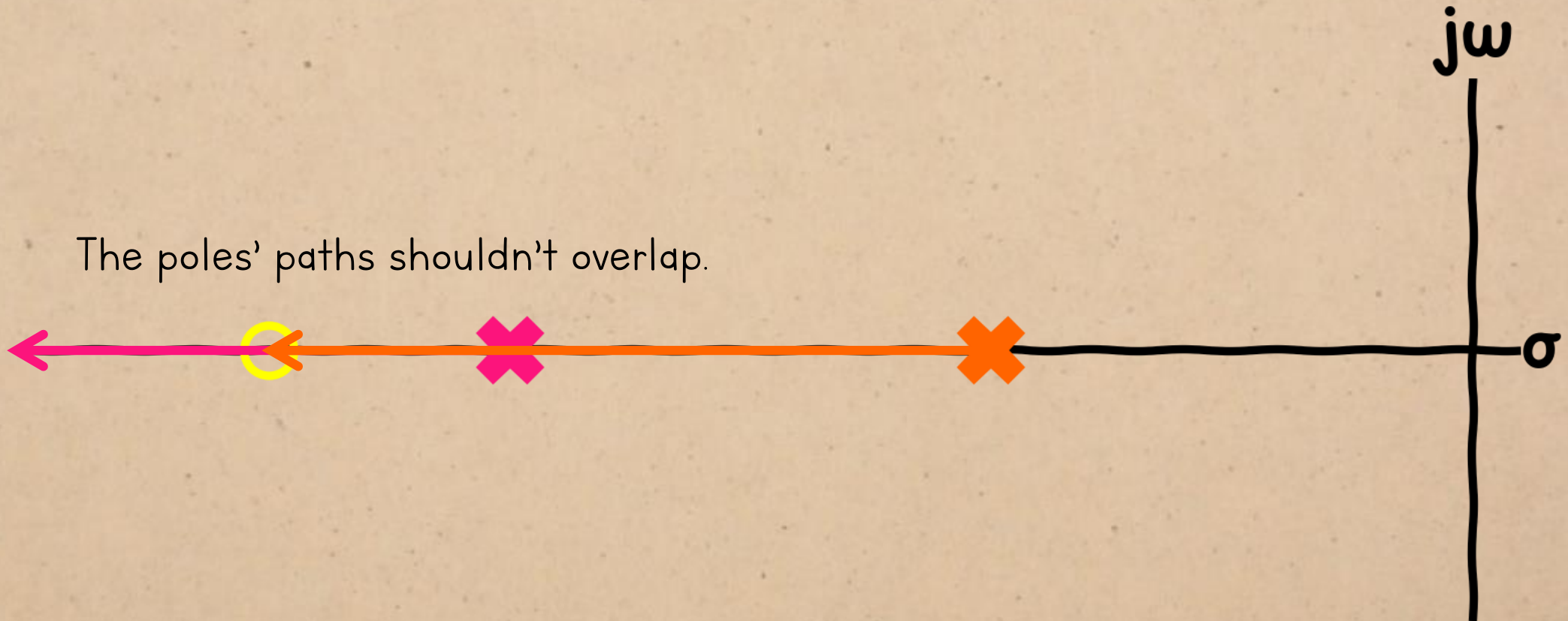
When the gain is very large, the characteristic equation is dominated by the open-loop numerator $\therefore$ the closed-loop poles move to the zeros

characteristic equation: $\infty N_Q = 0$

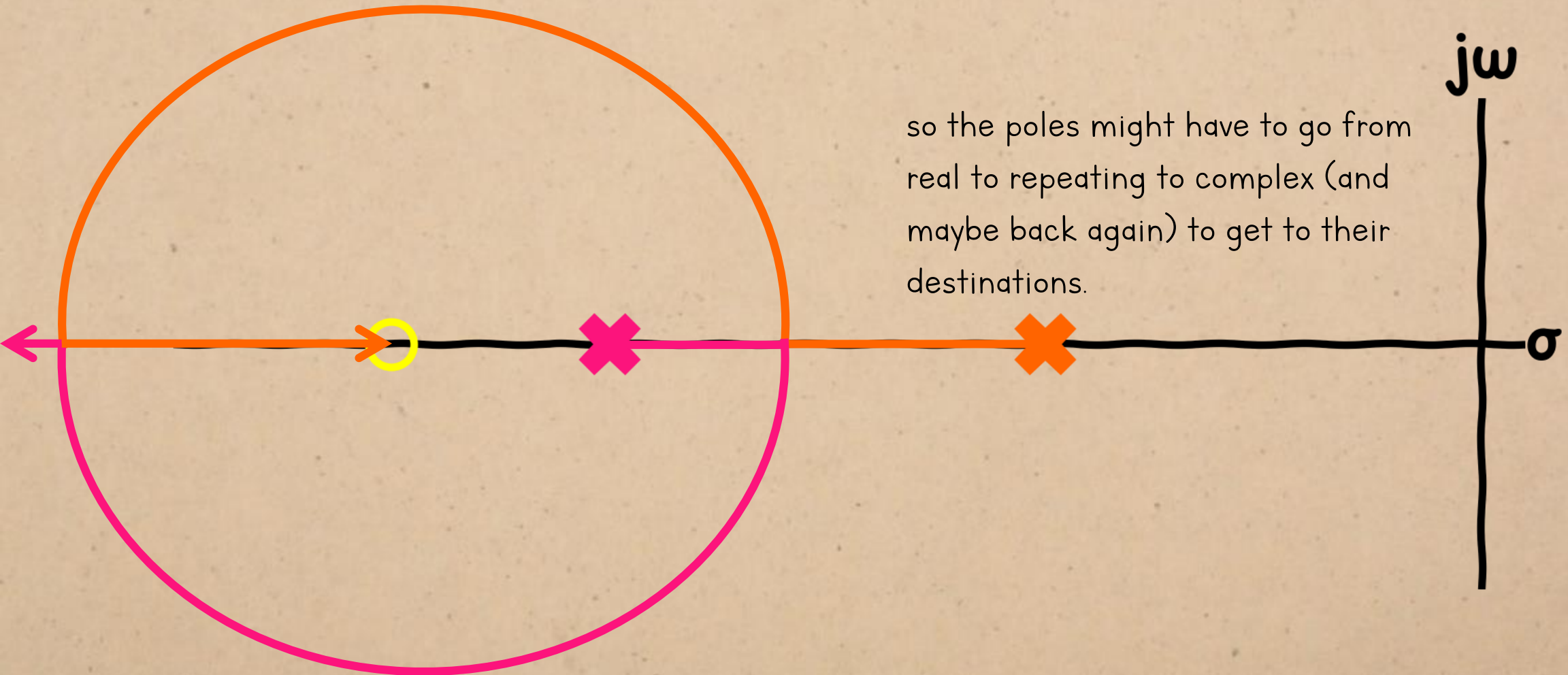


LAST TIME:

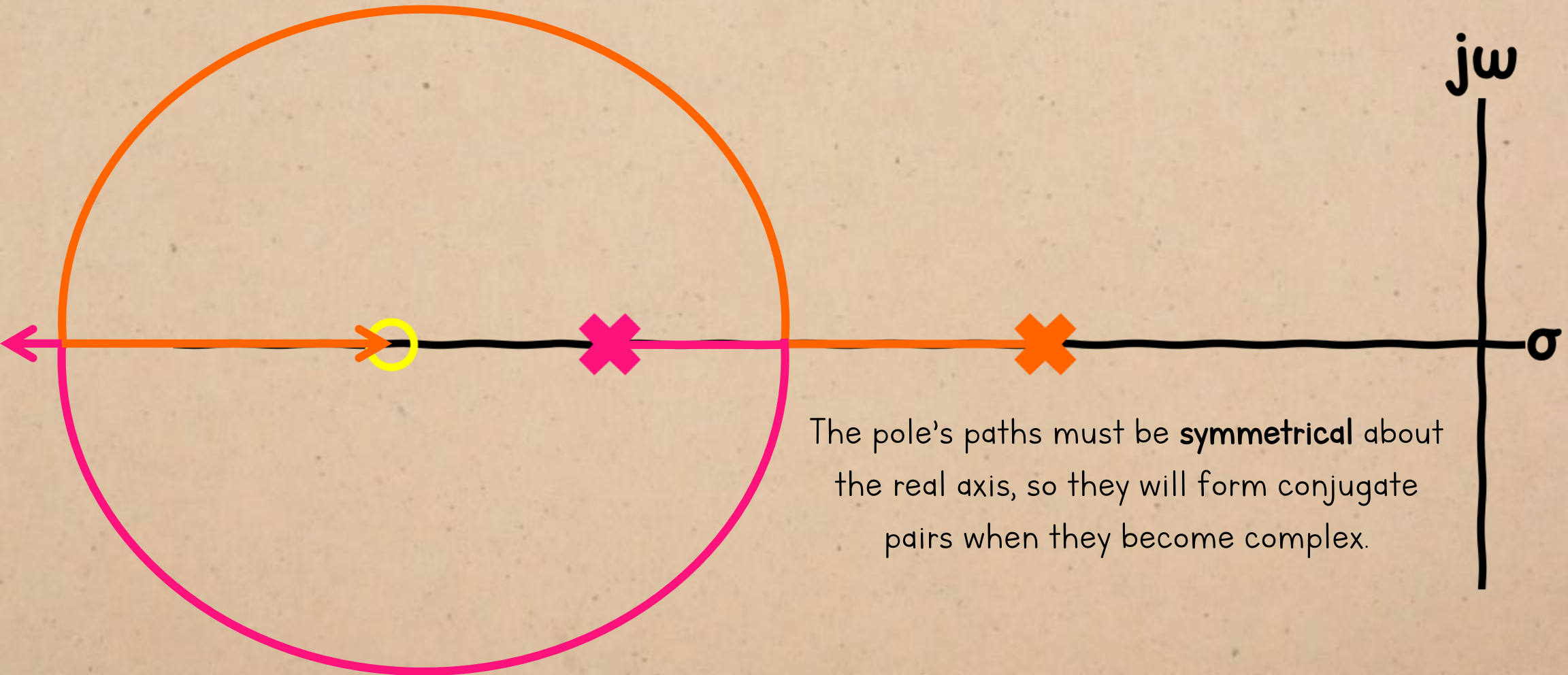
The poles' paths shouldn't overlap.



LAST TIME:

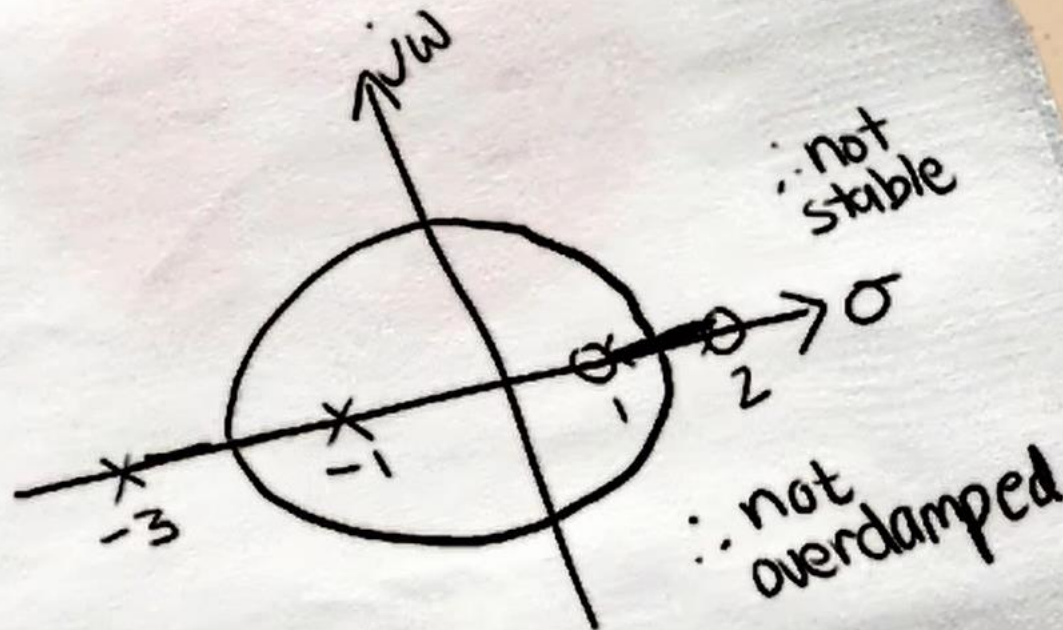


LAST TIME:



The pole's paths must be **symmetrical** about the real axis, so they will form conjugate pairs when they become complex.

If we put a system with poles at -1 and -3 , and zeros at 1 and 2 in closed loop, will it be **stable** and **overdamped** for any gain?

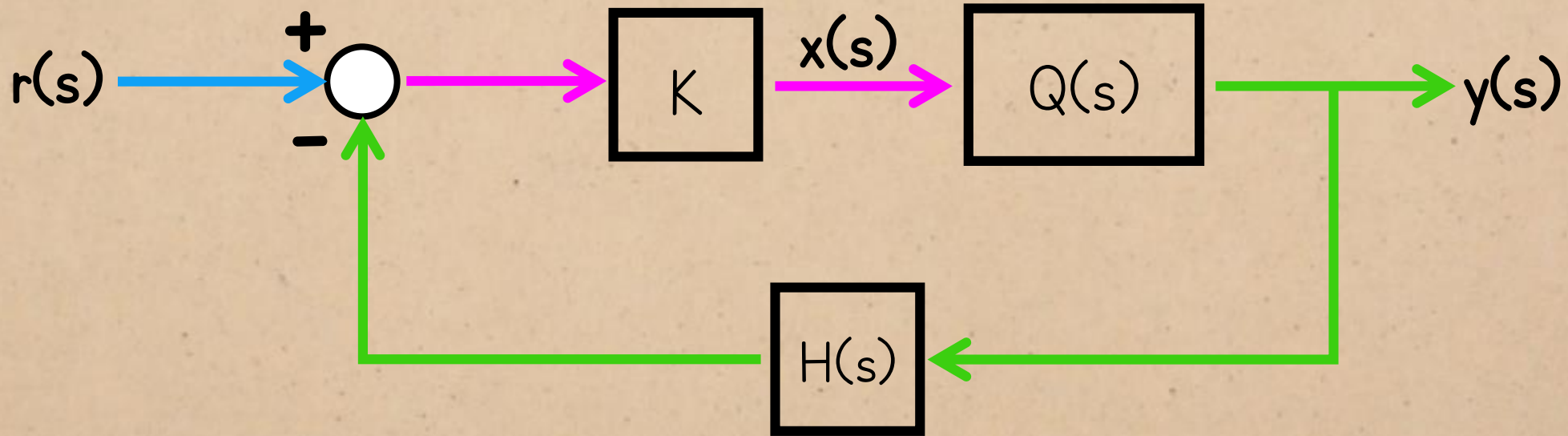


Using what we already know,
we can sketch a rough root
locus plot that will answer
this question.

-1 and -3 , and
it be **stable**
n?

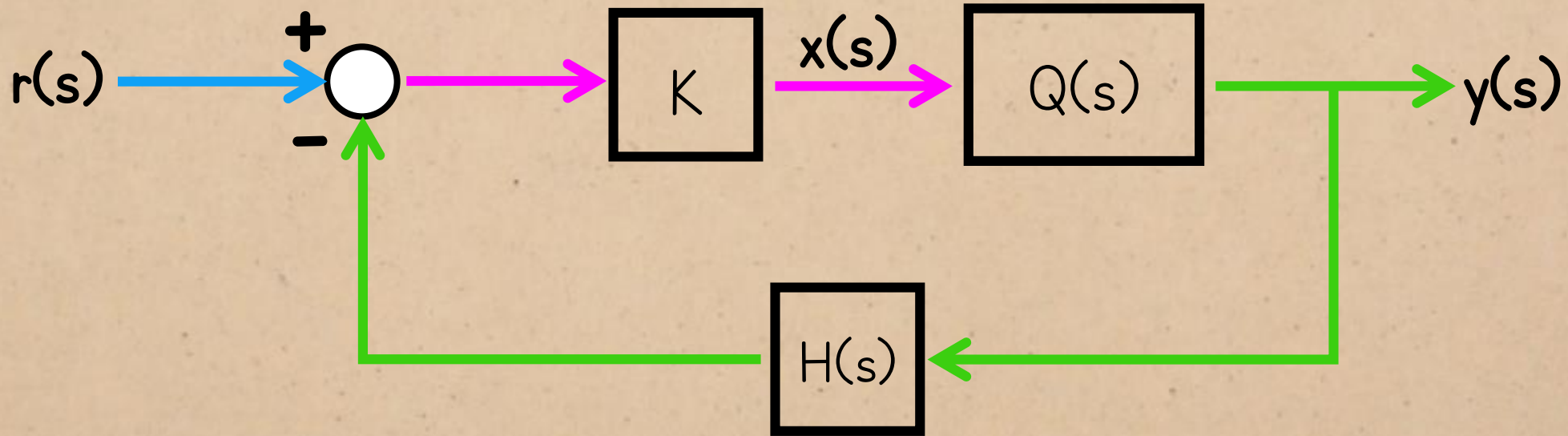
ROOT LOCUS GEOMETRY

$$G_{CL} = \frac{KQ(s)}{1 + KQ(s)H(s)}$$



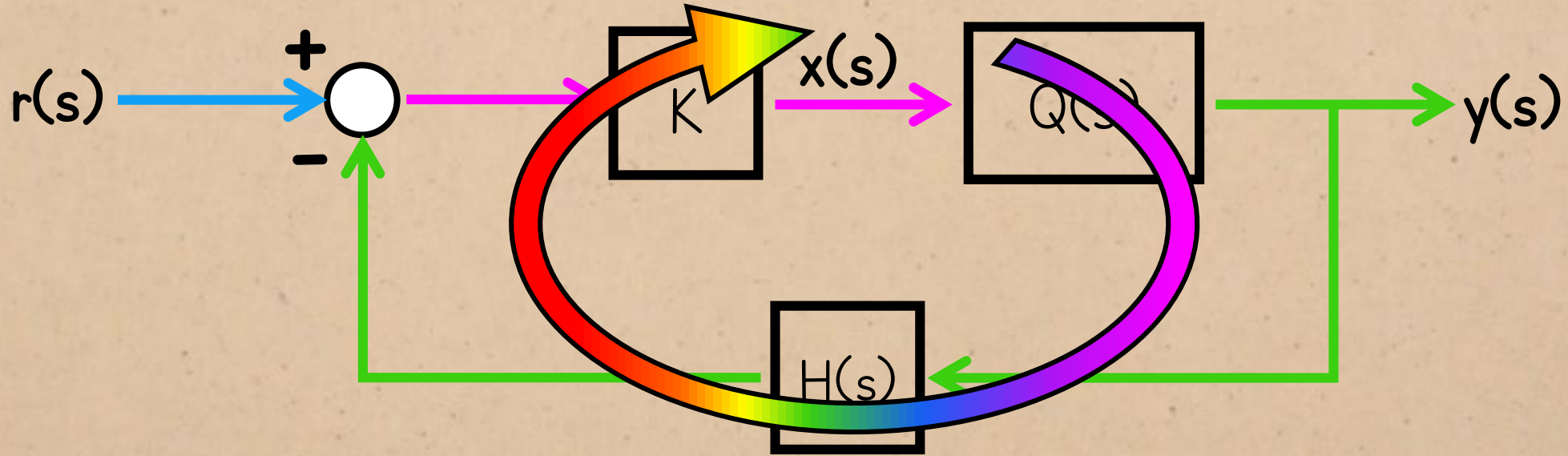
ROOT LOCUS GEOMETRY

characteristic equation: $1 + KQ(s)H(s) = 0$



ROOT LOCUS GEOMETRY

characteristic equation: $1 + KL(s) = 0$

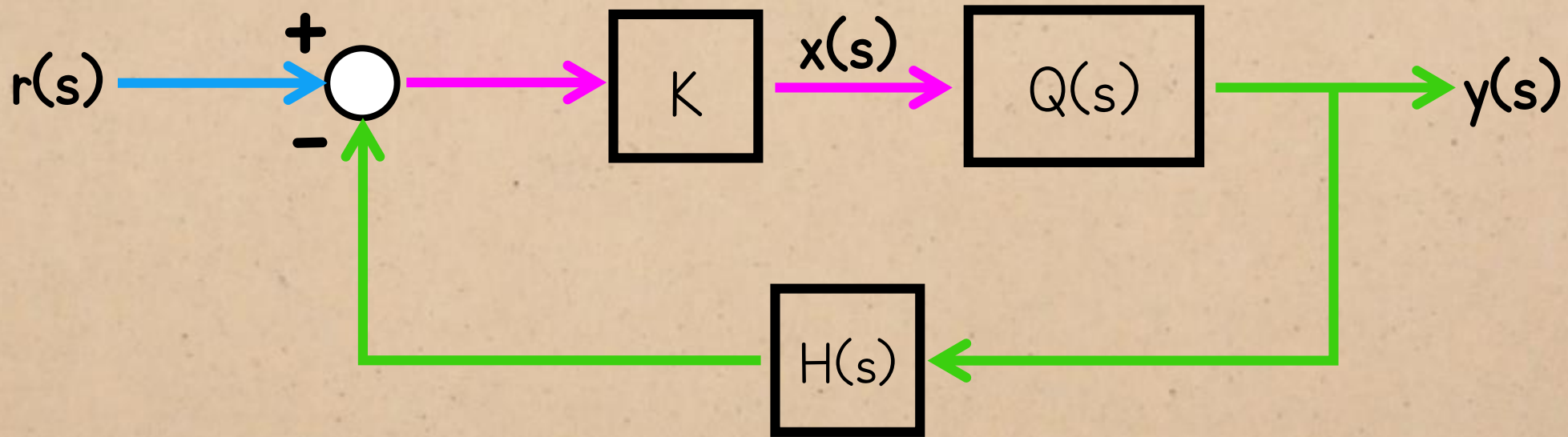


Loop transfer function: $L(s) = Q(s)H(s)$

ROOT LOCUS GEOMETRY

...so if $1 + KL(\lambda) = 0$,

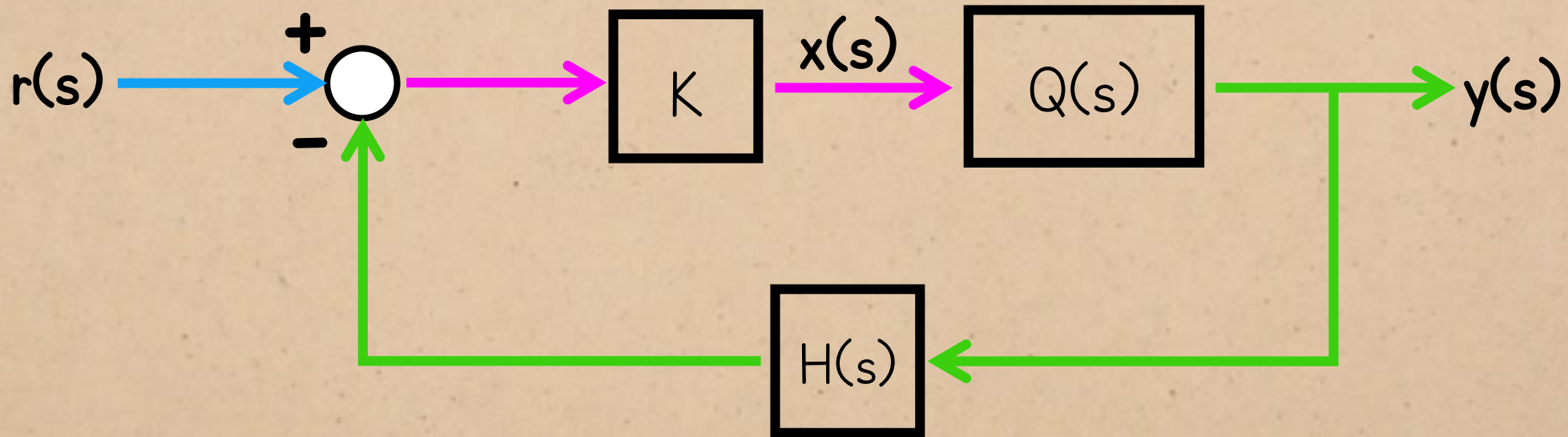
λ must be a closed-loop pole when the gain = K



ROOT LOCUS GEOMETRY

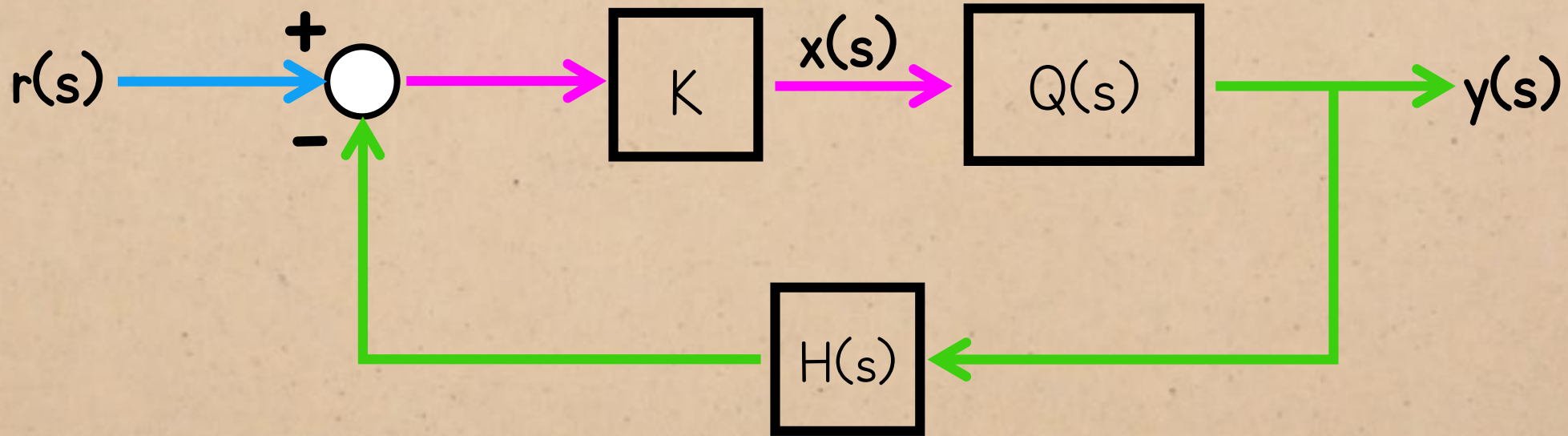
...so if $KL(\lambda) = -1$,

λ must be a closed-loop pole when the gain = K



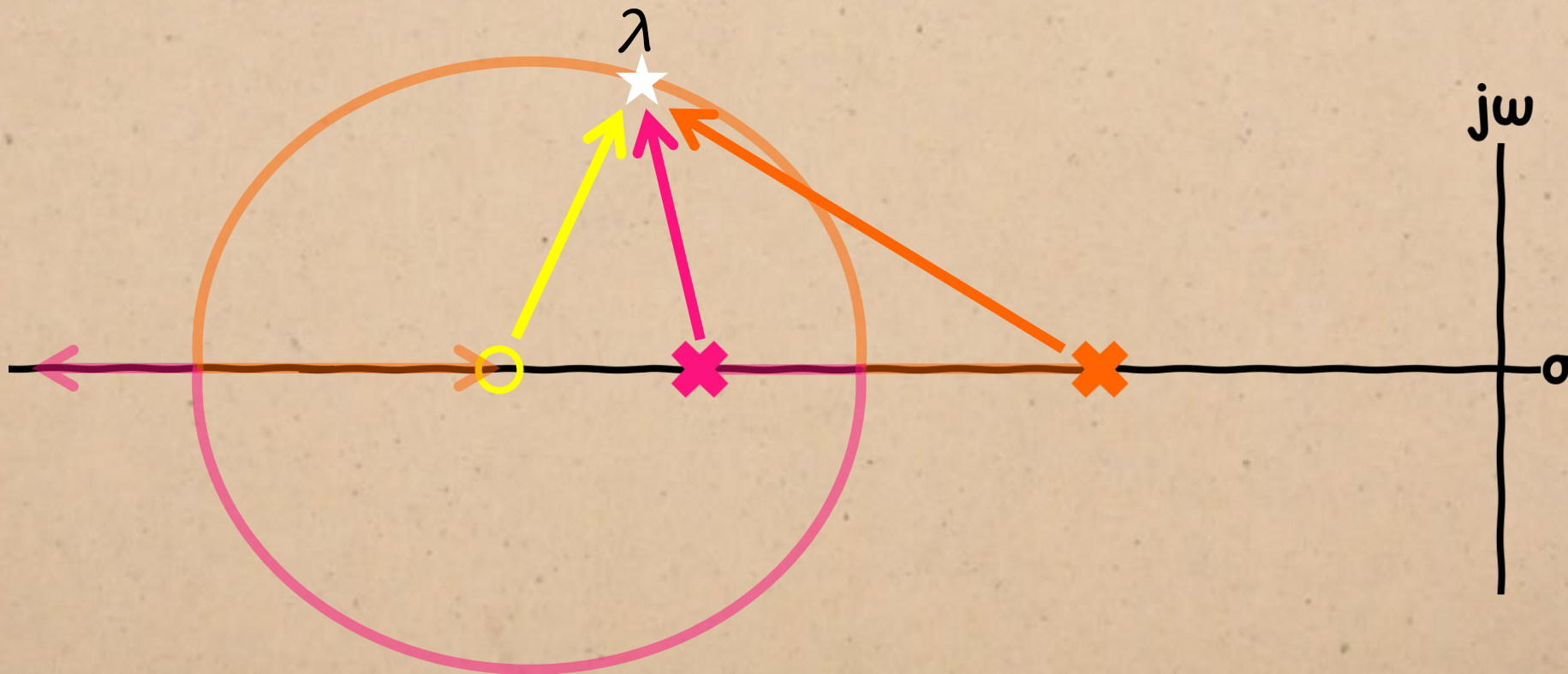
ROOT LOCUS GEOMETRY

$$\frac{K(\lambda - z_1)(\lambda - z_2) \dots (\lambda - z_m)}{(\lambda - p_1)(\lambda - p_2) \dots (\lambda - p_n)} = -1$$



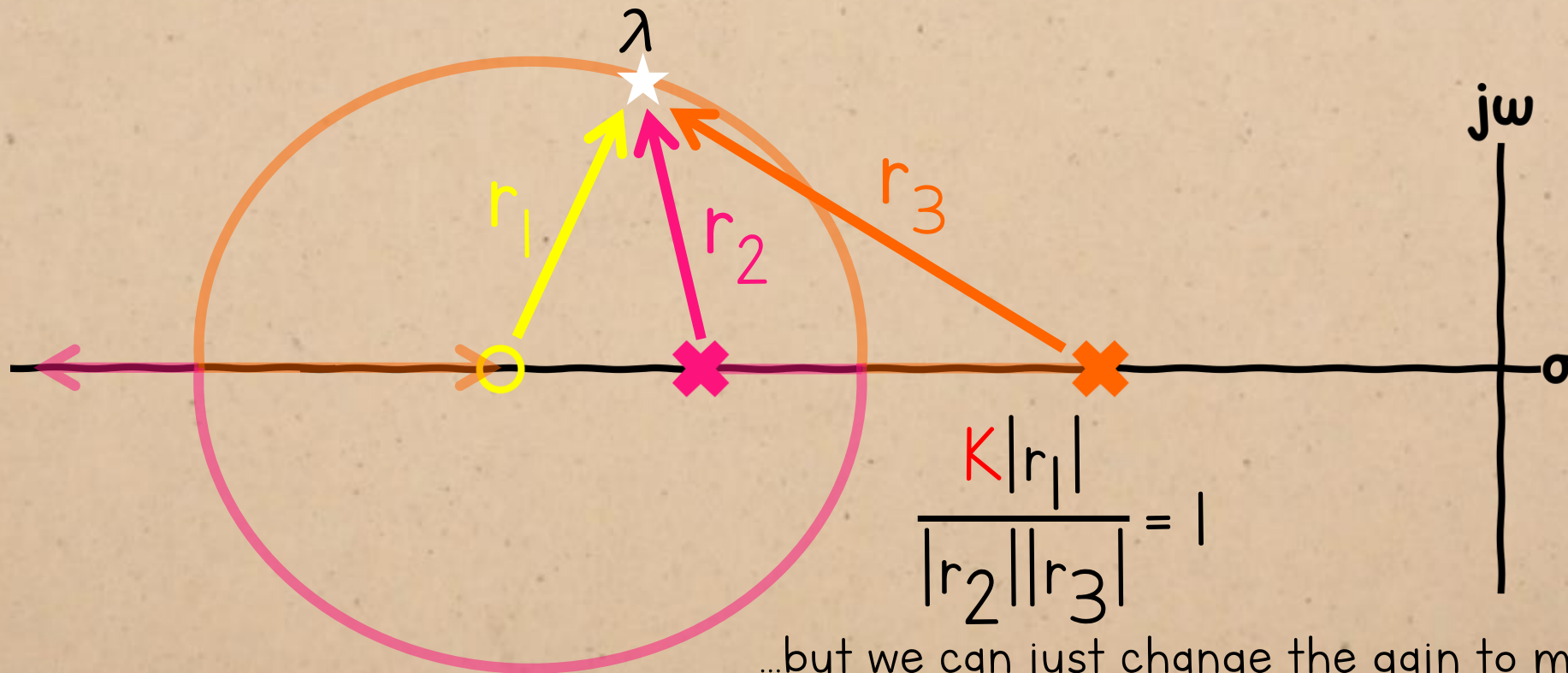
ROOT LOCUS GEOMETRY

$$\frac{K[\text{product of vectors from all zeros to } \lambda]}{[\text{product of vectors from all poles to } \lambda]} = -1$$



ROOT LOCUS GEOMETRY

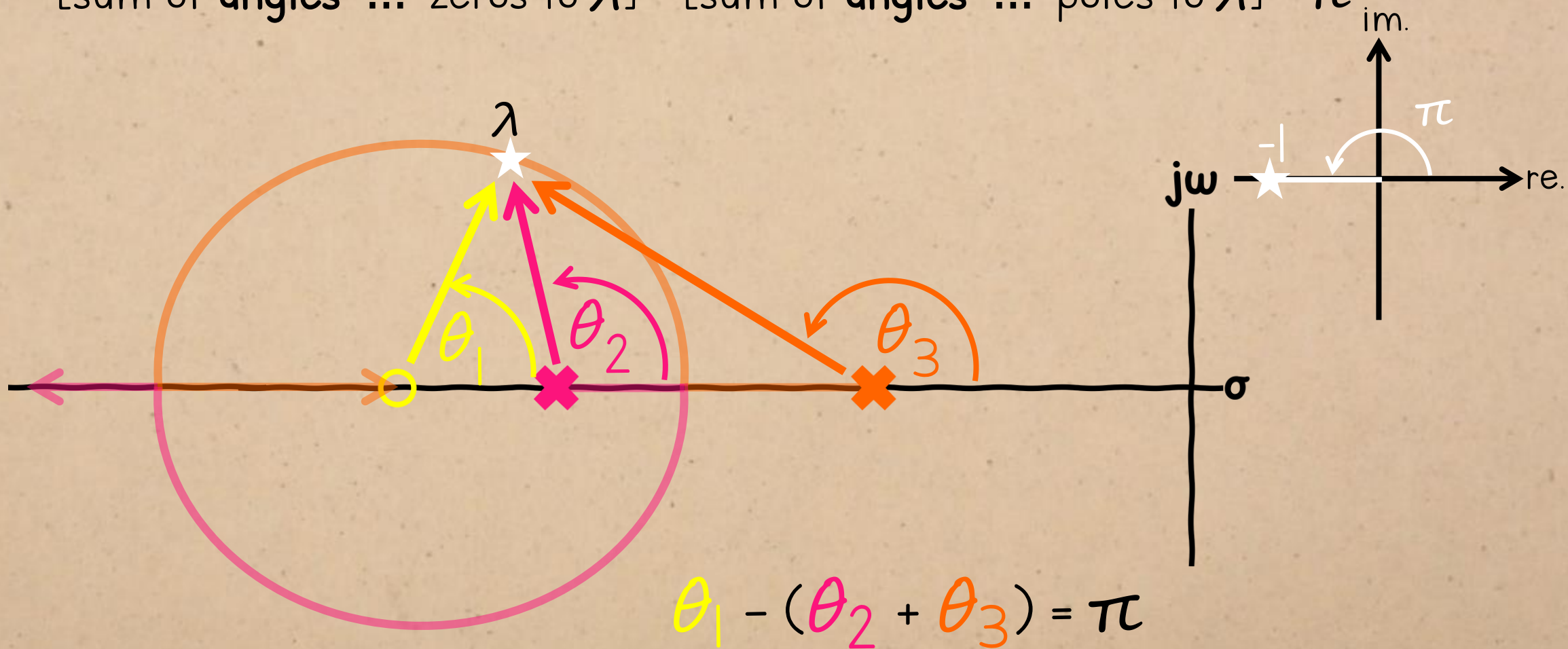
$$\frac{K[\text{product of magnitudes of vectors from all zeros to } \lambda]}{[\text{product of magnitudes of vectors from all poles to } \lambda]} = 1$$



...but we can just change the gain to make this true

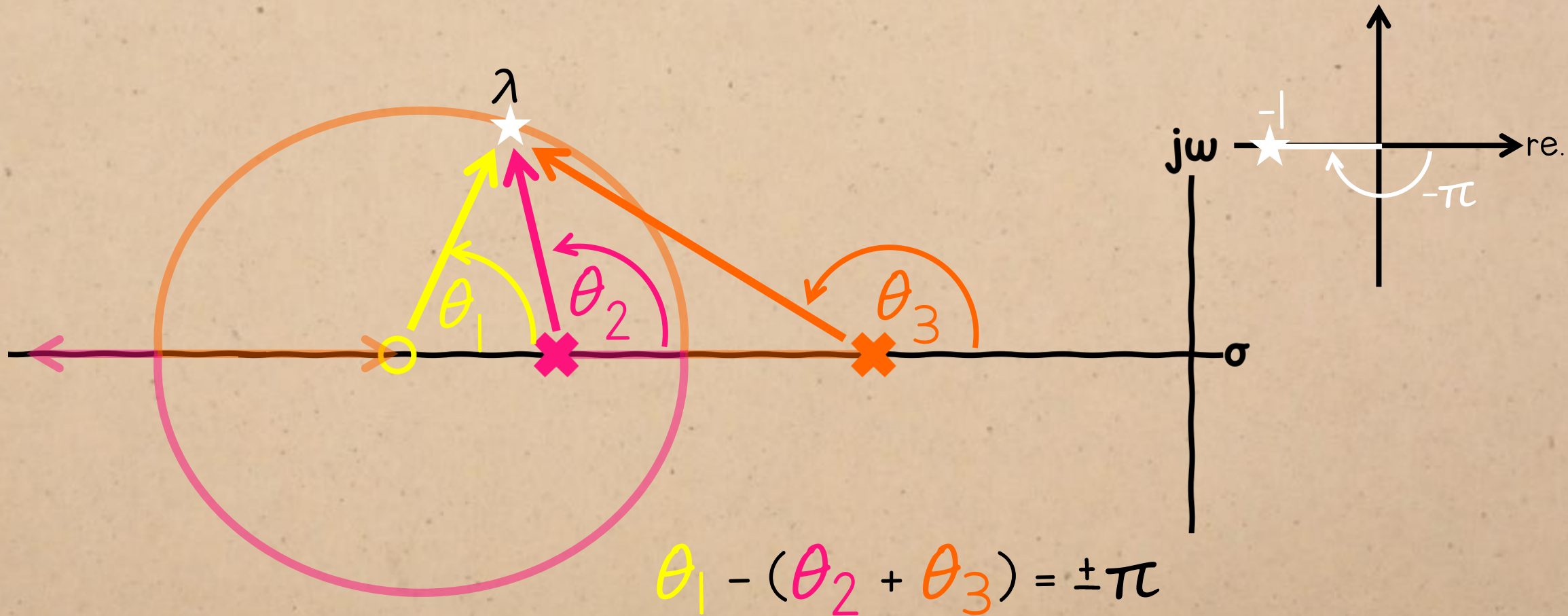
ROOT LOCUS GEOMETRY

[sum of angles ... zeros to λ] - [sum of angles ... poles to λ] = π



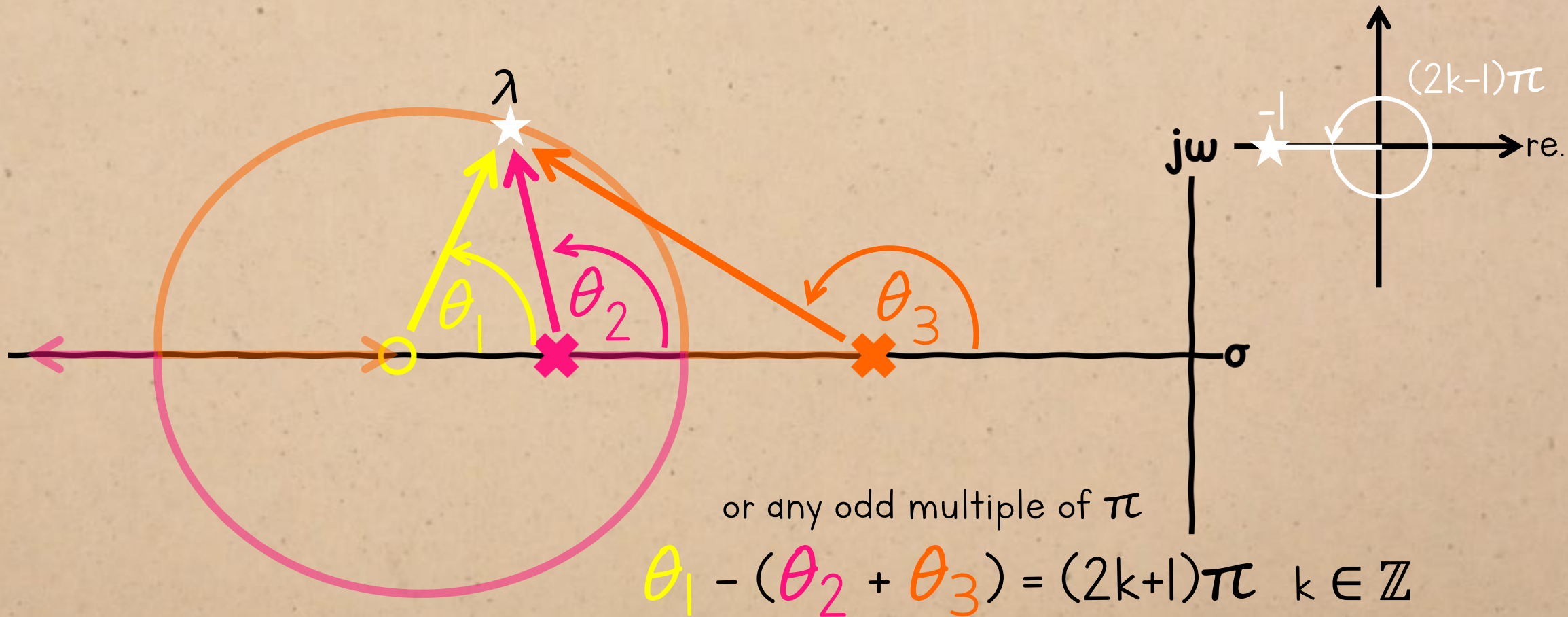
ROOT LOCUS GEOMETRY

[sum of angles ... zeros to λ] - [sum of angles ... poles to λ] = $\pm\pi$ _{im.}



ROOT LOCUS GEOMETRY

$$[\text{sum of angles ... zeros to } \lambda] - [\text{sum of angles ... poles to } \lambda] = (2k+1)\pi_{\text{im.}}$$

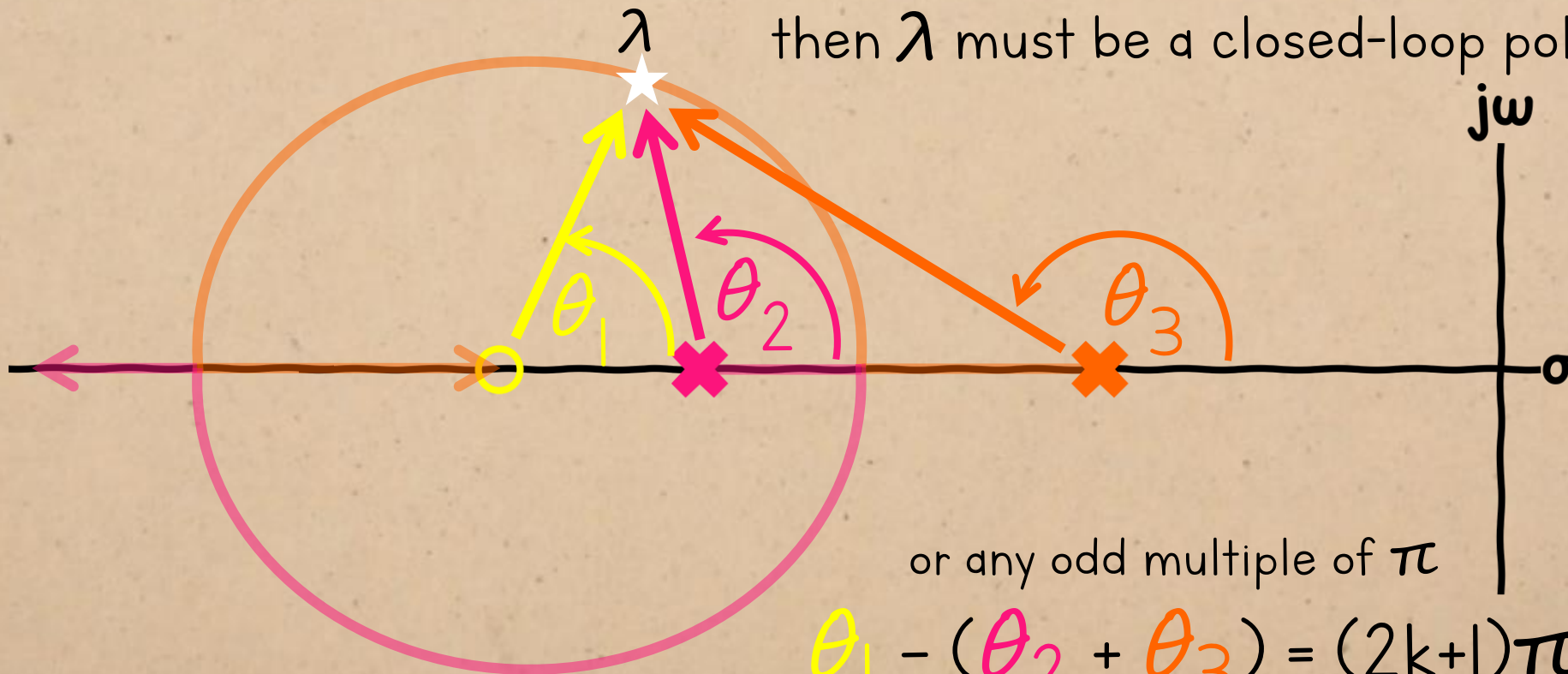


ROOT LOCUS GEOMETRY

Angle criterion:

if [sum of angles ... zeros to λ] - [sum of angles ... poles to λ] = $(2k+1)\pi$

then λ must be a closed-loop pole for some gain.

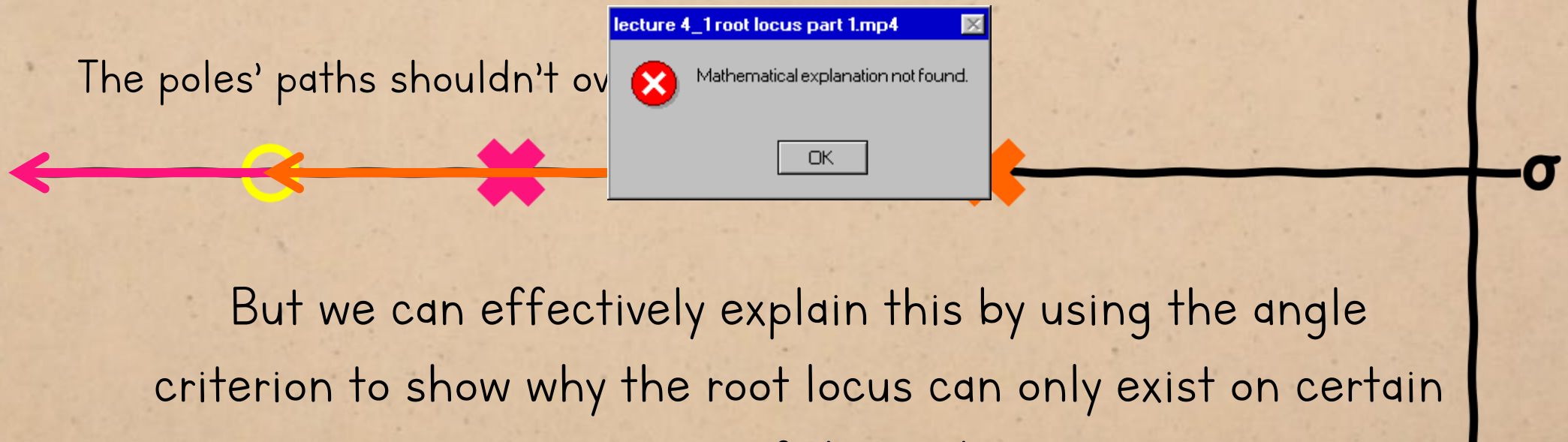


or any odd multiple of π

$$\theta_1 - (\theta_2 + \theta_3) = (2k+1)\pi \quad k \in \mathbb{Z}$$

LAST TIME:

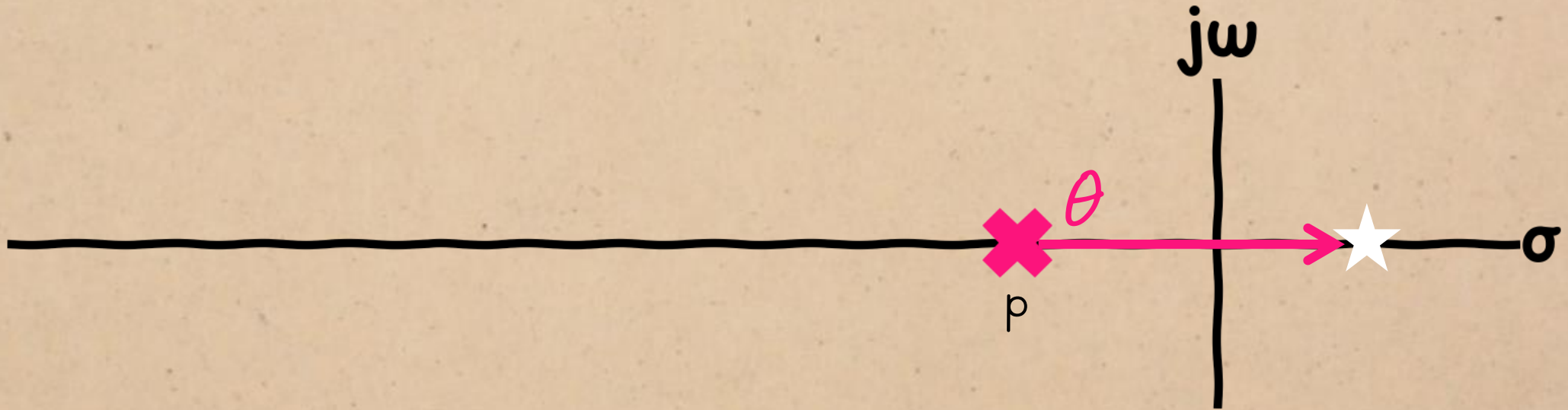
We didn't have an explanation for why the poles' paths don't overlap.



But we can effectively explain this by using the angle criterion to show why the root locus can only exist on certain segments of the real axis.

REAL AXIS SEGMENTS

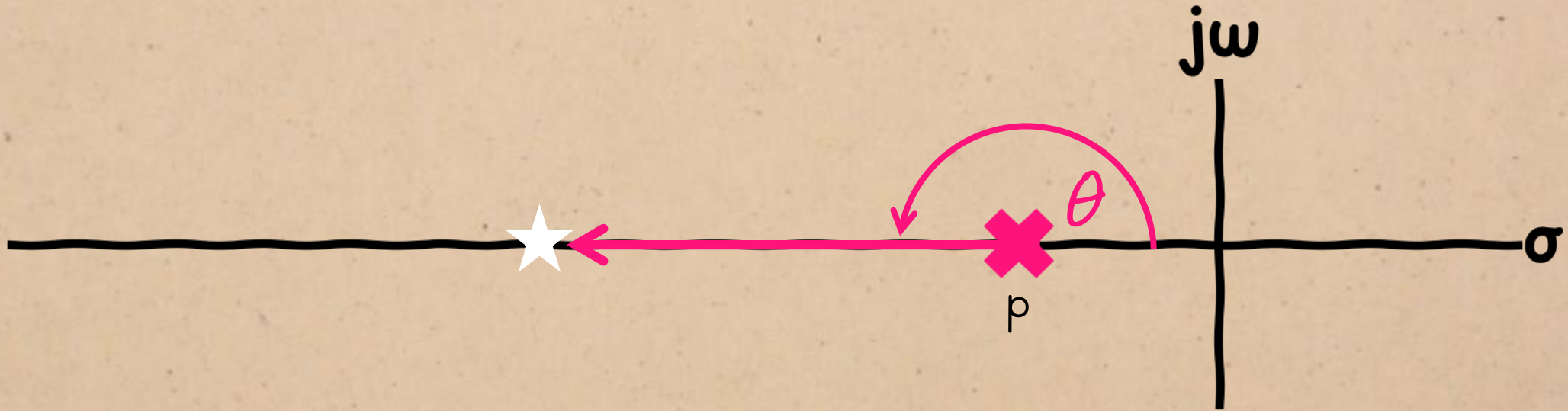
open-loop transfer function: $\frac{1}{s - p}$



The angle sum = 0 for all points to the right $\therefore$ they can't be closed-loop poles.

REAL AXIS SEGMENTS

open-loop transfer function: $\frac{1}{s - p}$



The angle sum = 0 for all points to the right $\therefore$ they can't be closed-loop poles.

The angle sum = π for all points to the left $\therefore$ they are on the root locus.

REAL AXIS SEGMENTS

open-loop transfer function: $\frac{1}{s - p}$



... so even if it wasn't already clear from this,

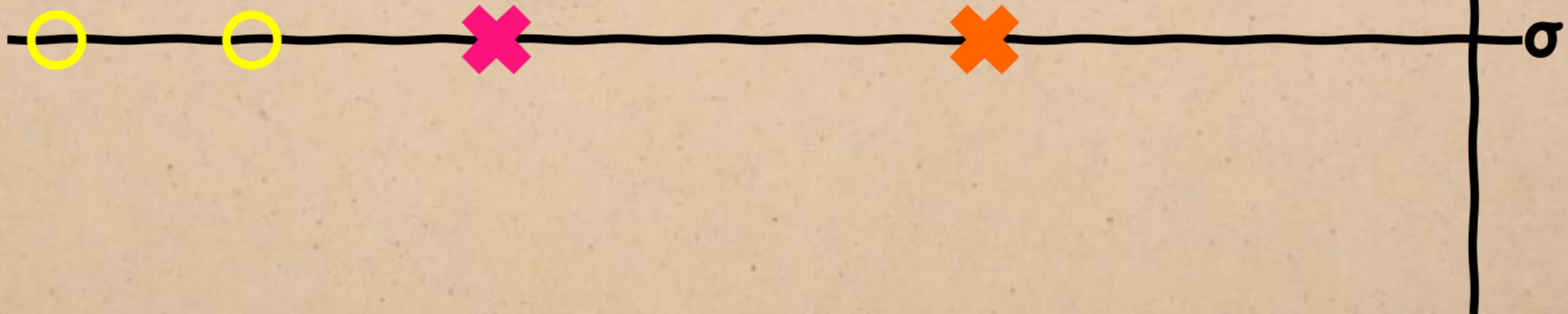
characteristic equation: $K + s - p = 0$

the angles show where the closed-loop poles can go.

REAL AXIS SEGMENTS

open-loop transfer function: $\frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)}$

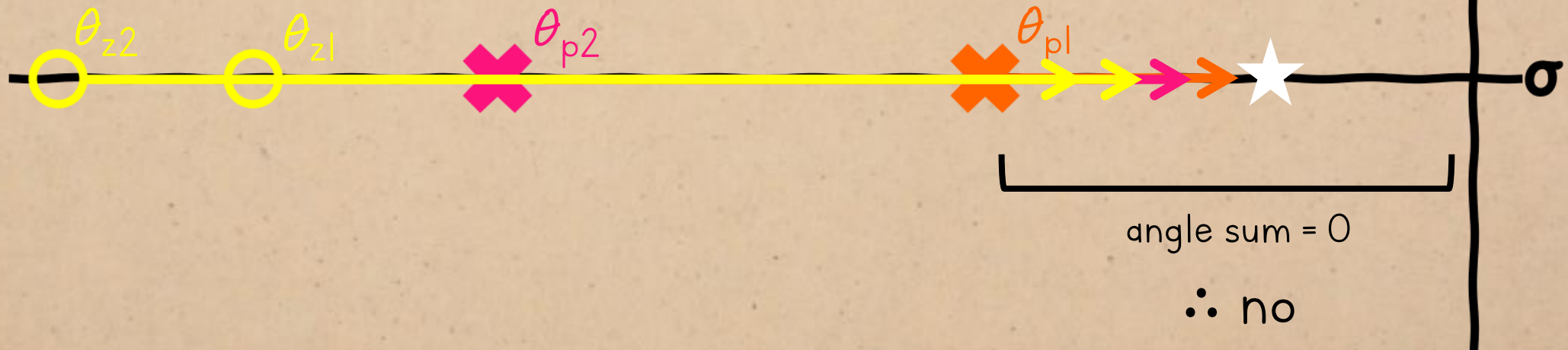
We can tell if a section of the real axis is on the root locus based on how many **critical frequencies** (poles or zeros) are on its **right**.



REAL AXIS SEGMENTS

open-loop transfer function: $\frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)}$

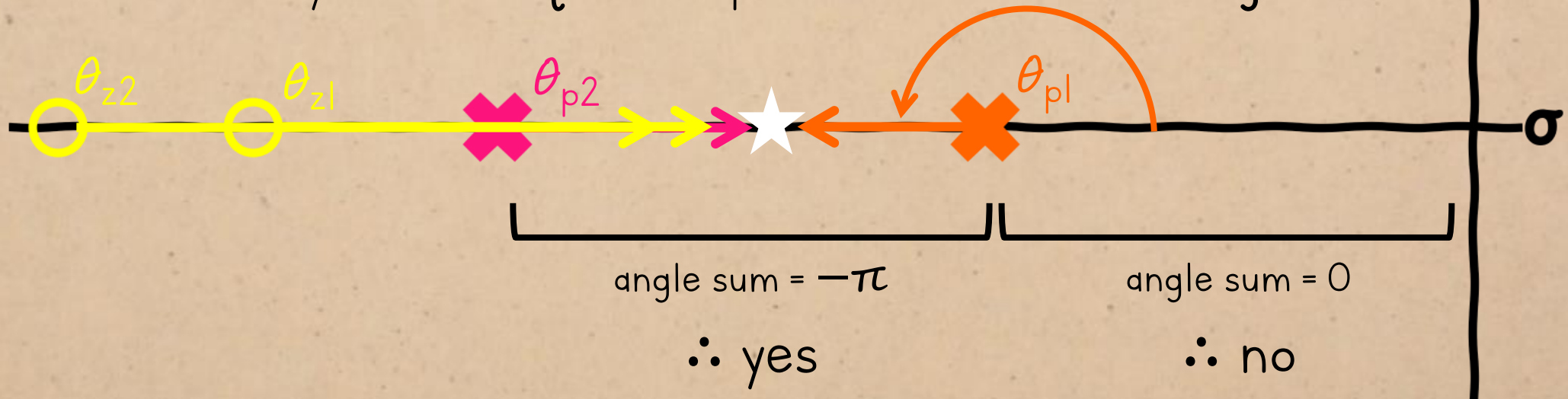
We can tell if a section of the real axis is on the root locus based on how many **critical frequencies** (poles or zeros) are on its **right**.



REAL AXIS SEGMENTS

open-loop transfer function: $\frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)}$

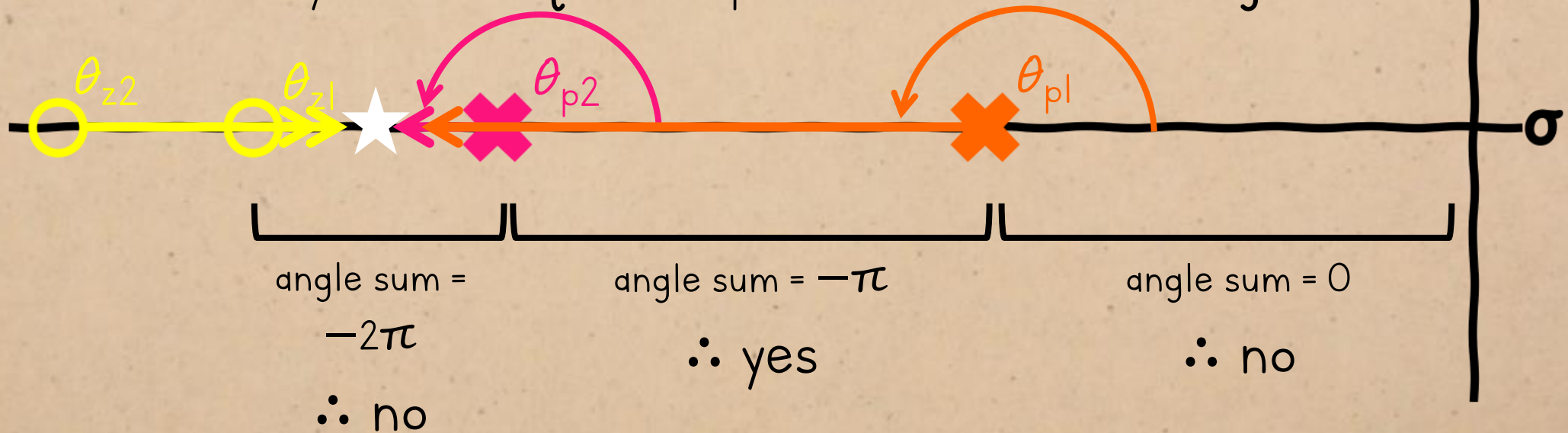
We can tell if a section of the real axis is on the root locus based on how many **critical frequencies** (poles or zeros) are on its **right**.



REAL AXIS SEGMENTS

open-loop transfer function: $\frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)}$

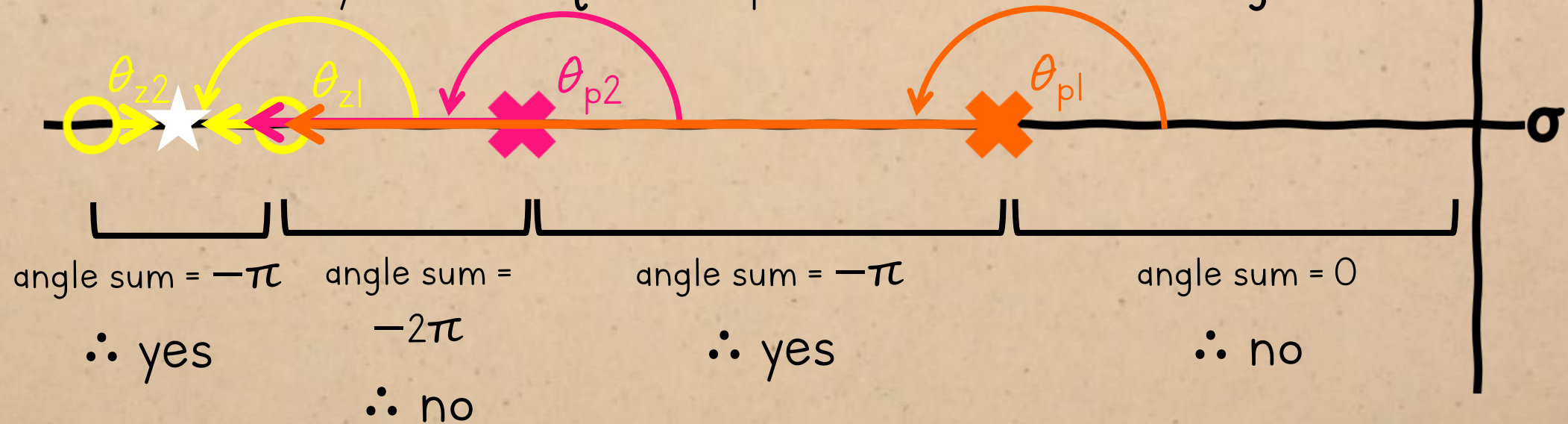
We can tell if a section of the real axis is on the root locus based on how many **critical frequencies** (poles or zeros) are on its **right**.



REAL AXIS SEGMENTS

open-loop transfer function: $\frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)}$

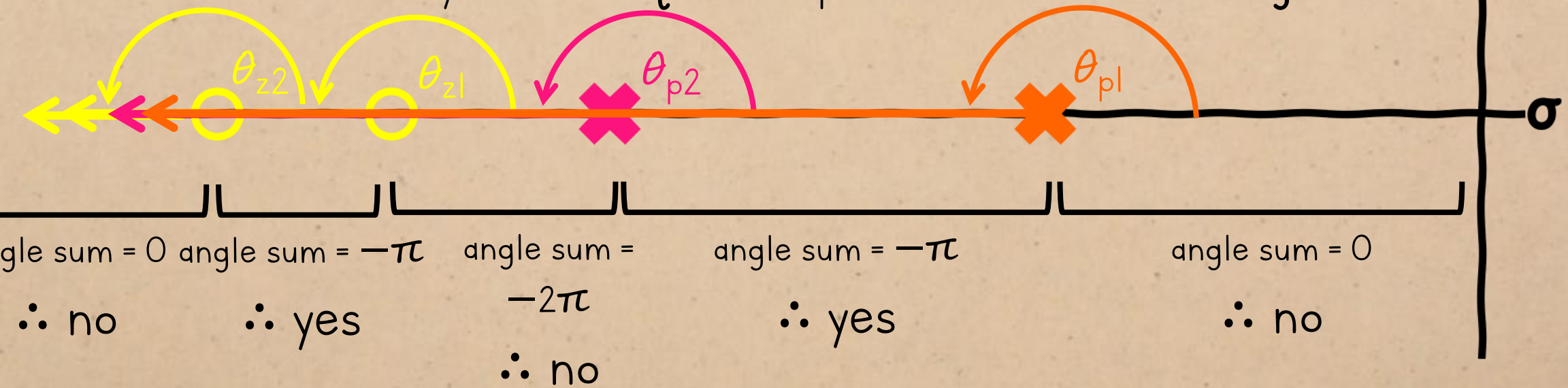
We can tell if a section of the real axis is on the root locus based on how many **critical frequencies** (poles or zeros) are on its **right**.



REAL AXIS SEGMENTS

open-loop transfer function: $\frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)}$

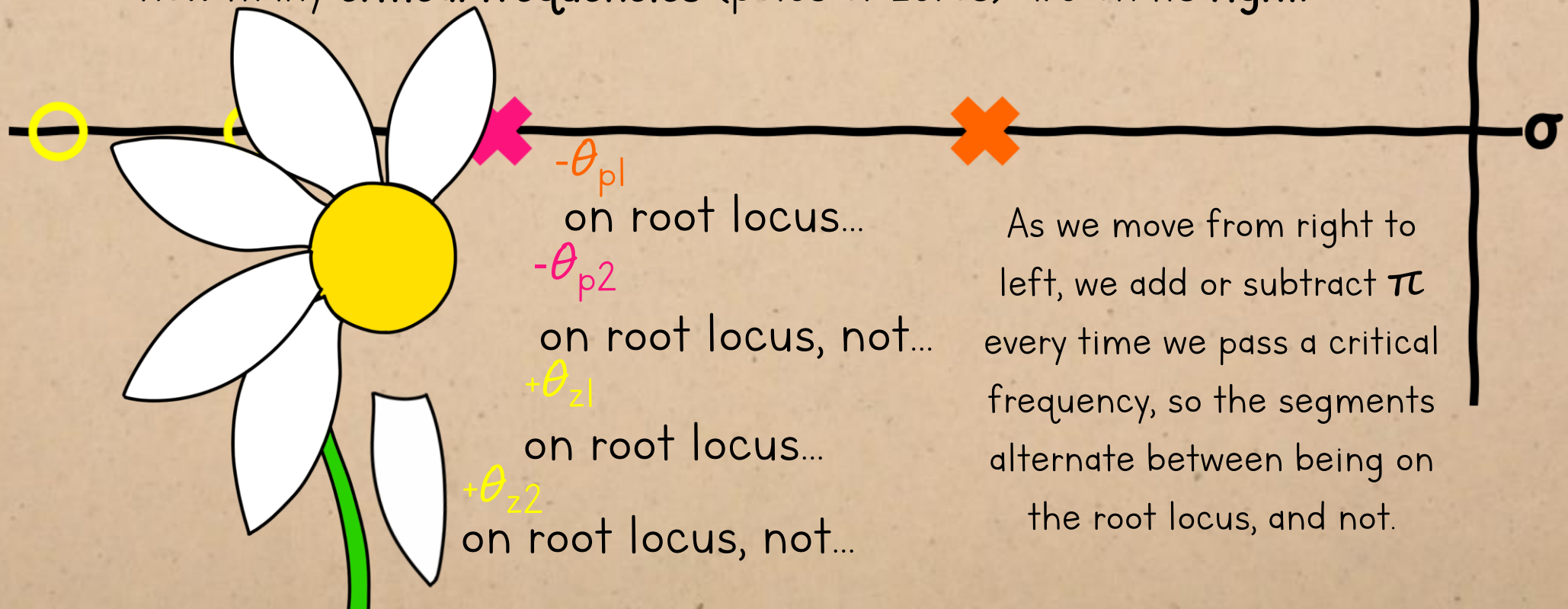
We can tell if a section of the real axis is on the root locus based on how many **critical frequencies** (poles or zeros) are on its **right**.



REAL AXIS SEGMENTS

open-loop transfer function: $\frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)}$

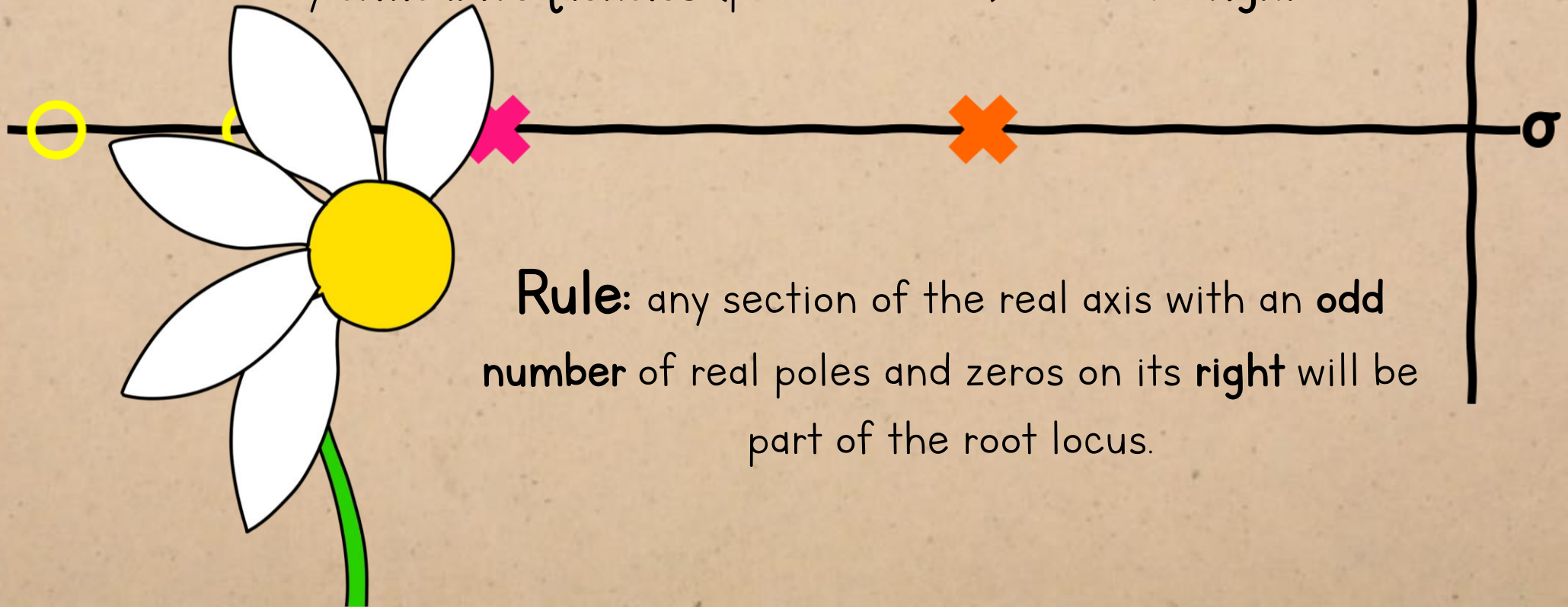
We can tell if a section of the real axis is on the root locus based on how many **critical frequencies** (poles or zeros) are on its **right**.



REAL AXIS SEGMENTS

open-loop transfer function: $\frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)}$

We can tell if a section of the real axis is on the root locus based on how many **critical frequencies** (poles or zeros) are on its **right**.



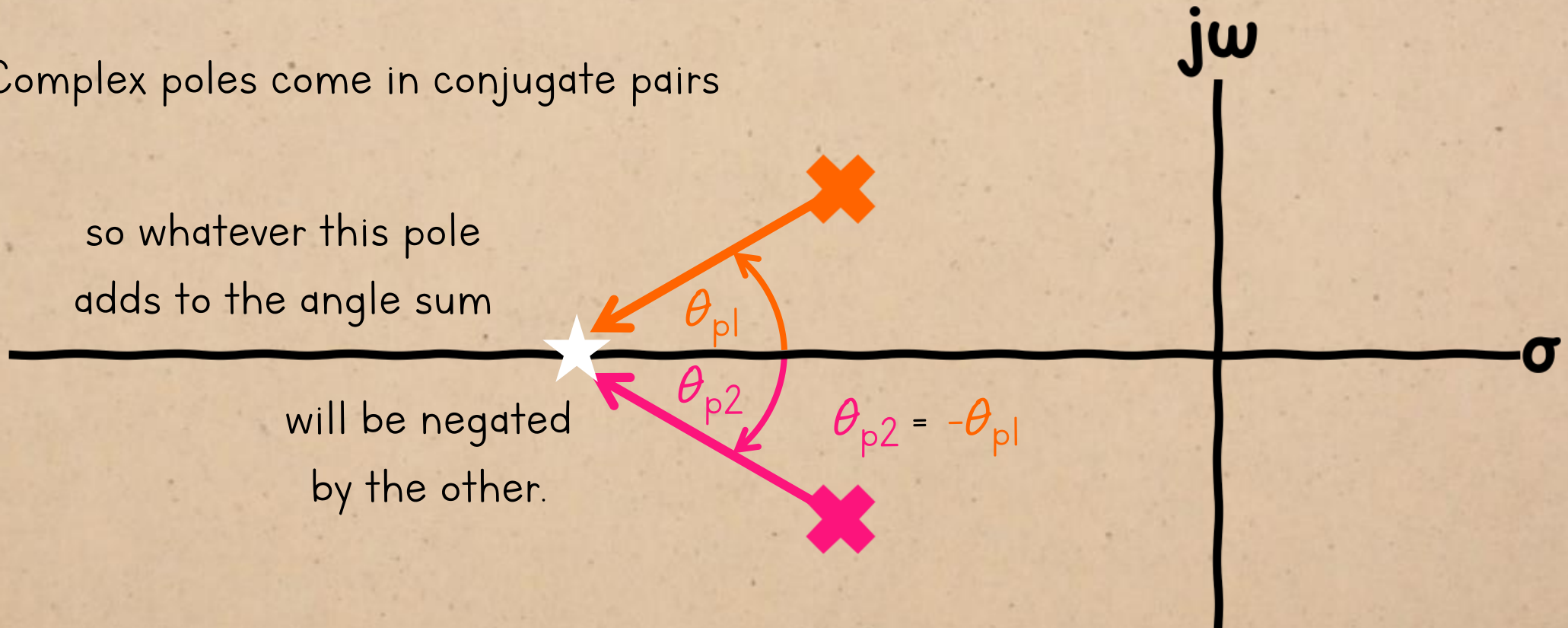
Rule: any section of the real axis with an **odd** number of real poles and zeros on its **right** will be part of the root locus.

REAL AXIS SEGMENTS

Complex poles come in conjugate pairs

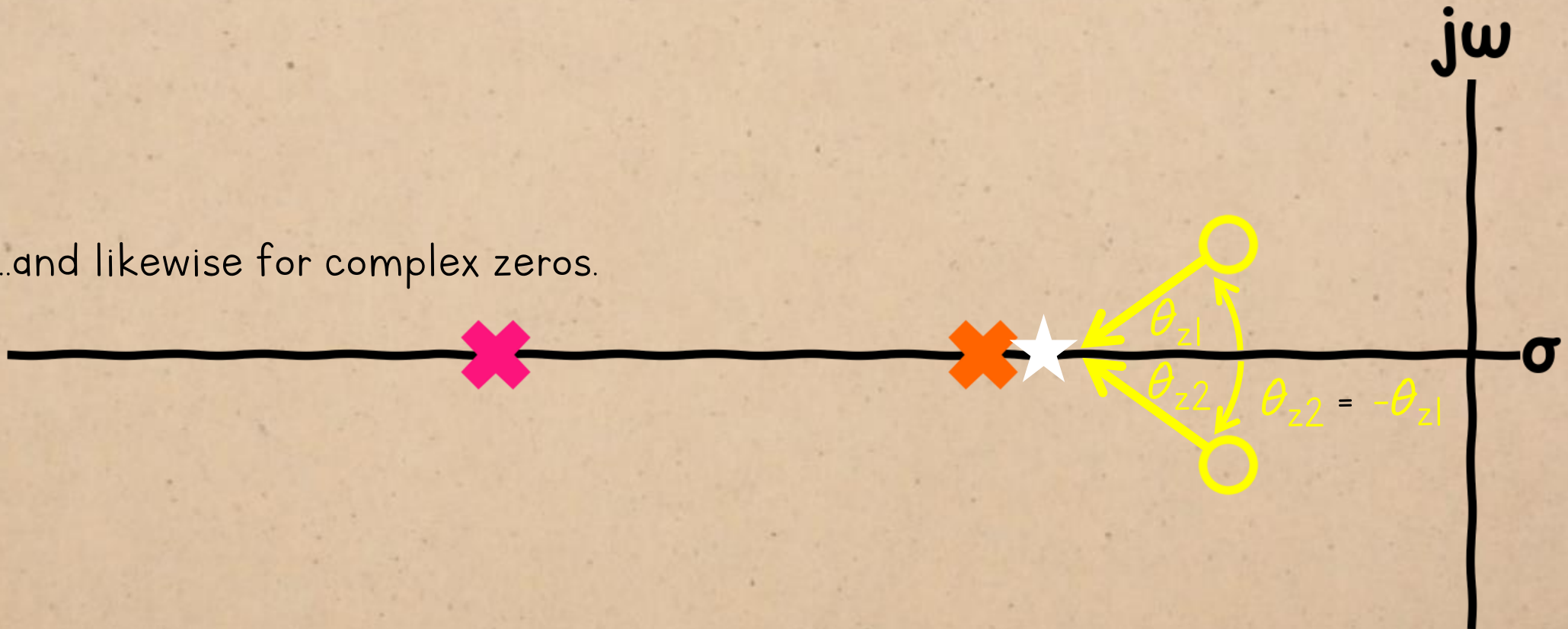
so whatever this pole
adds to the angle sum

will be negated
by the other.

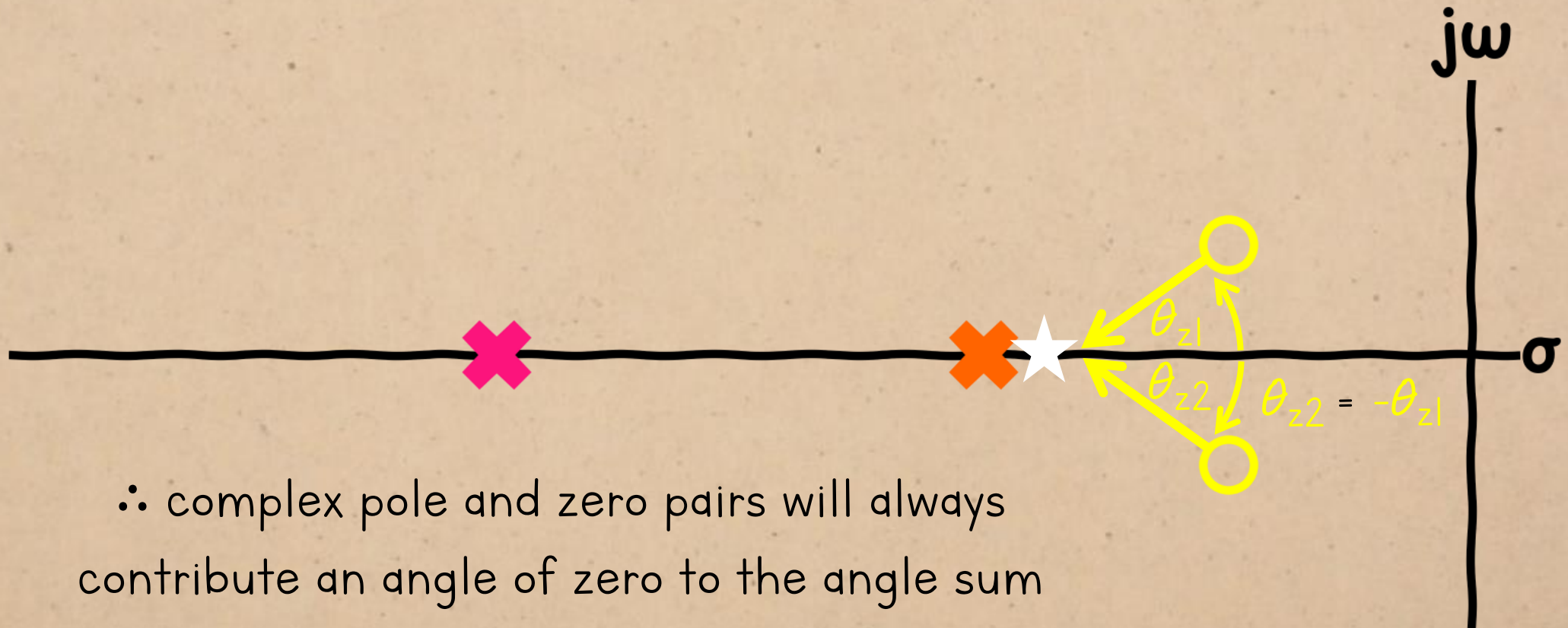


REAL AXIS SEGMENTS

...and likewise for complex zeros.



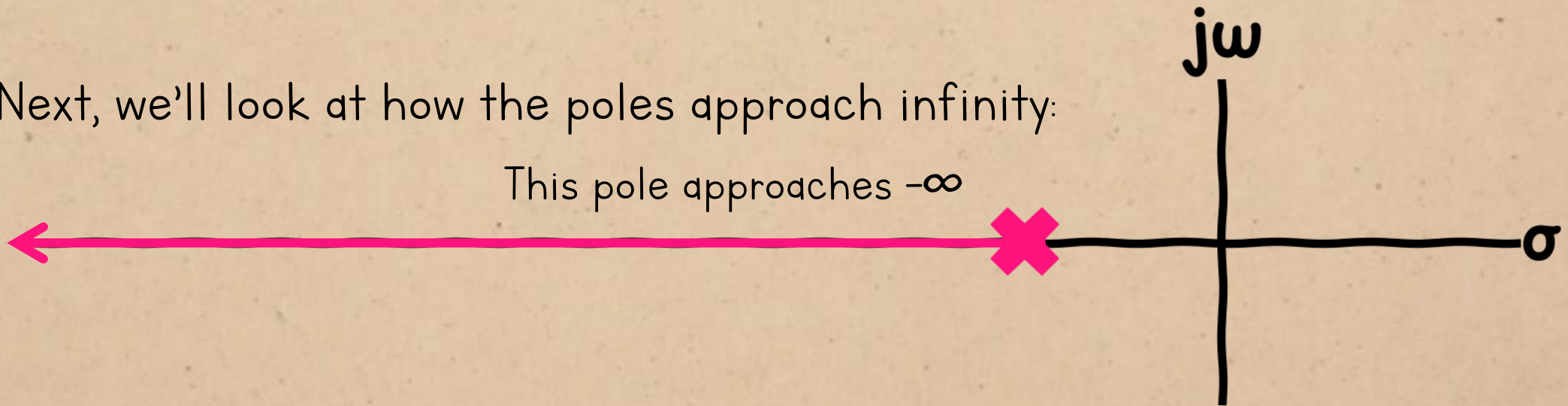
REAL AXIS SEGMENTS

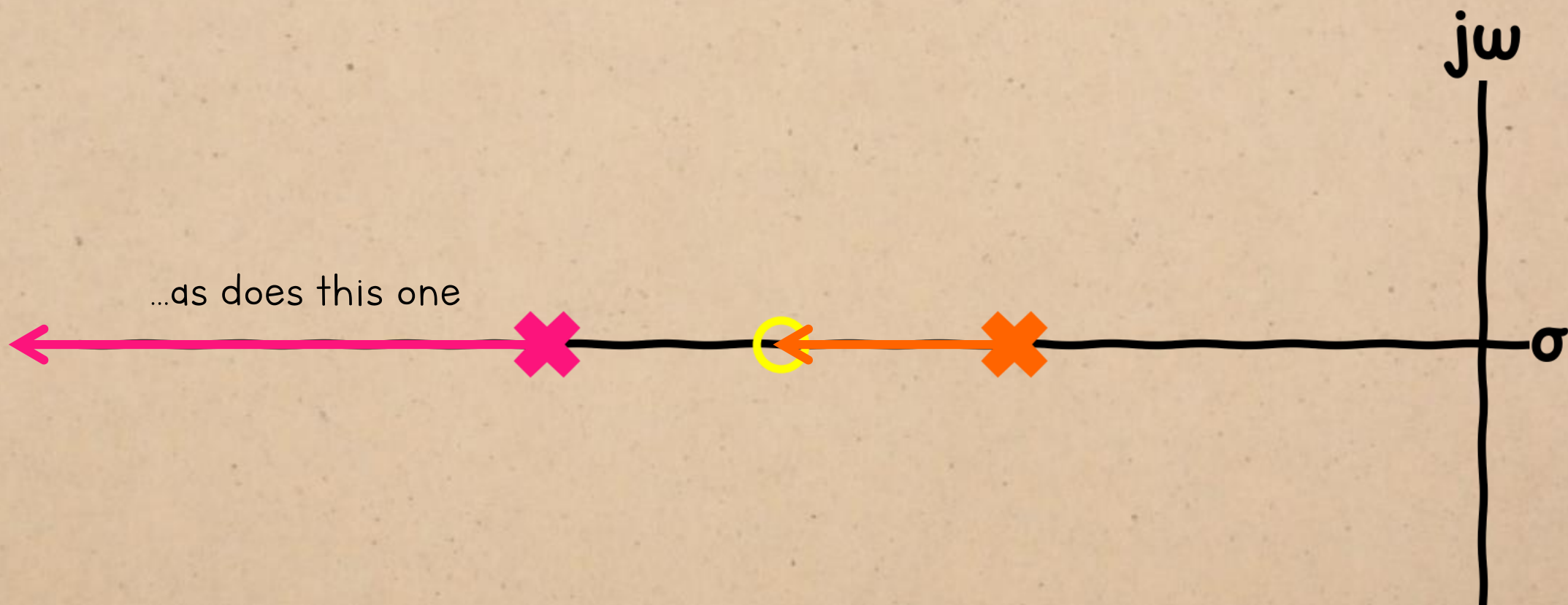


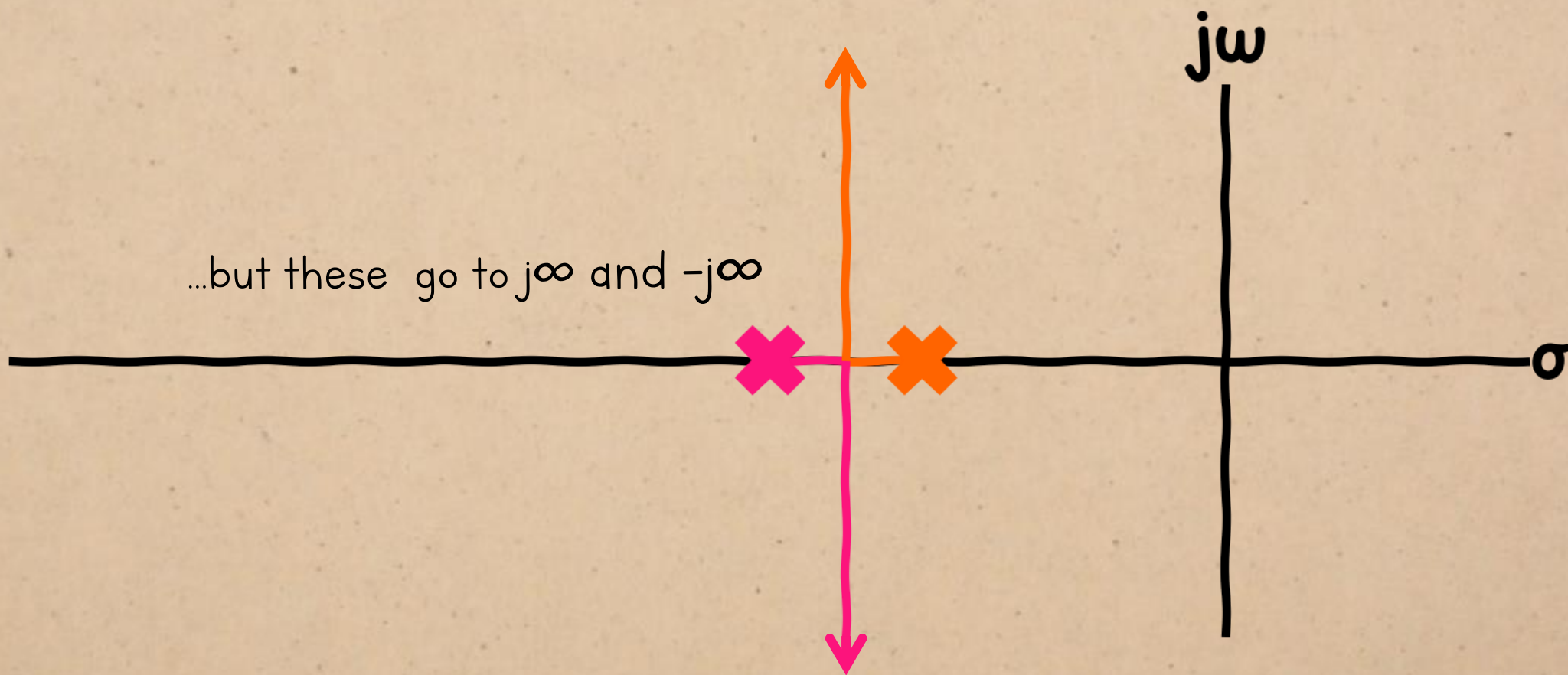
$\therefore$ complex pole and zero pairs will always contribute an angle of zero to the angle sum for points on the real axis.

Next, we'll look at how the poles approach infinity:

This pole approaches $-\infty$









...but these go to $j\infty$ and $-j\infty$

INFINITY

INFINITY

INFINITY

INFINITY

$j\omega$

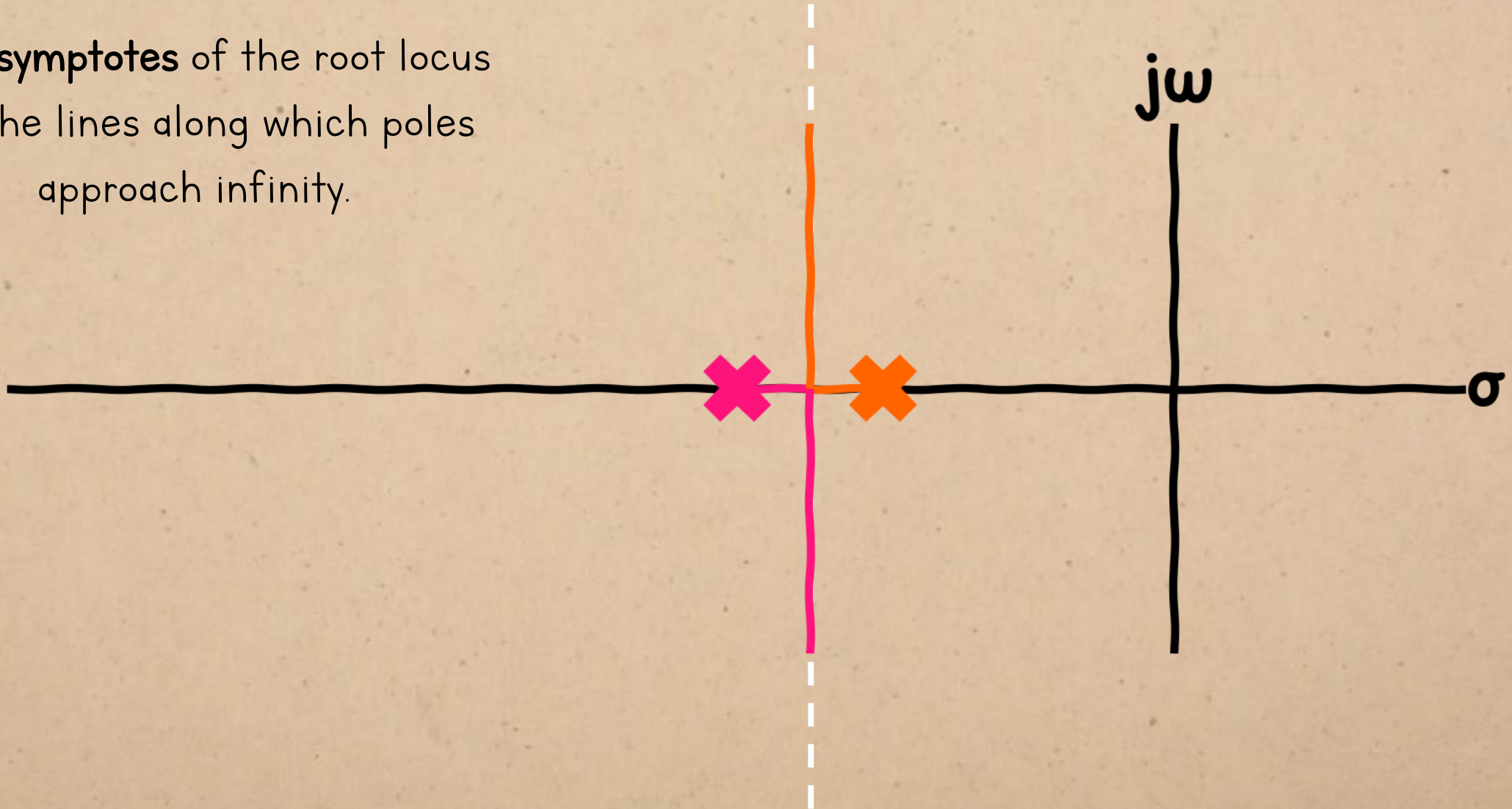
σ

so how do we know which way
each pole must go to infinity?



ASYMPTOTES

The **asymptotes** of the root locus are the lines along which poles approach infinity.



ASYMPTOTES

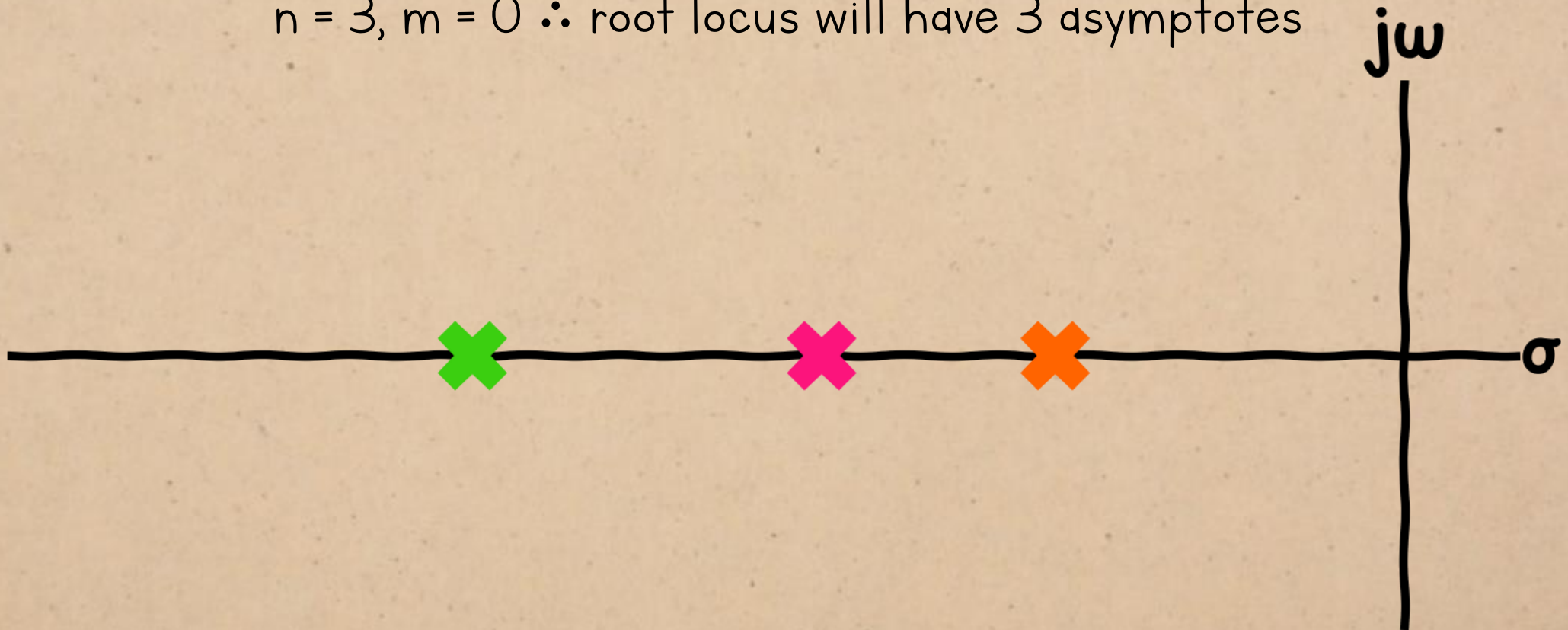
Only the poles that don't have zeros to go to will travel to infinity,
so a system with m zeros and n poles

$$\frac{(s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)}$$

will have **$n - m$** asymptotes.

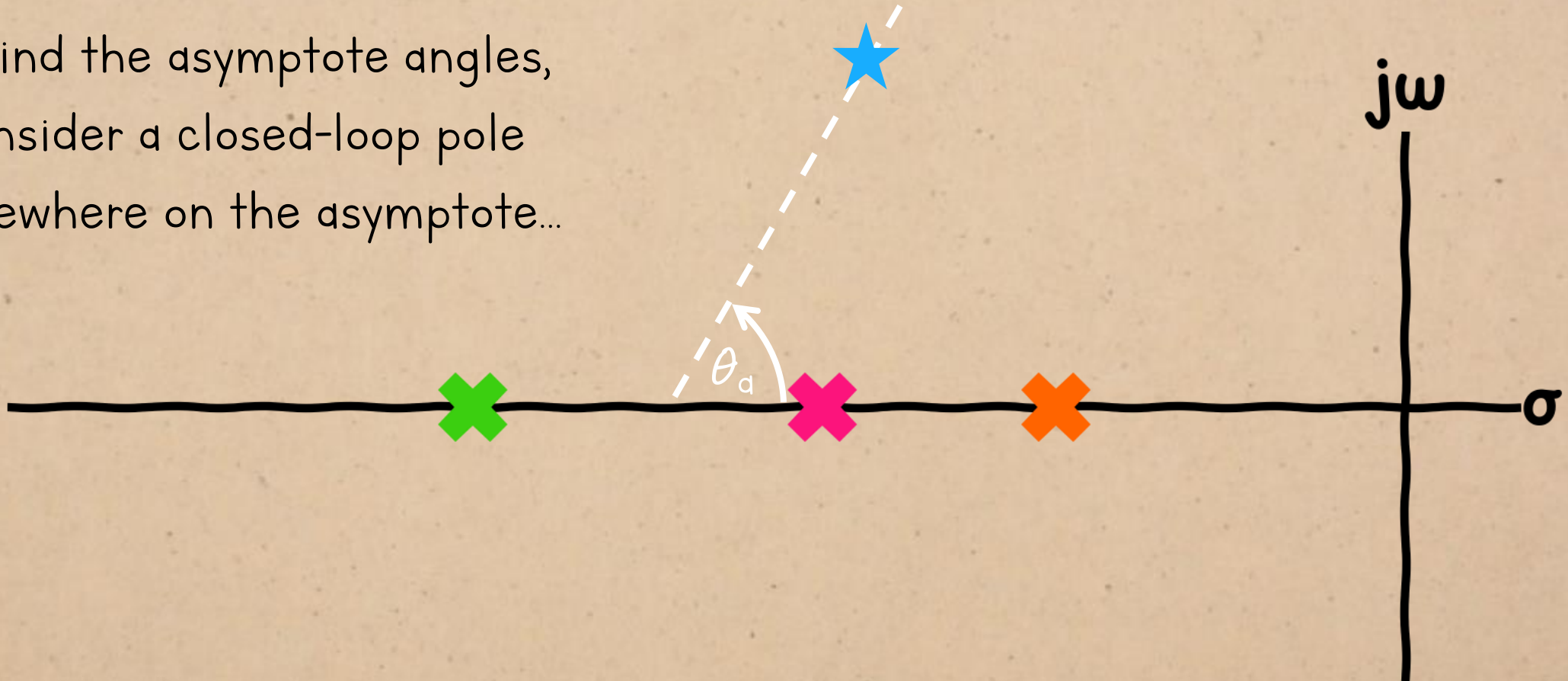
ASYMPTOTES

$n = 3, m = 0 \therefore$ root locus will have 3 asymptotes

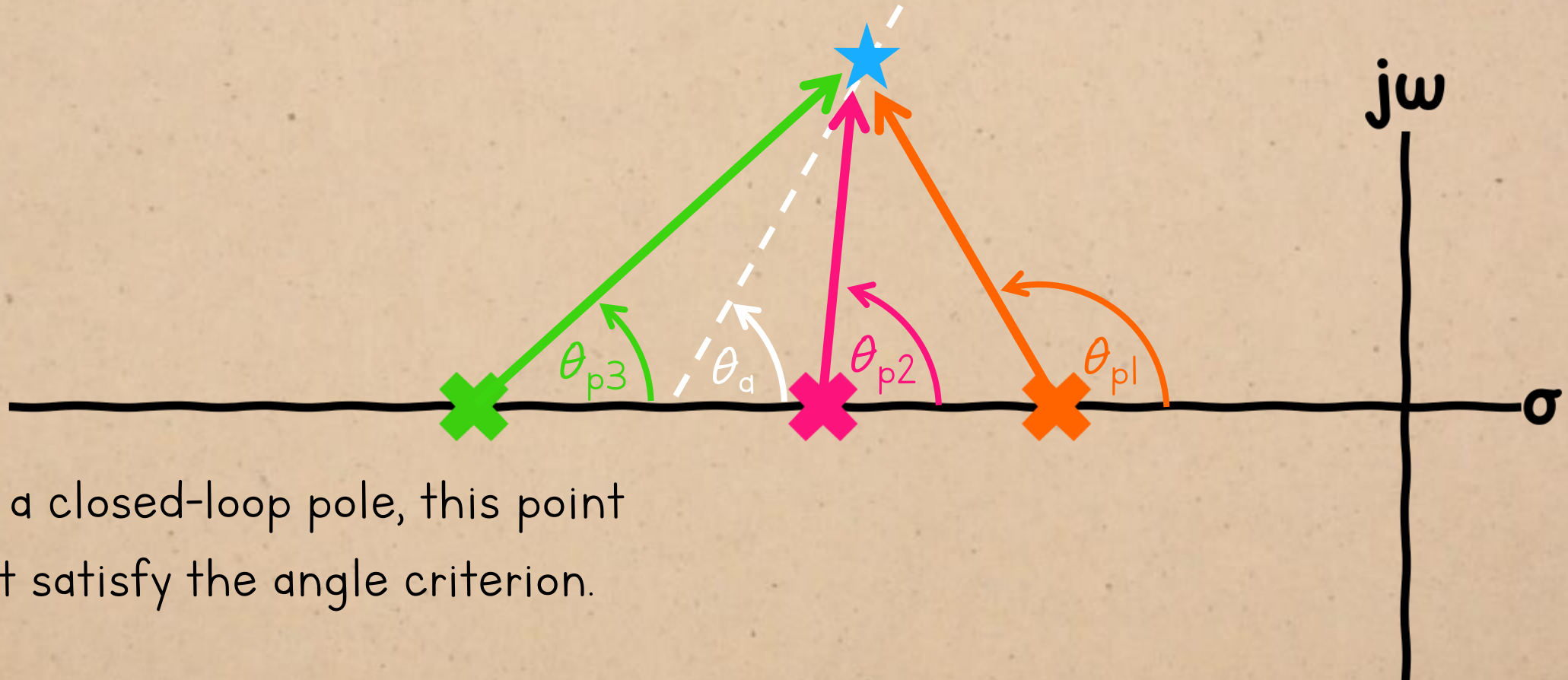


ASYMPTOTES

To find the asymptote angles,
consider a closed-loop pole
somewhere on the asymptote...



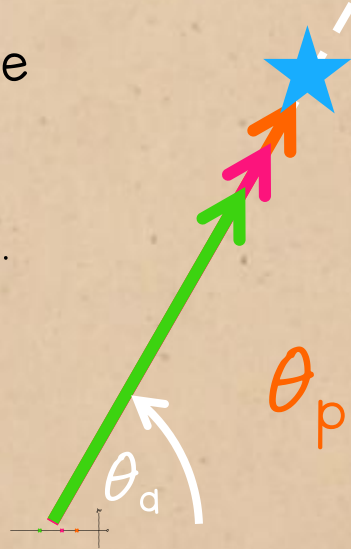
ASYMPTOTES



To be a closed-loop pole, this point must satisfy the angle criterion.

ASYMPTOTES

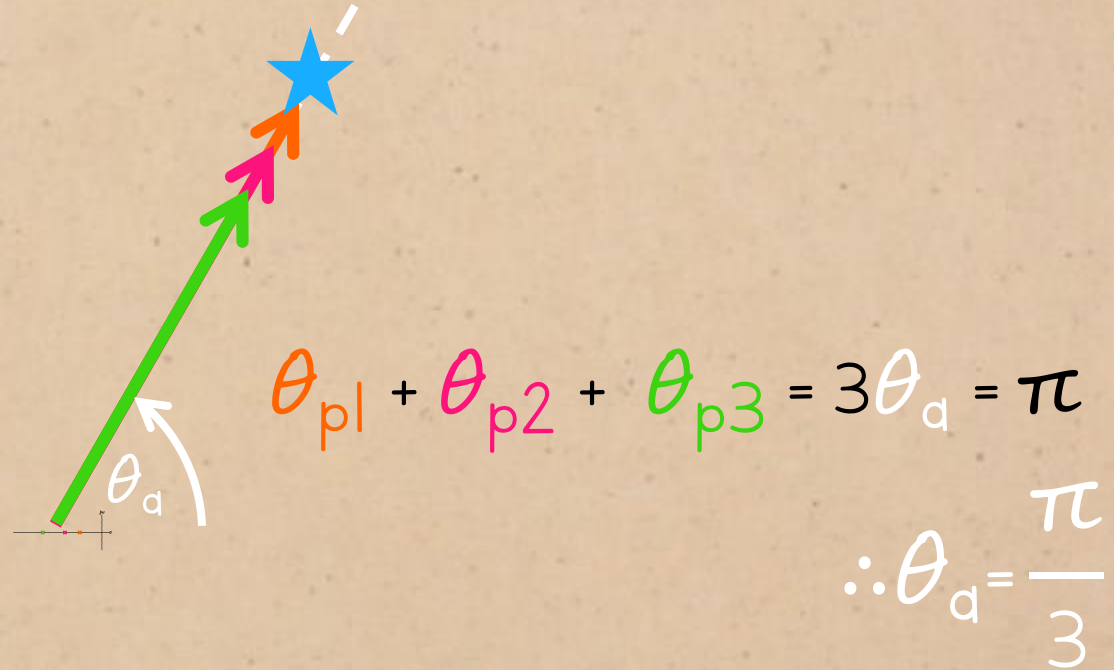
As the point approaches infinity along the asymptote, the distances between the open-loop poles appear negligibly small...



$$\theta_{p1} = \theta_{p2} = \theta_{p3} = \theta_a$$

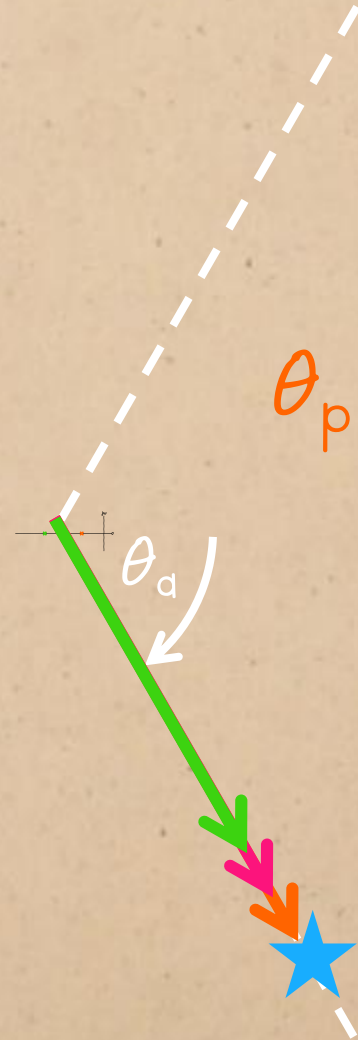
so all the vectors from the poles
tend to the same angle
(which is also the asymptote angle)

ASYMPTOTES



(if you're wondering why the pole angles are positive: because of symmetry, it doesn't actually matter if we add the zero angles and subtract the pole angles or add the pole angles and subtract the zero angles.)

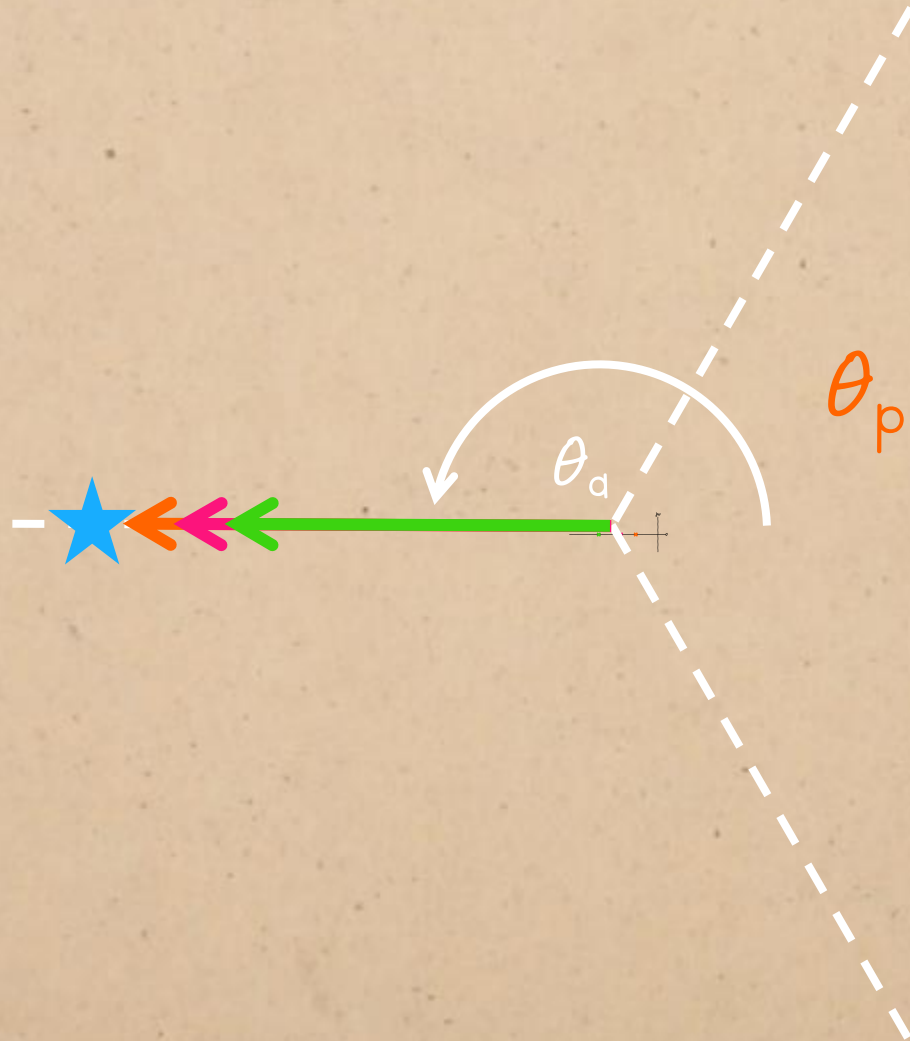
ASYMPTOTES



$$\theta_{p1} + \theta_{p2} + \theta_{p3} = 3\theta_d = -\pi$$

$$\therefore \theta_d = -\frac{\pi}{3}$$

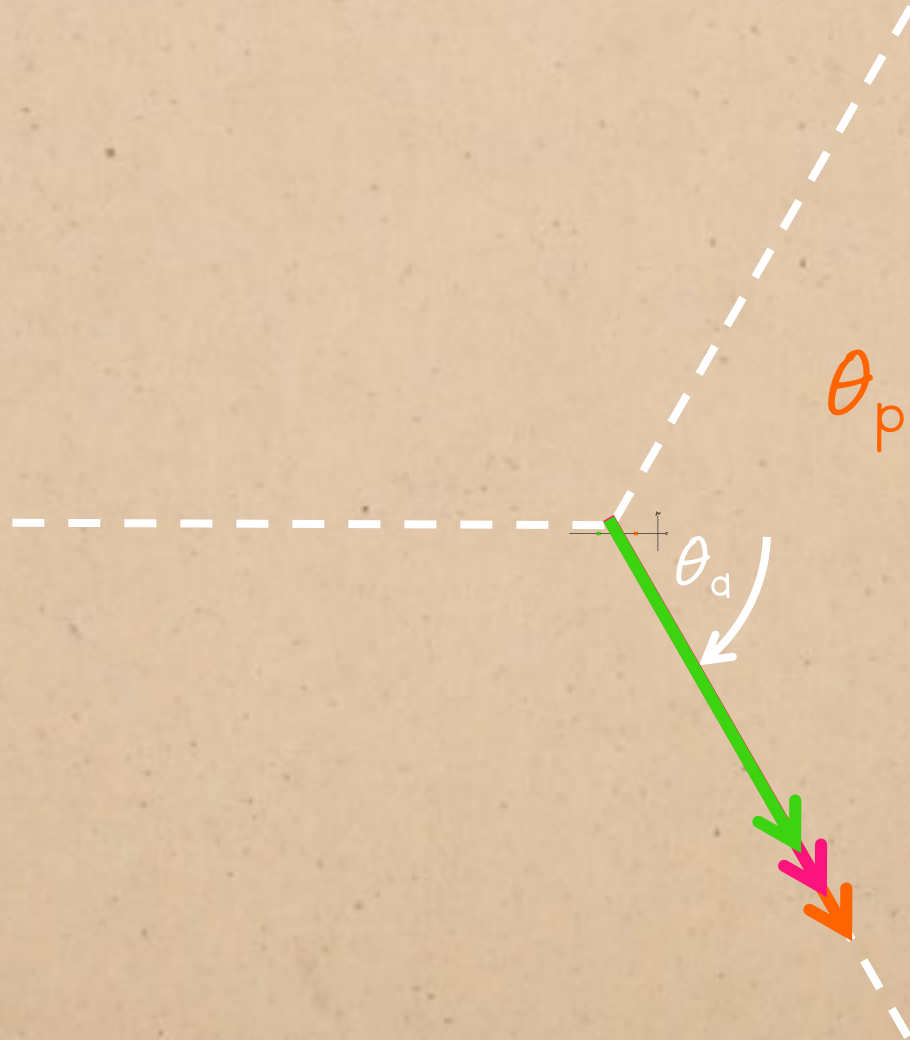
ASYMPTOTES



$$\theta_{p1} + \theta_{p2} + \theta_{p3} = 3\theta_a = 3\pi$$

$$\therefore \theta_a = \pi$$

ASYMPTOTES

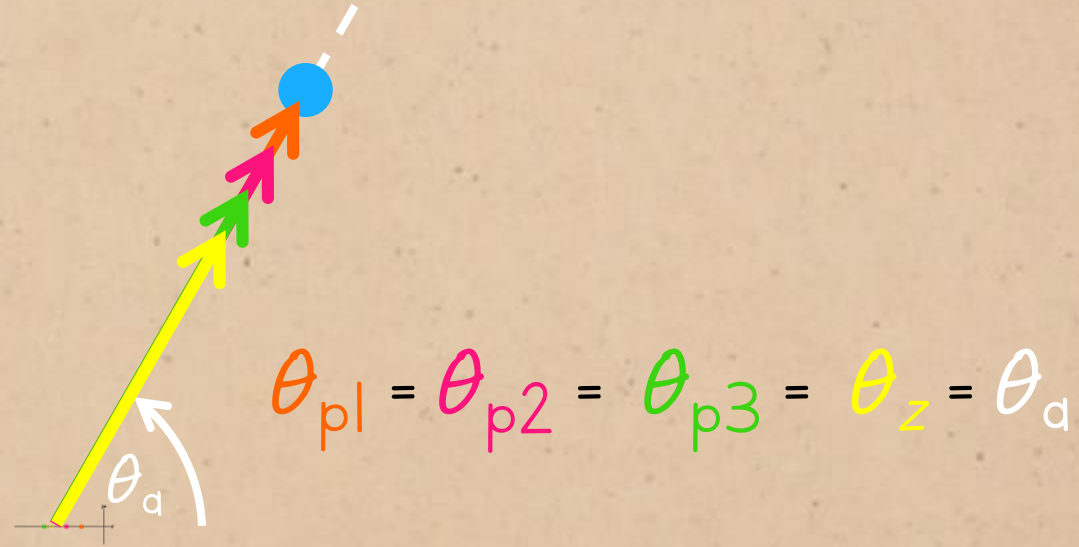


$$\theta_{p1} + \theta_{p2} + \theta_{p3} = 3\theta_d = 5\pi$$

$$\therefore \theta_d = \frac{5\pi}{3}$$

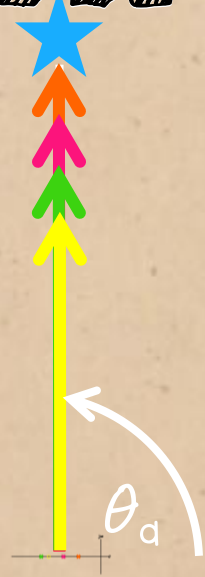
(any larger multiples of π will just give one of the angles we already have)

ASYMPTOTES



If there are zeros, their angles will also be the same as the asymptote angle.

ASYMPTOTES



...but they get subtracted:

$$\theta_{p1} + \theta_{p2} + \theta_{p3} - \theta_z = 2\theta_d = \pi$$

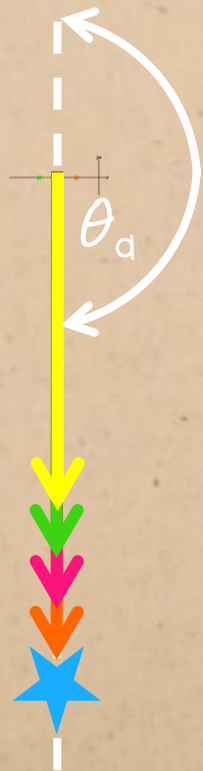
$$\therefore \theta_d = \frac{\pi}{2}$$

ASYMPTOTES

...but they get subtracted:

$$\theta_{p1} + \theta_{p2} + \theta_{p3} - \theta_z = 2\theta_d = -\pi$$

$$\therefore \theta_d = -\frac{\pi}{2}$$



ASYMPTOTES

Rule: the angles of the asymptotes can be determined using

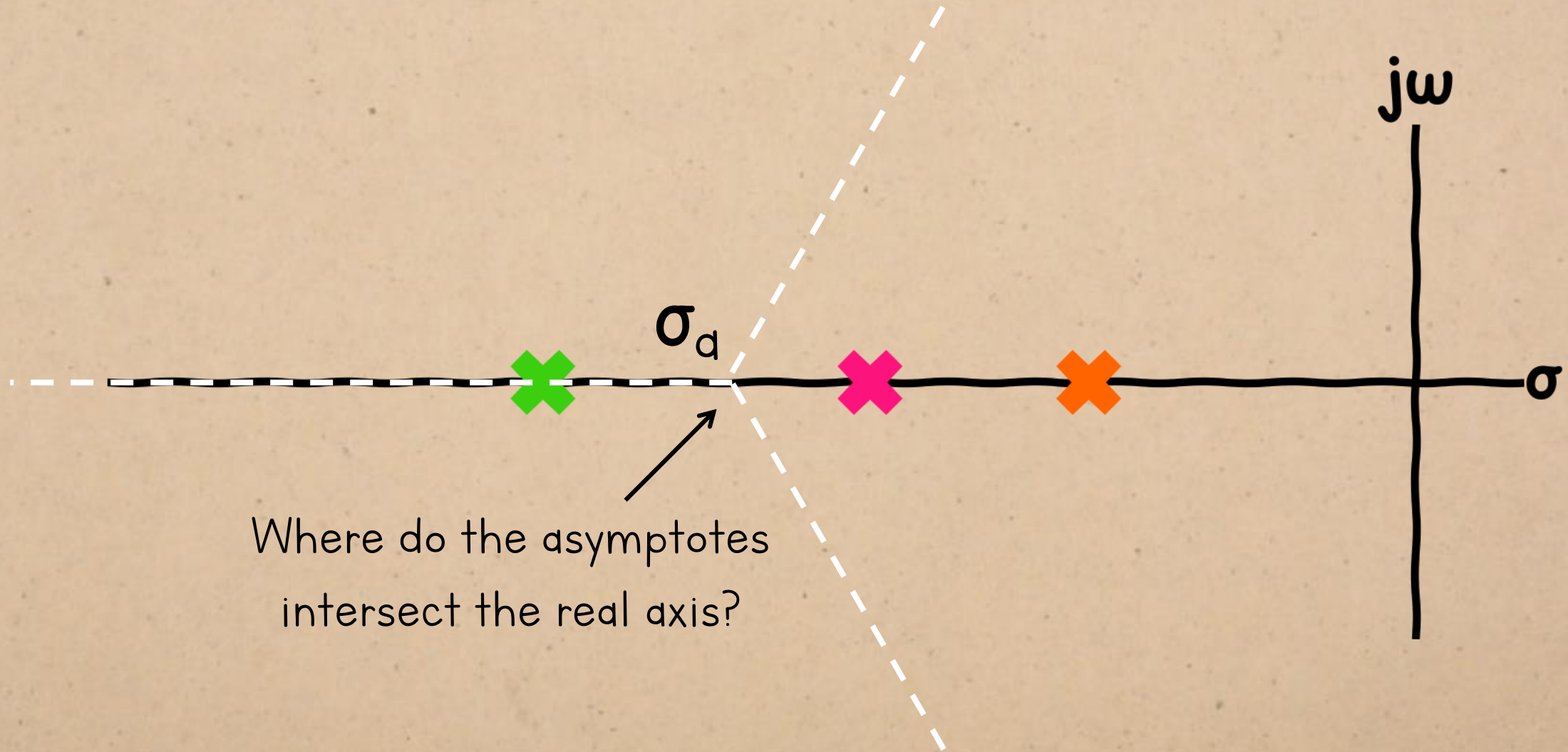
$$\theta_d = \frac{(2k - 1)\pi}{n - m} \quad k = \pm 1, \pm 2 \dots$$

ASYMPTOTES

Rule: the angles of the asymptotes can be determined using

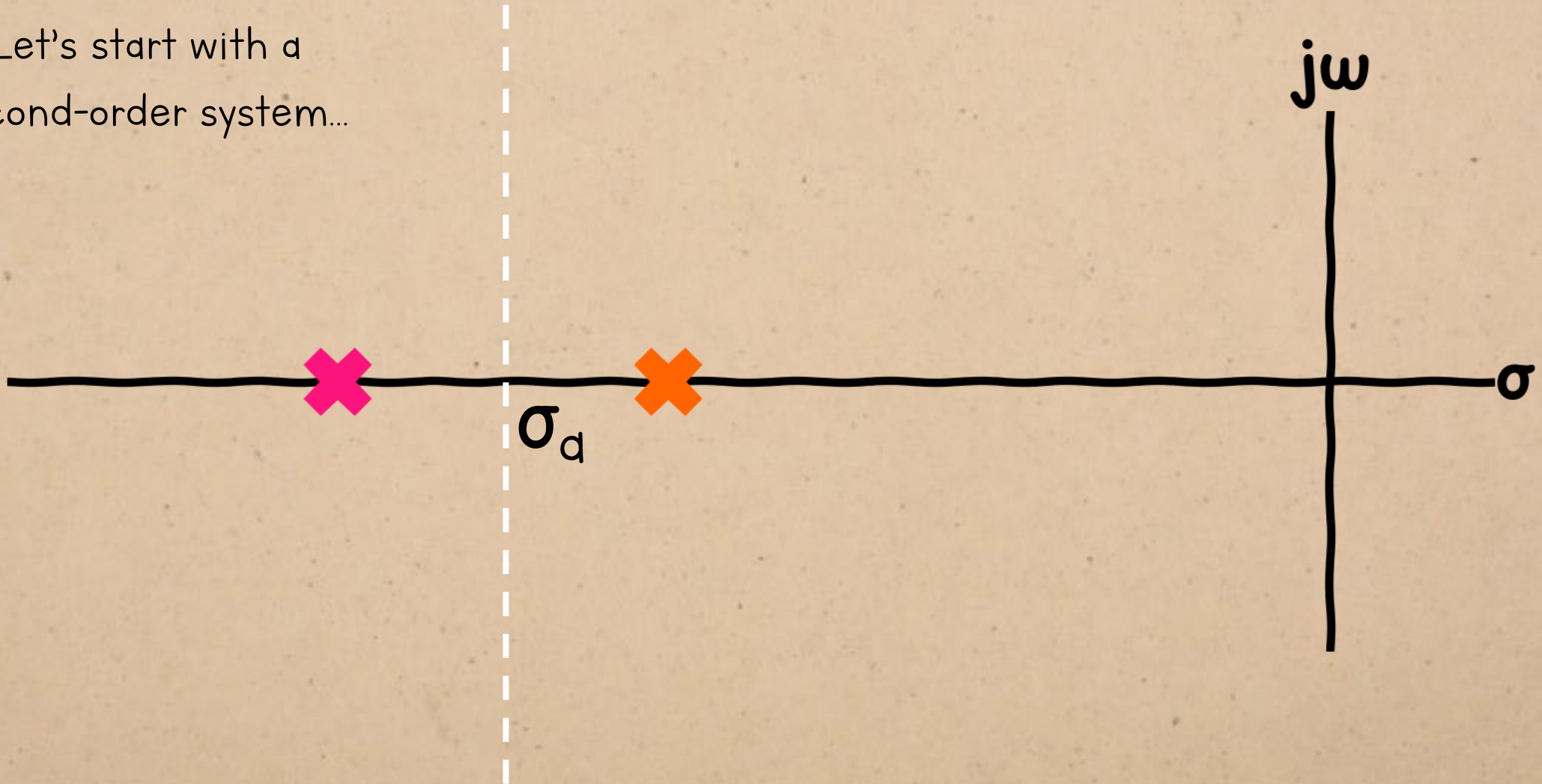
$$\theta_d = \frac{(2k - 1)\pi}{(\text{number of asymptotes})} \quad k = \pm 1, \pm 2, \dots$$

ASYMPTOTES



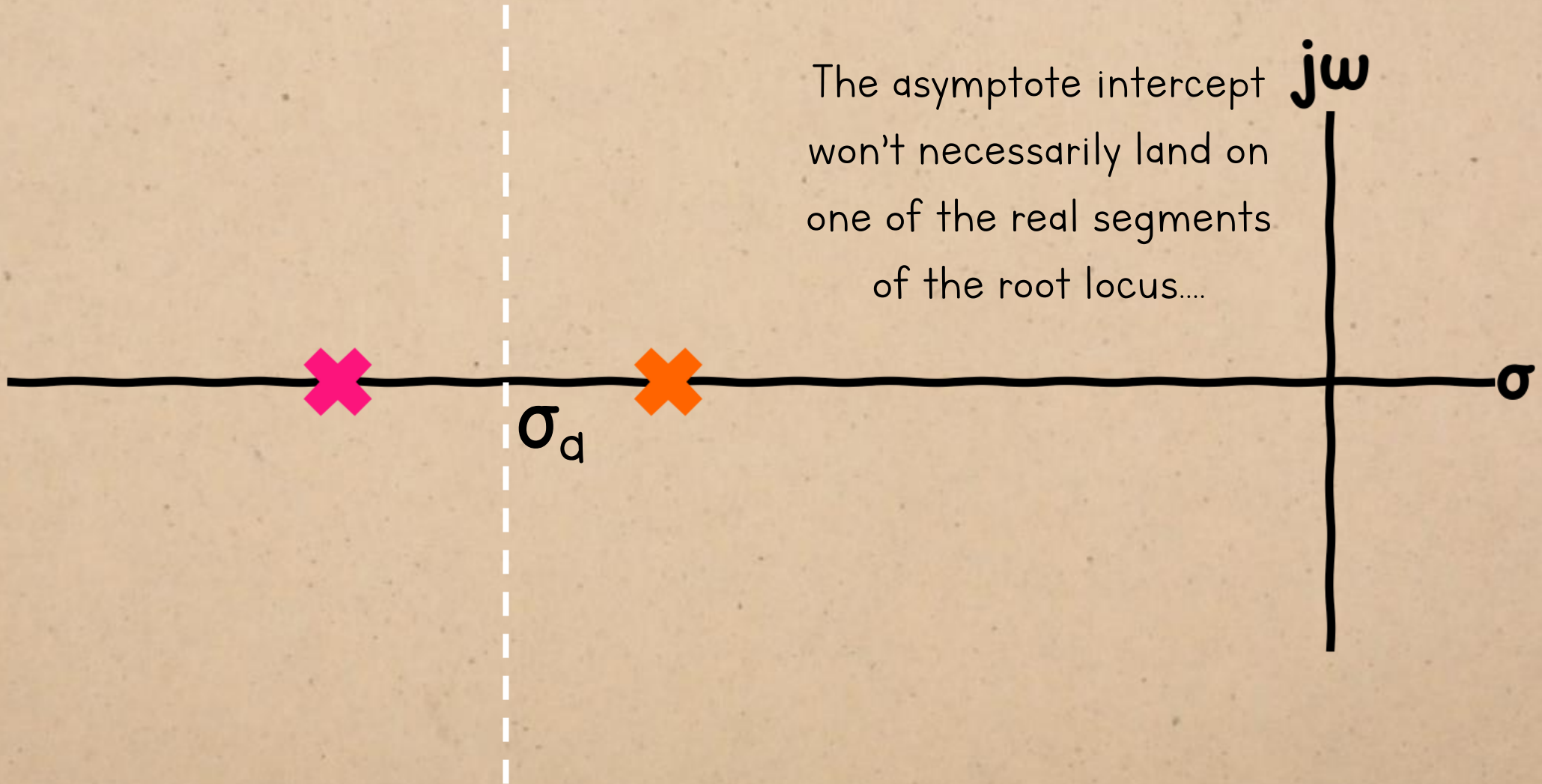
ASYMPTOTES

Let's start with a
second-order system...

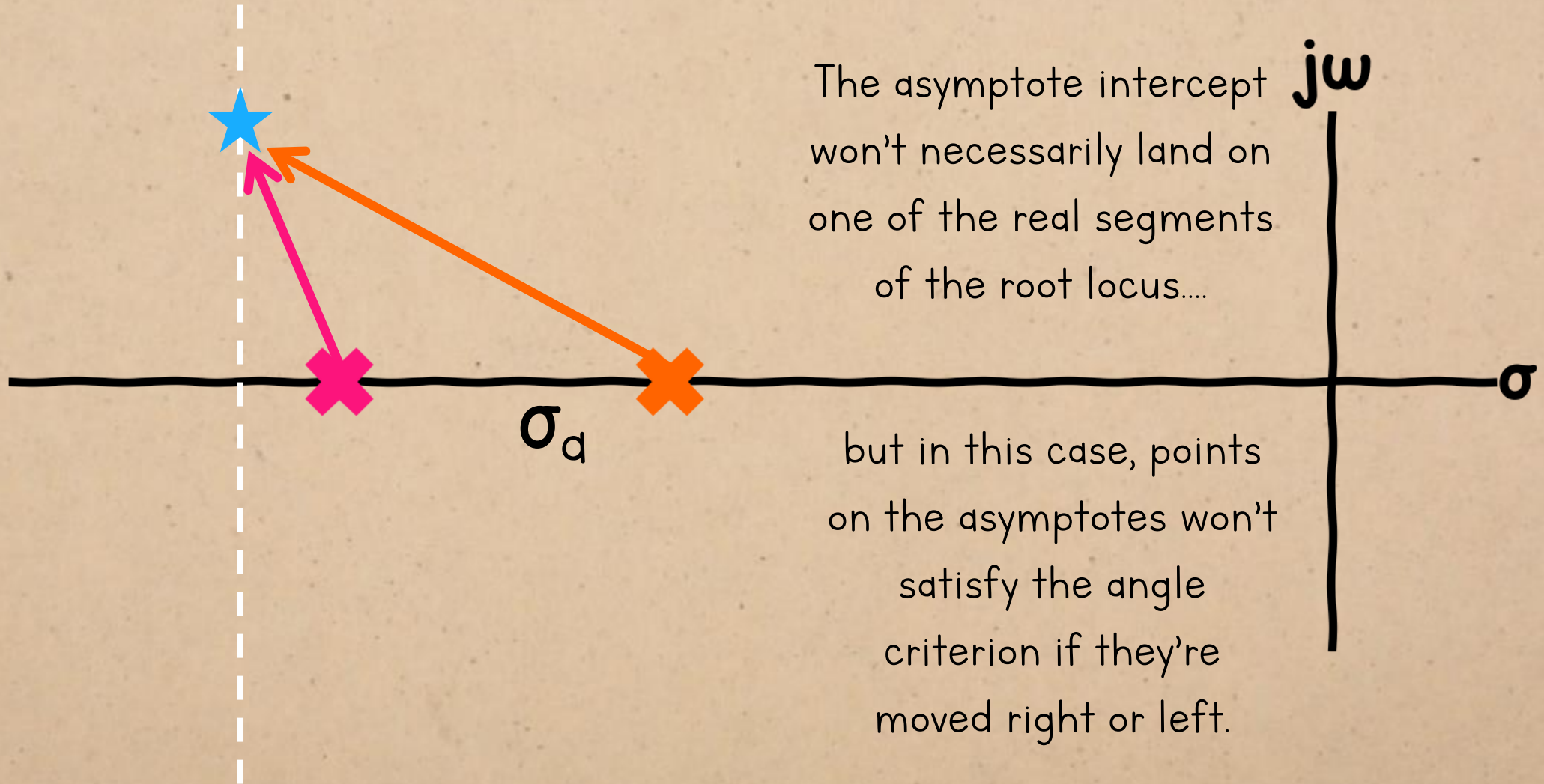


ASYMPTOTES

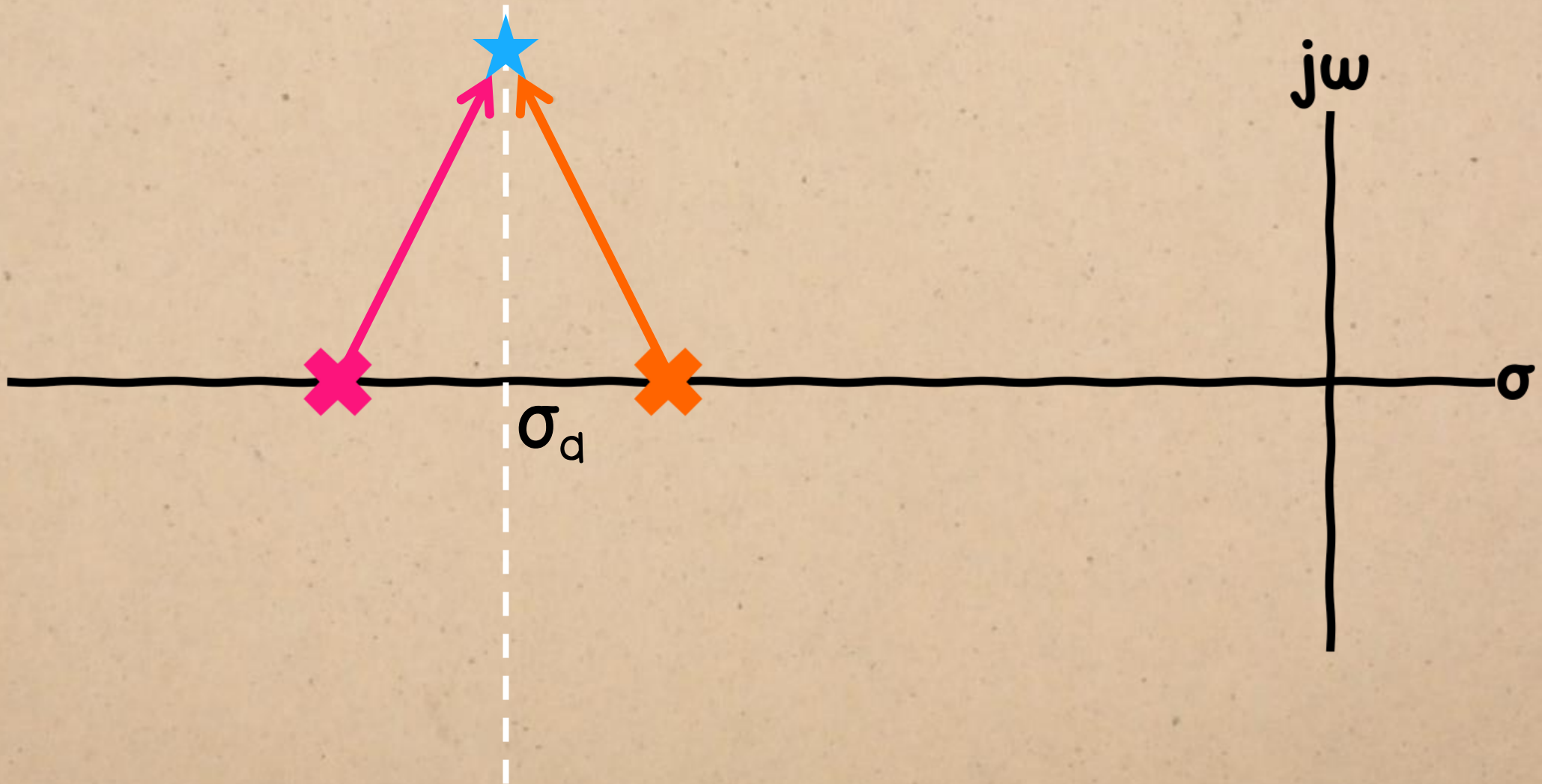
The asymptote intercept $j\omega$
won't necessarily land on
one of the real segments
of the root locus....



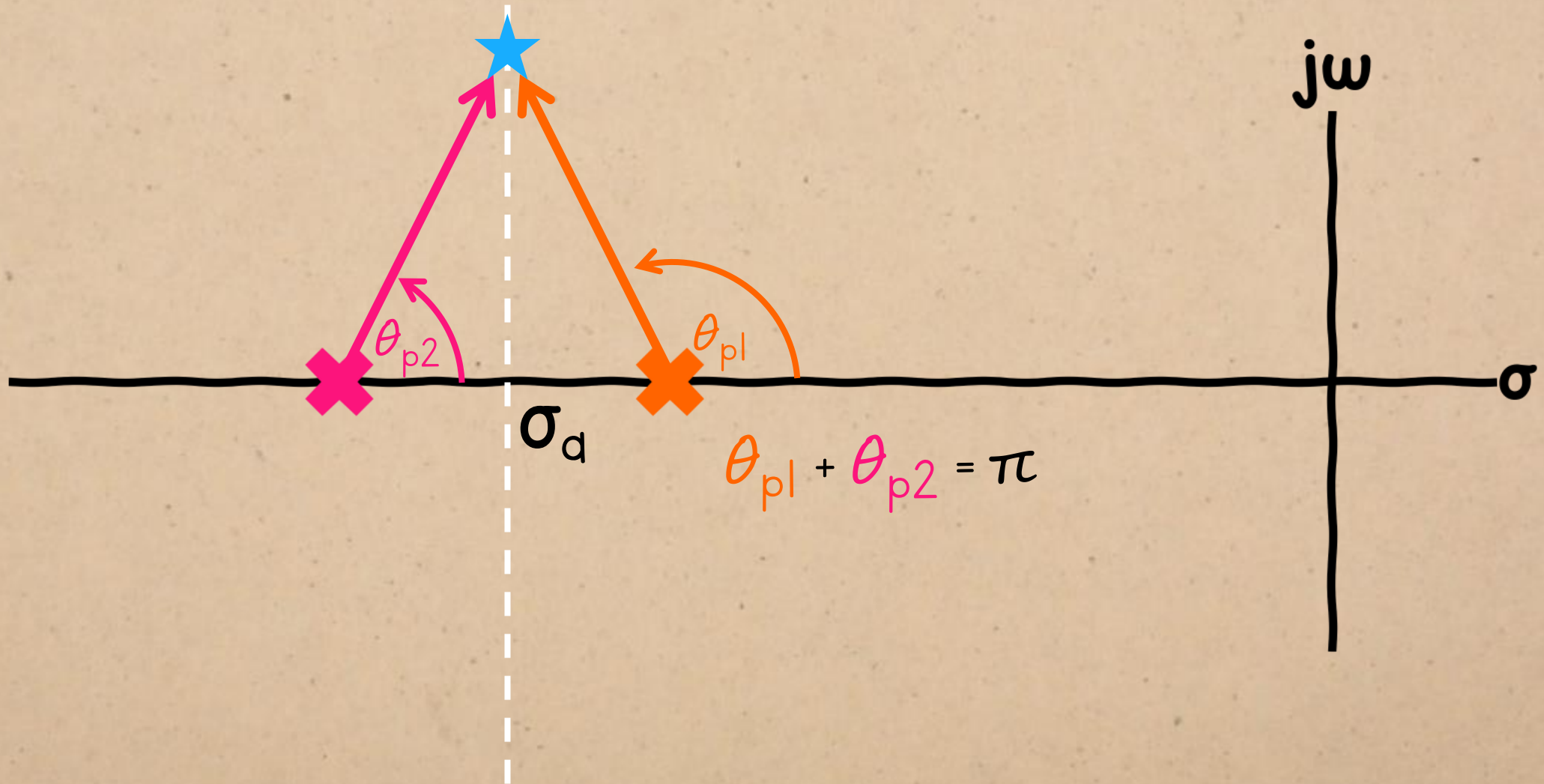
ASYMPTOTES



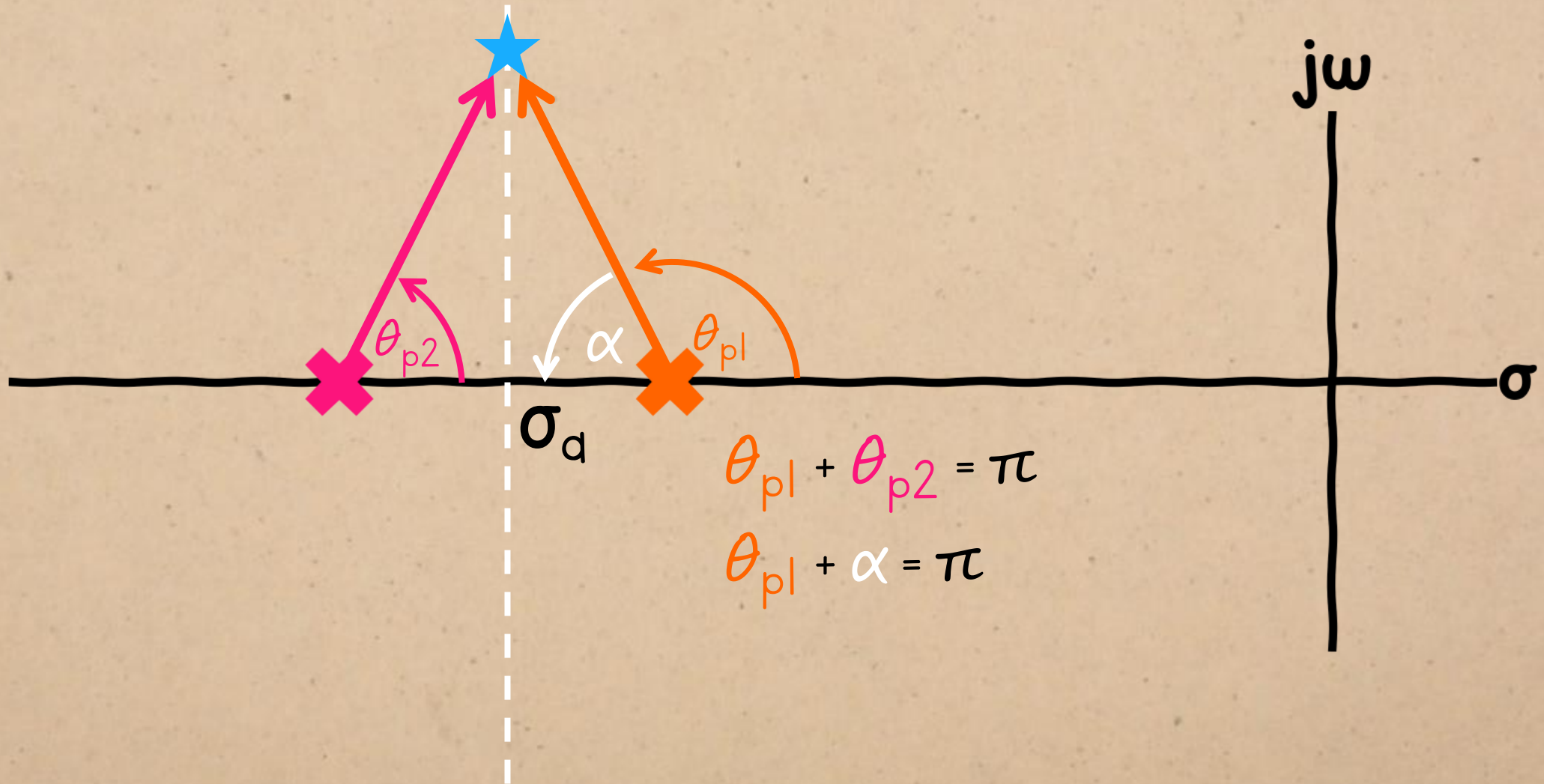
ASYMPTOTES



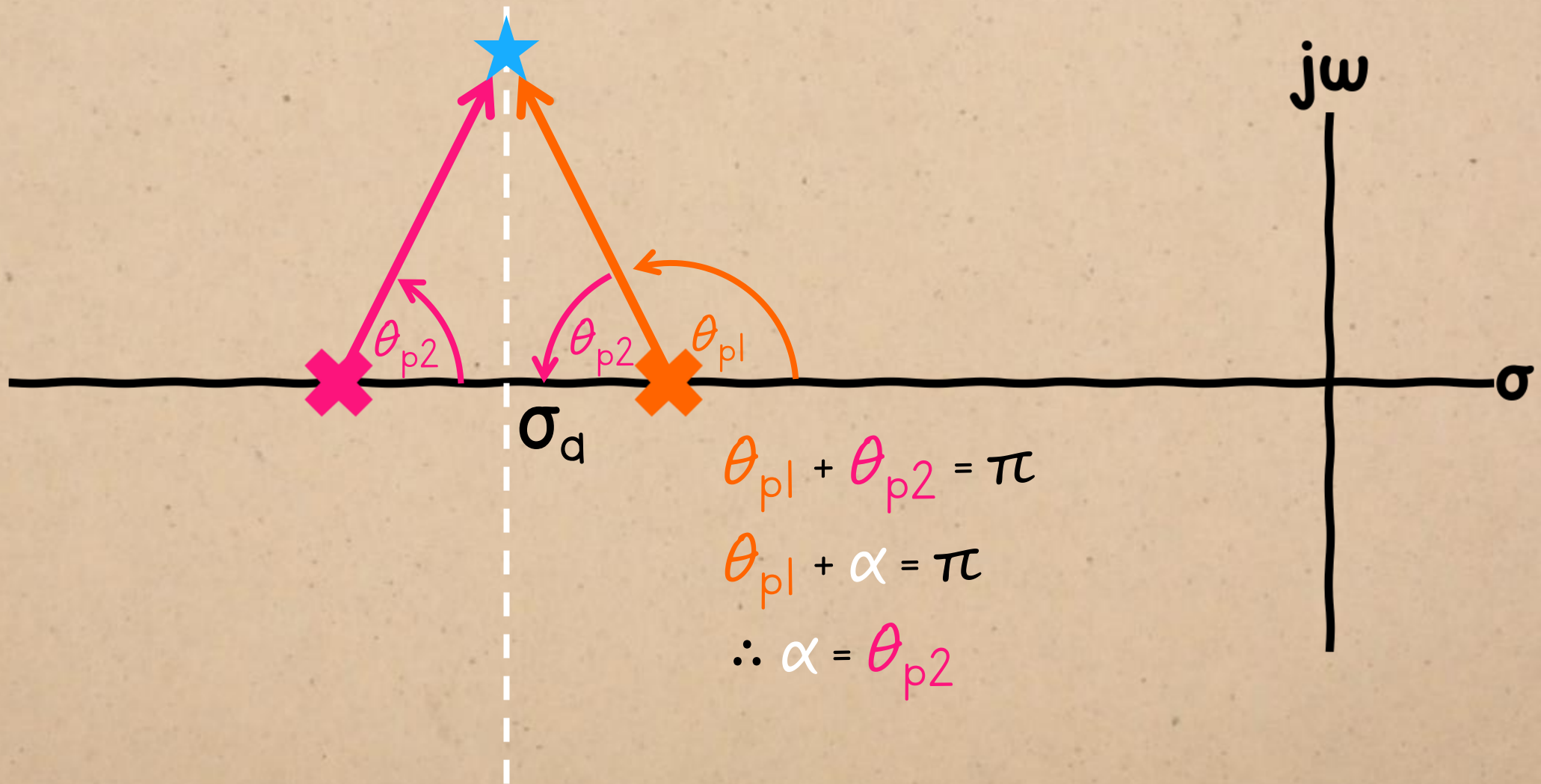
ASYMPTOTES



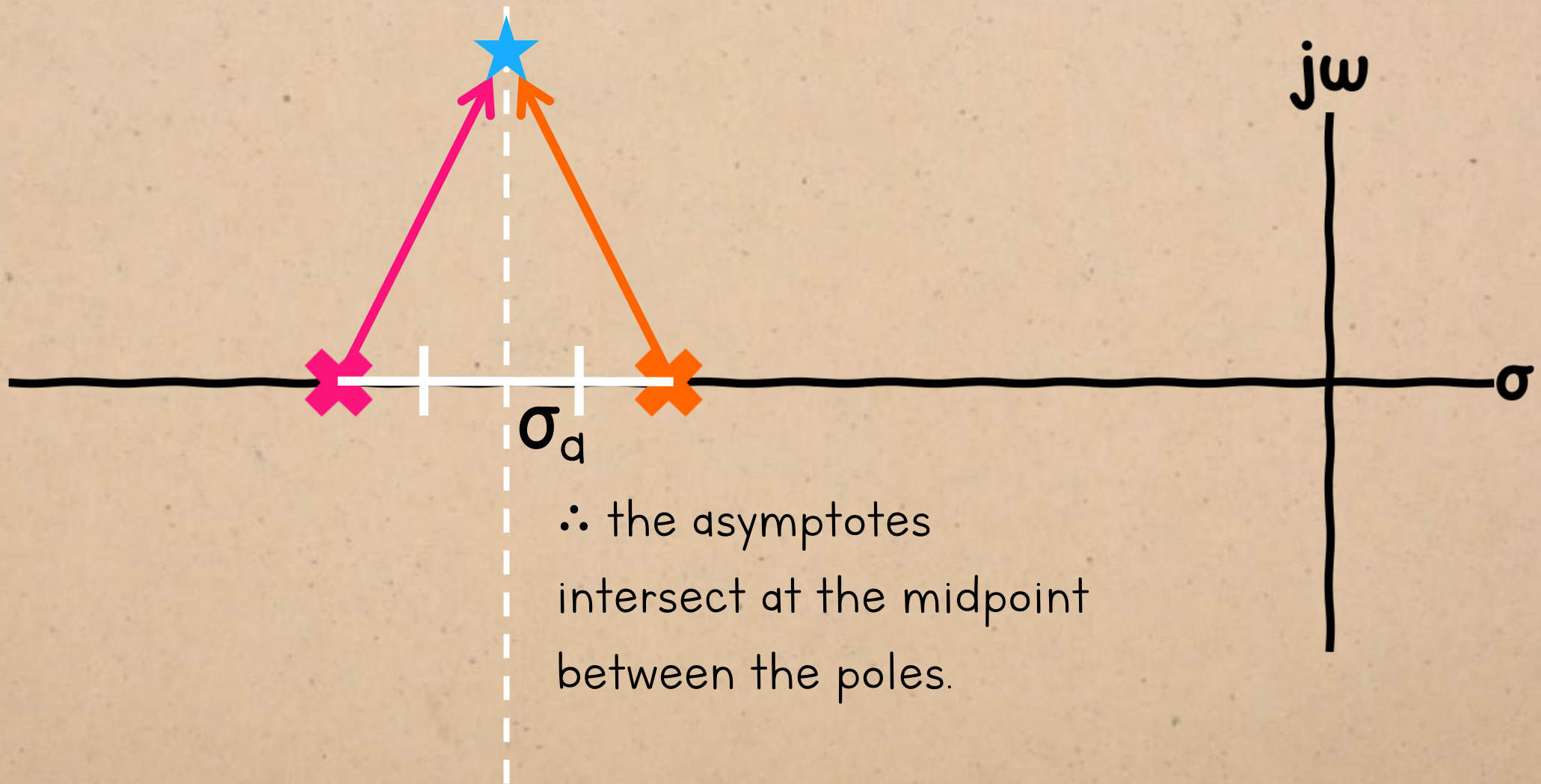
ASYMPTOTES



ASYMPTOTES

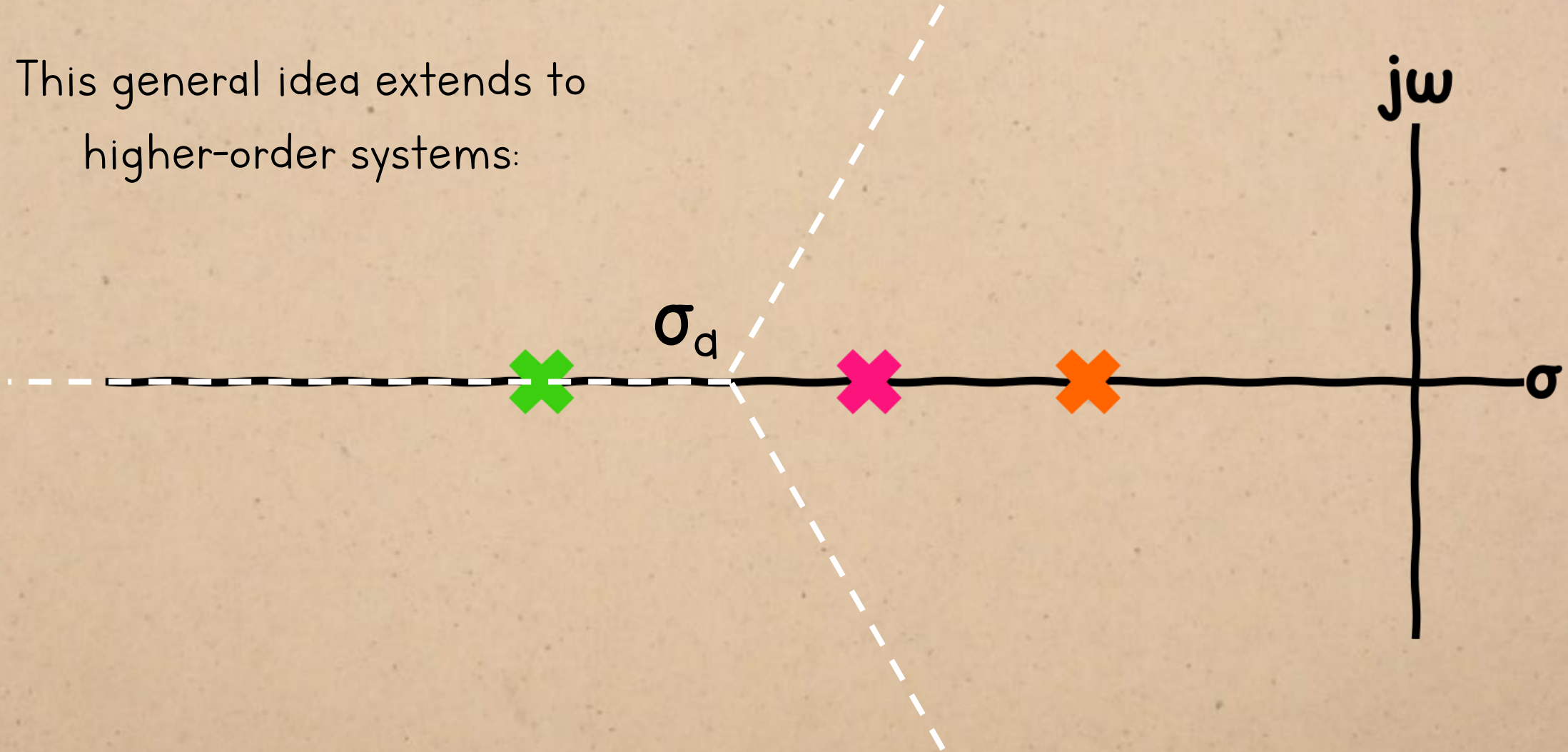


ASYMPTOTES

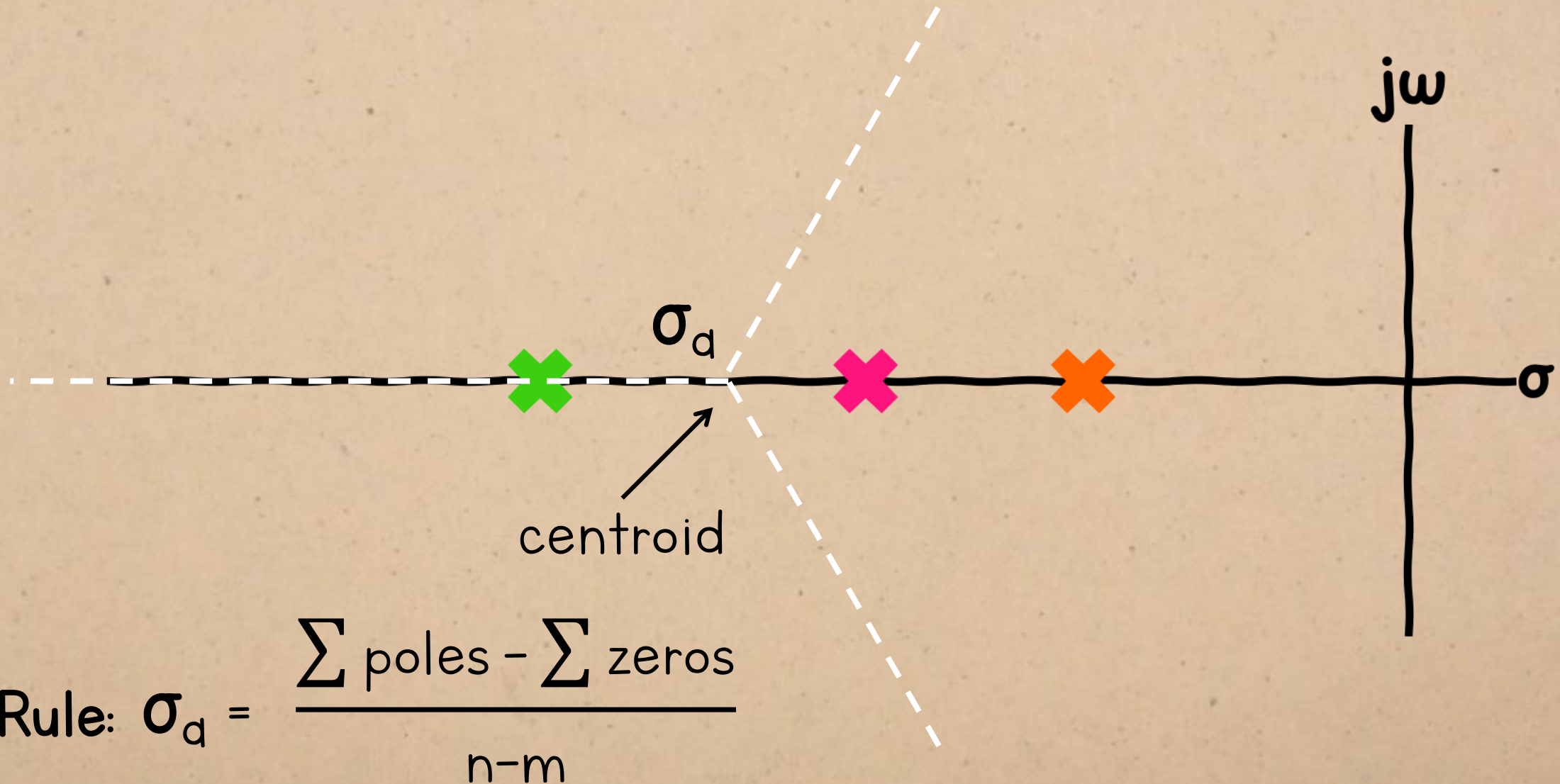


ASYMPTOTES

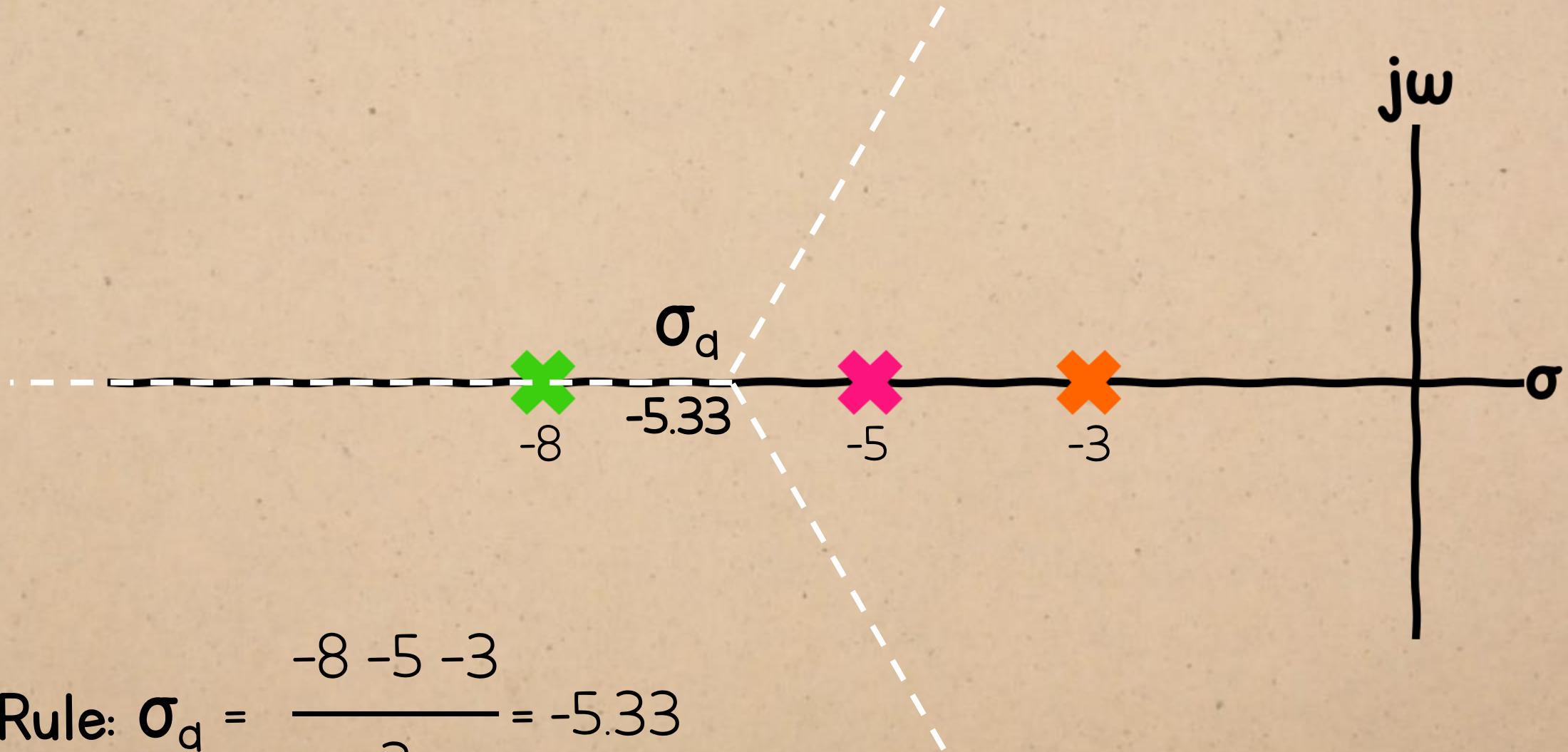
This general idea extends to
higher-order systems:



ASYMPTOTES

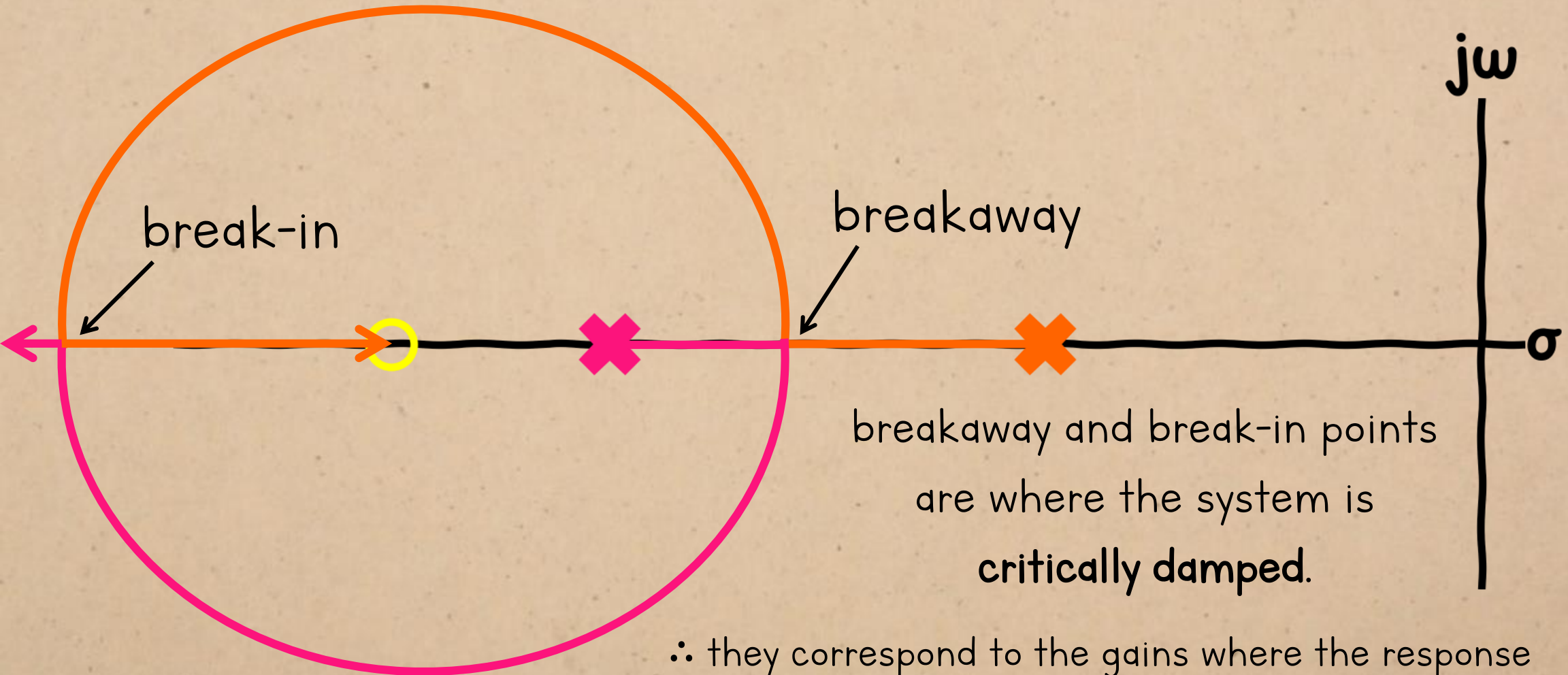


ASYMPTOTES



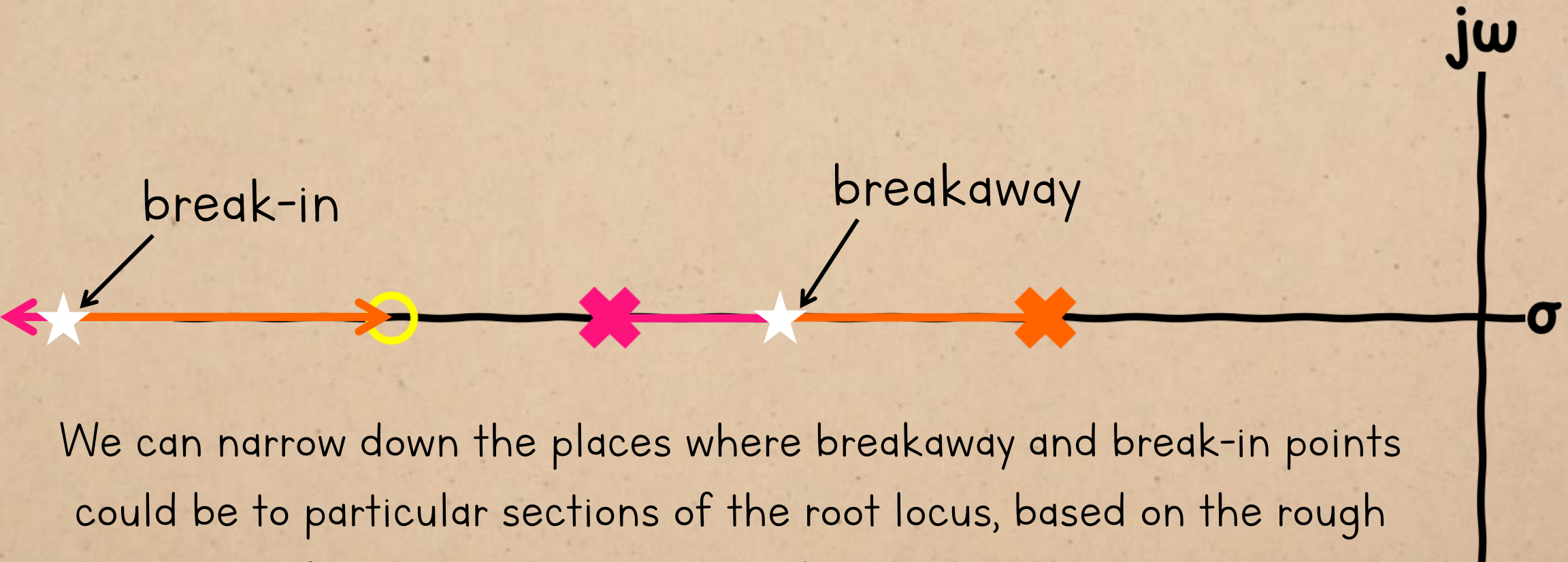
Rule: $\sigma_d = \frac{-8 - 5 - 3}{3} = -5.33$

BREAKAWAY/IN POINTS



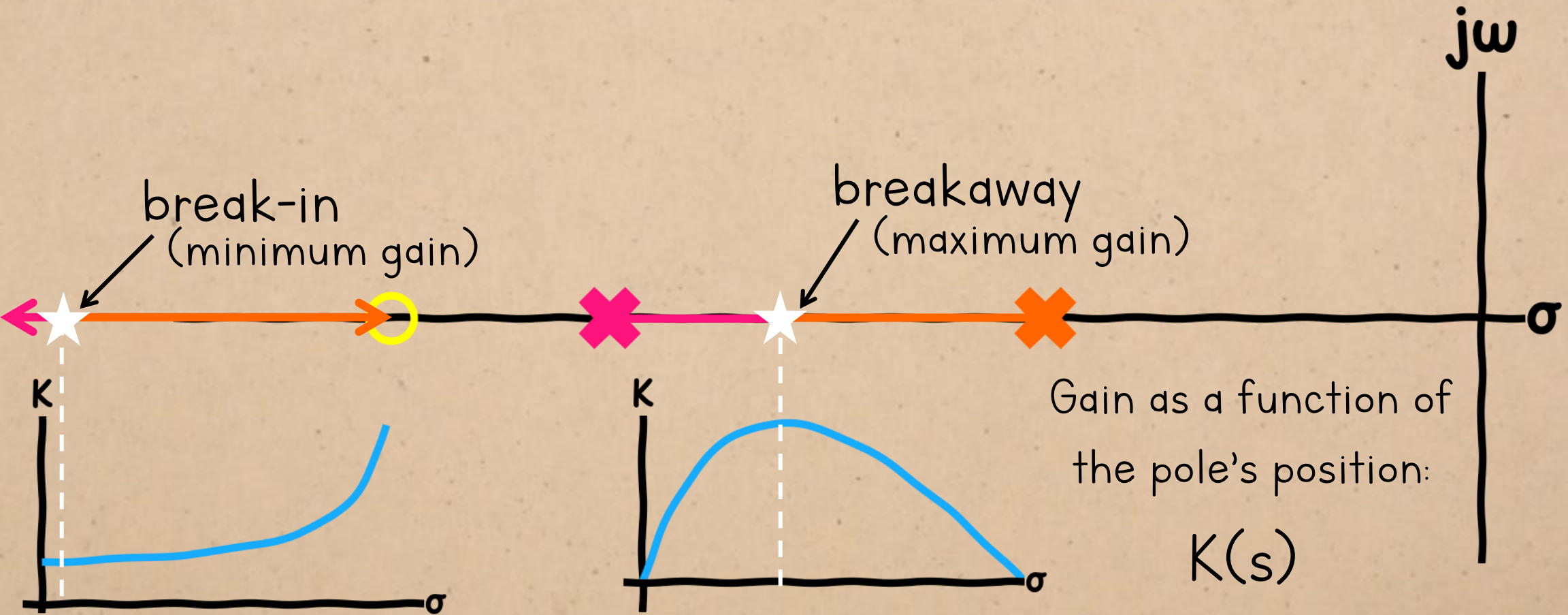
$\therefore$ they correspond to the gains where the response transitions from over- to underdamped and back.

BREAKAWAY/IN POINTS

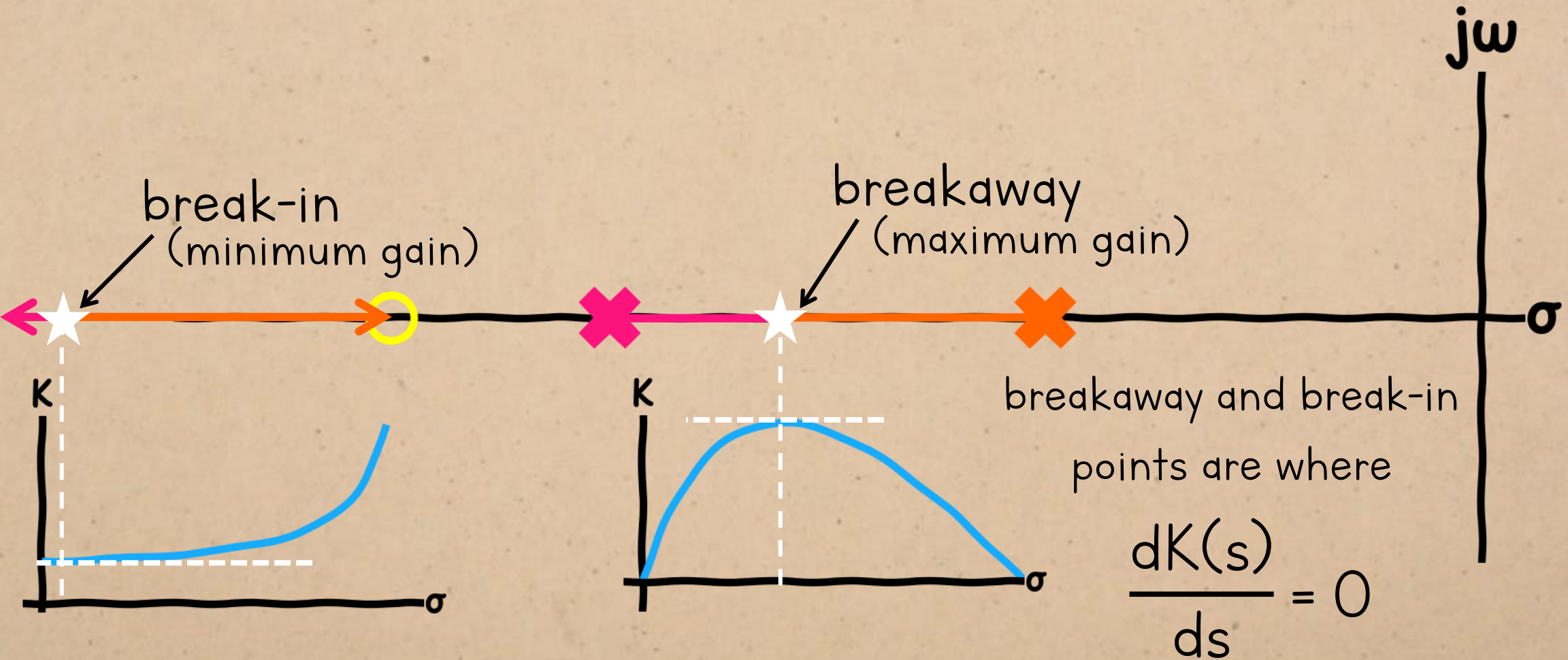


We can narrow down the places where breakaway and break-in points could be to particular sections of the root locus, based on the rough sketching rules we covered in the last lecture.

BREAKAWAY/IN POINTS



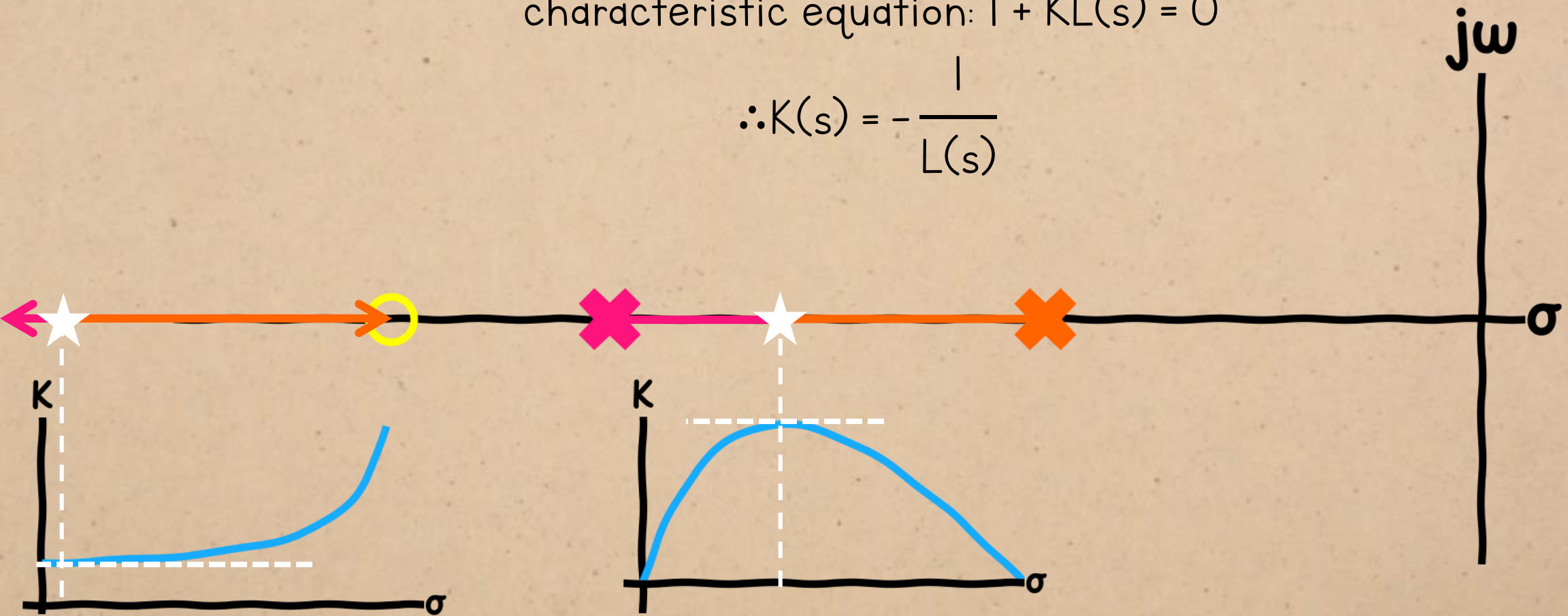
BREAKAWAY/IN POINTS



BREAKAWAY/IN POINTS

characteristic equation: $1 + KL(s) = 0$

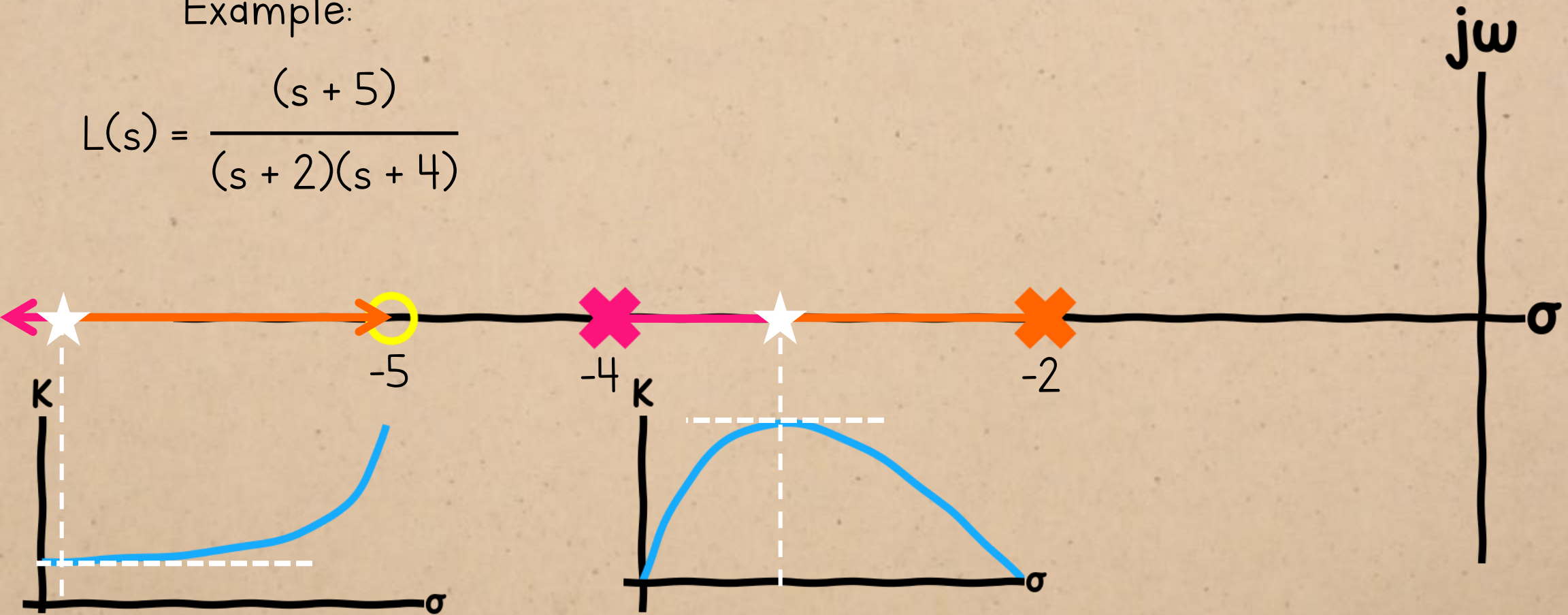
$$\therefore K(s) = -\frac{1}{L(s)}$$



BREAKAWAY/IN POINTS

Example:

$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)}$$

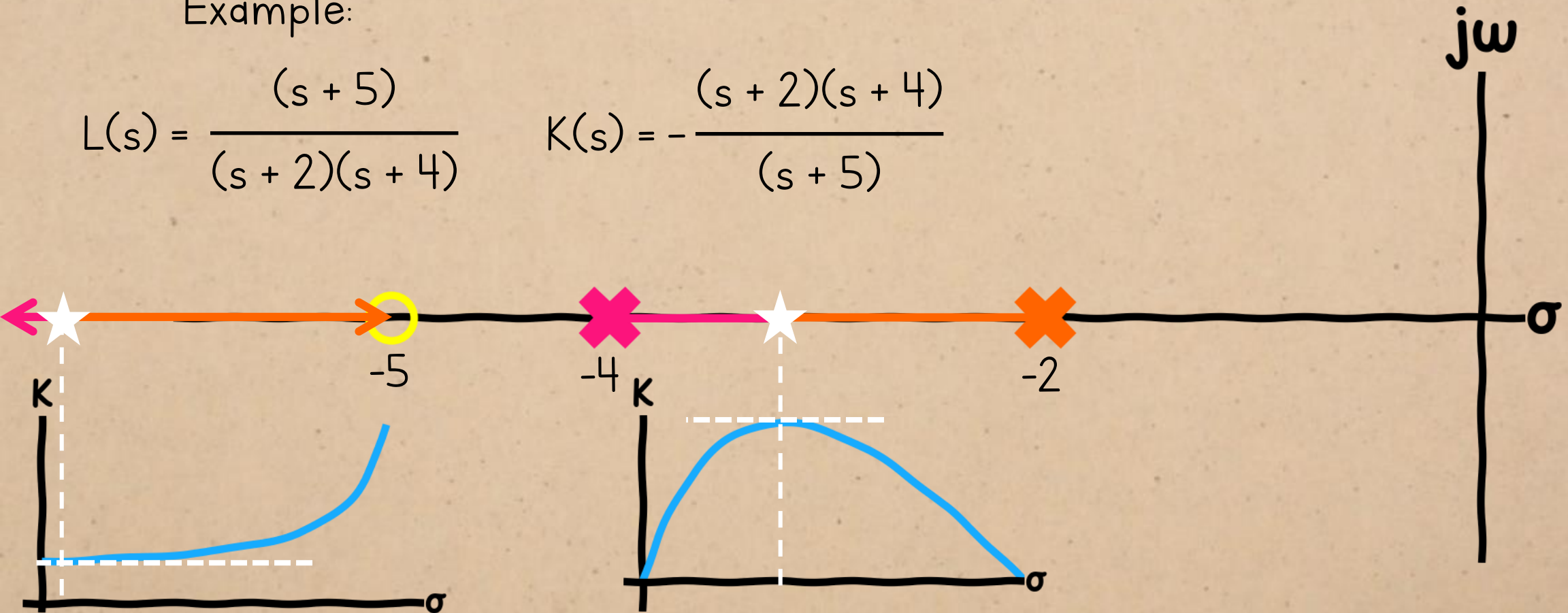


BREAKAWAY/IN POINTS

Example:

$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)}$$

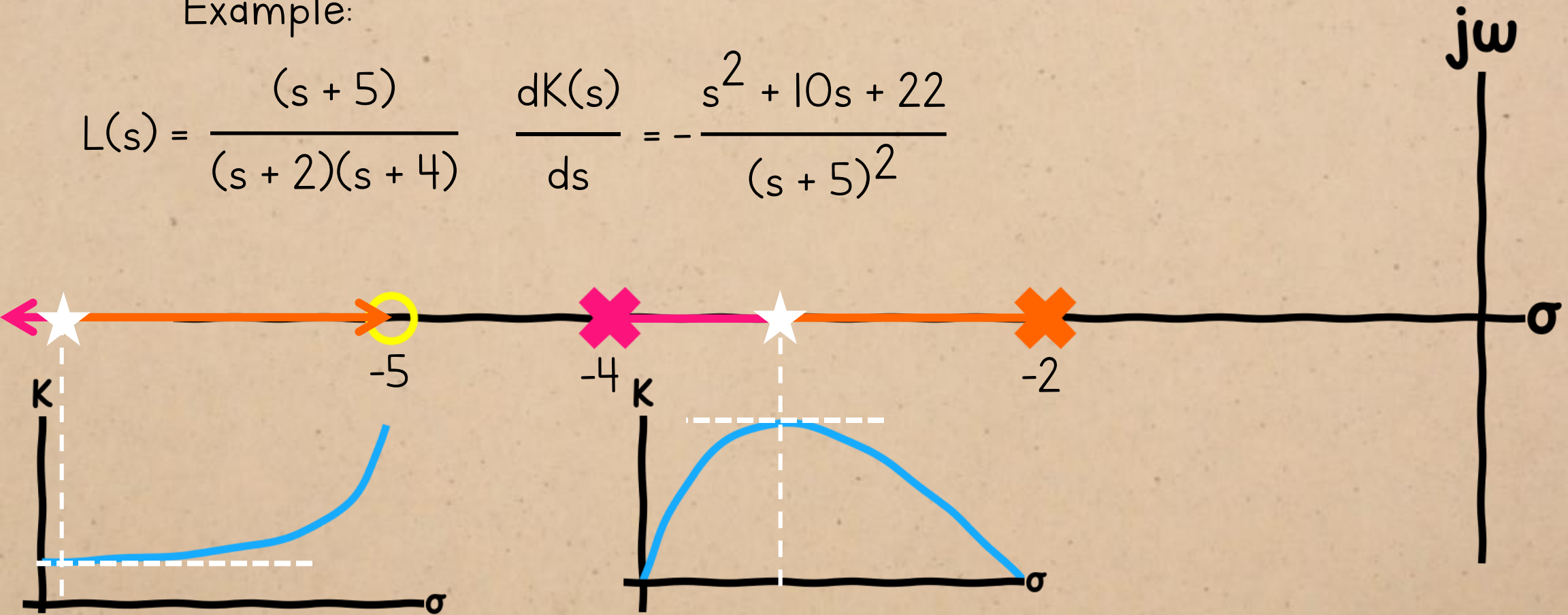
$$K(s) = -\frac{(s + 2)(s + 4)}{(s + 5)}$$



BREAKAWAY/IN POINTS

Example:

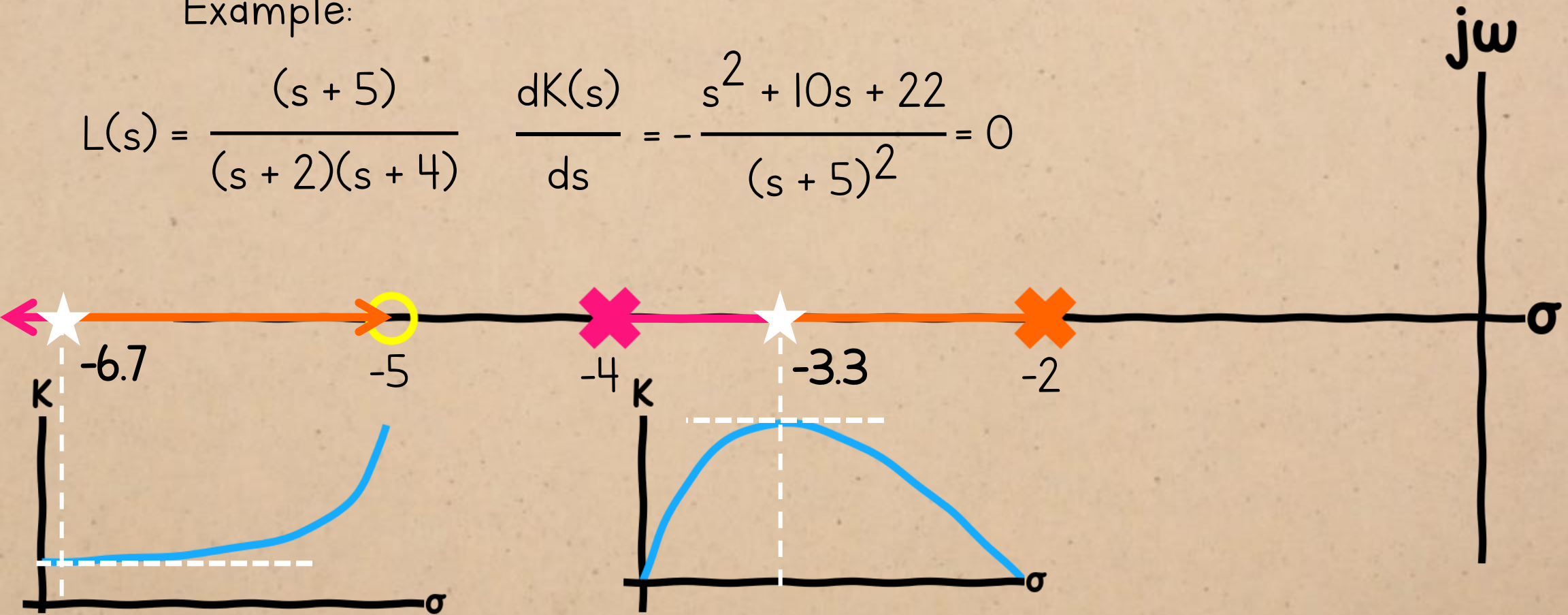
$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)} \quad \frac{dK(s)}{ds} = -\frac{s^2 + 10s + 22}{(s + 5)^2}$$



BREAKAWAY/IN POINTS

Example:

$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)} \quad \frac{dK(s)}{ds} = -\frac{s^2 + 10s + 22}{(s + 5)^2} = 0$$



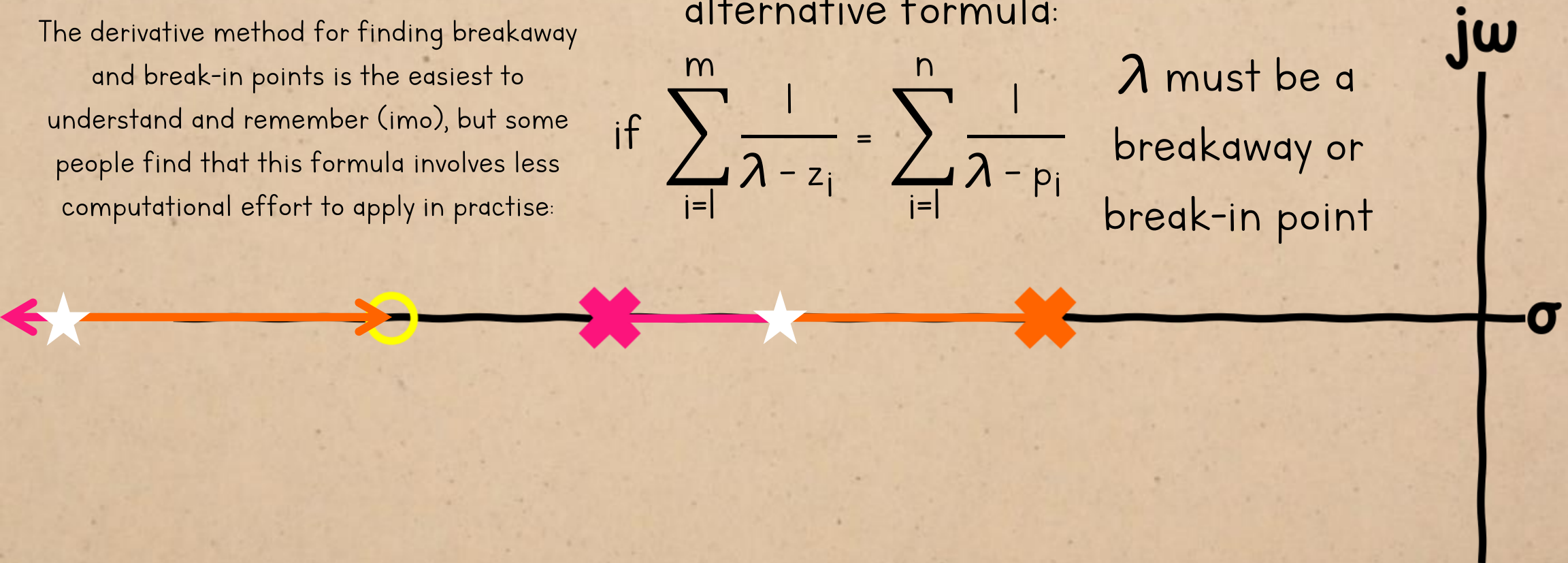
BREAKAWAY/IN POINTS

The derivative method for finding breakaway and break-in points is the easiest to understand and remember (imo), but some people find that this formula involves less computational effort to apply in practise:

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



BREAKAWAY/IN POINTS

alternative formula:

if
$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point

A quick derivation...



we're looking for
points where

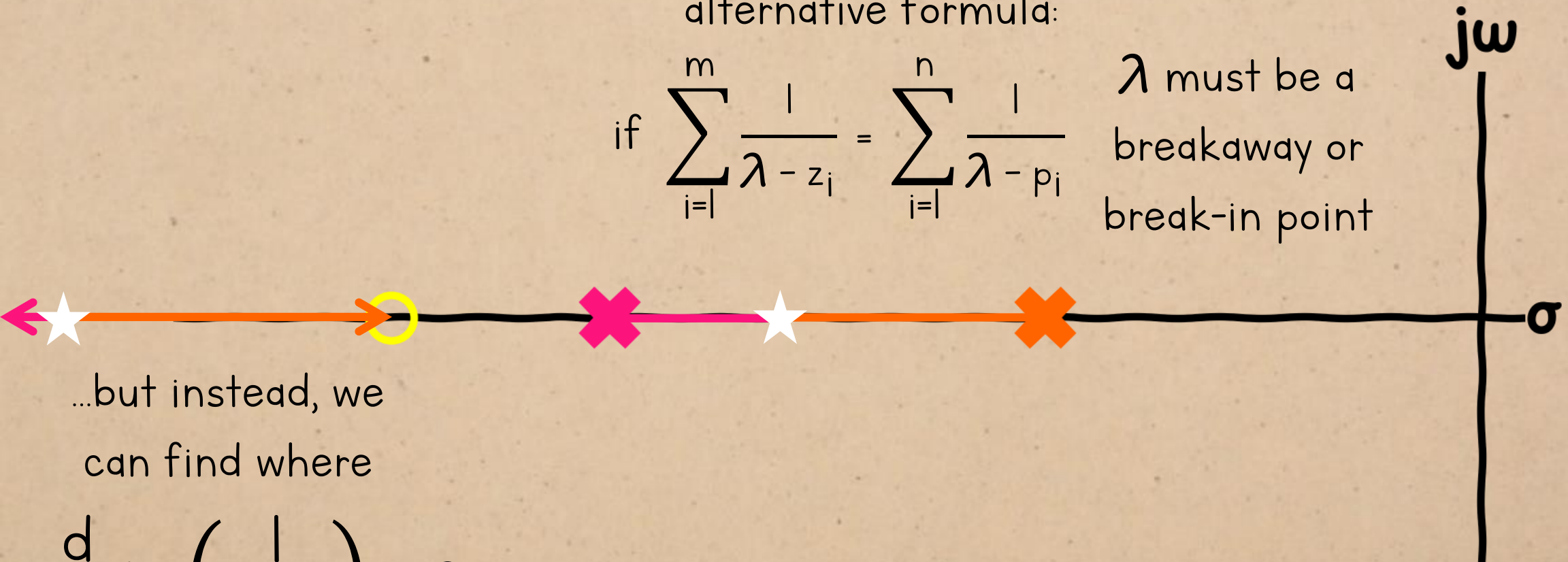
$$\frac{d}{ds} \frac{1}{L(s)} = 0$$

BREAKAWAY/IN POINTS

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



...but instead, we
can find where

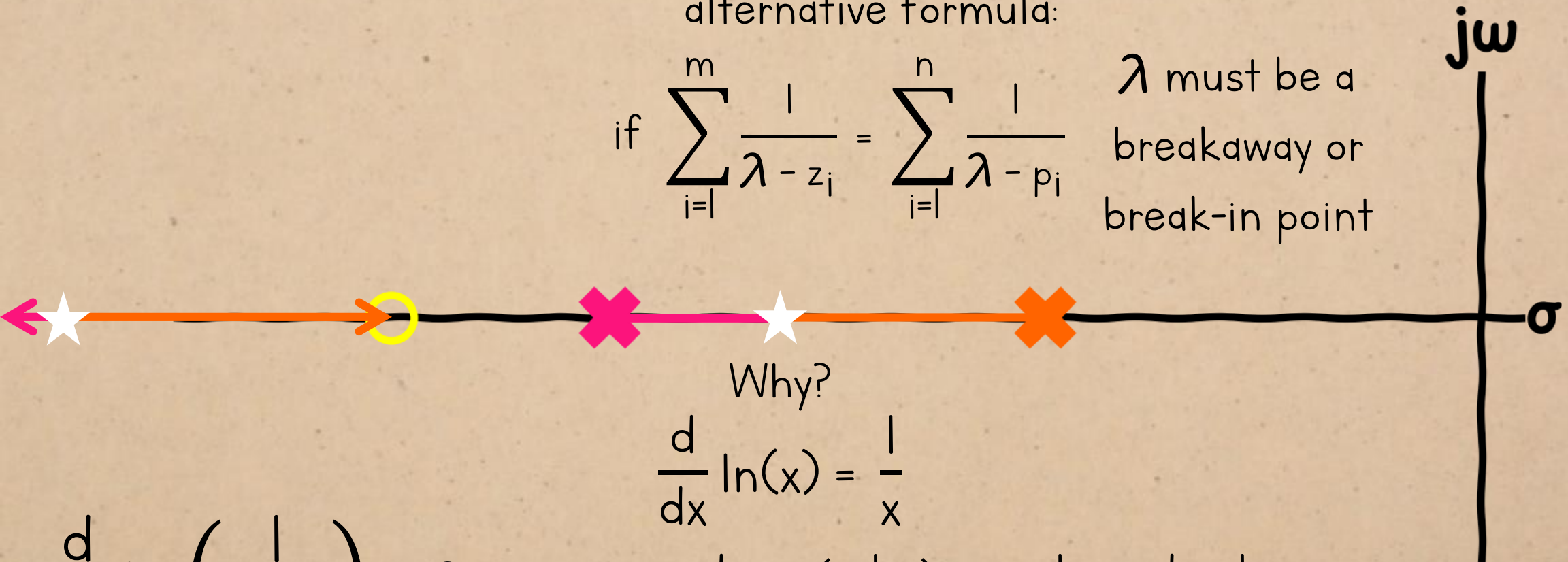
$$\frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = 0$$

because it will be at the same points

BREAKAWAY/IN POINTS

alternative formula:

if
$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$
 λ must be a breakaway or break-in point



Why?

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

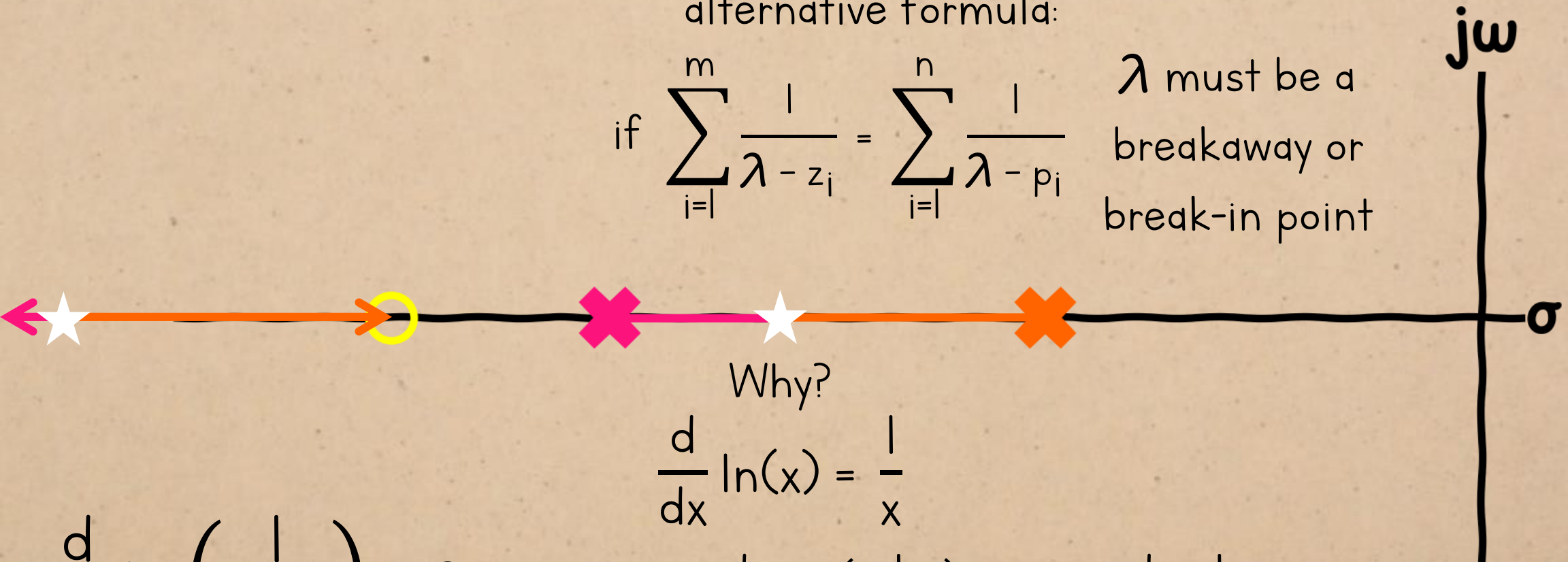
$$\frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = 0$$

$$\therefore \frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = \frac{1}{1/L(s)} \frac{d}{ds} \frac{1}{L(s)}$$

BREAKAWAY/IN POINTS

alternative formula:

if
$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$
 λ must be a breakaway or break-in point



Why?

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

$$\frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = 0$$

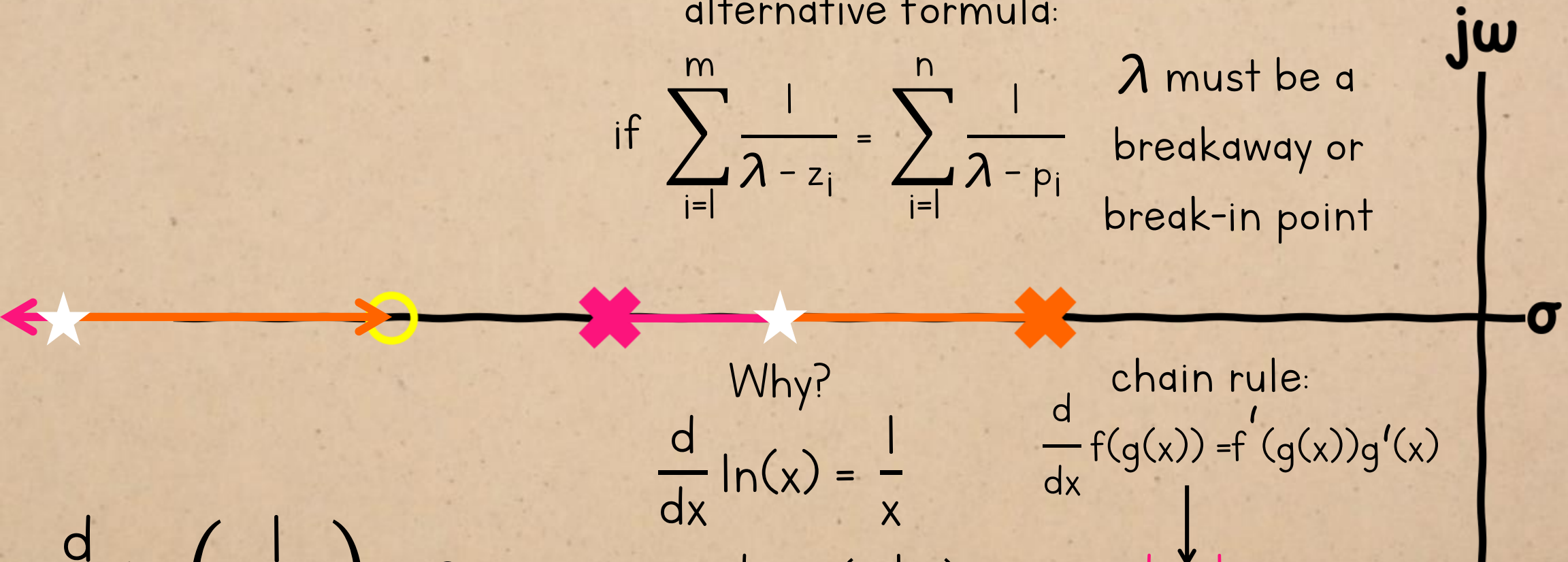
$$\therefore \frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = L(s) \frac{d}{ds} \frac{1}{L(s)}$$

BREAKAWAY/IN POINTS

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



Why?

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

$$\therefore \frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = L(s) \frac{d}{ds} \frac{1}{L(s)}$$

chain rule:

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

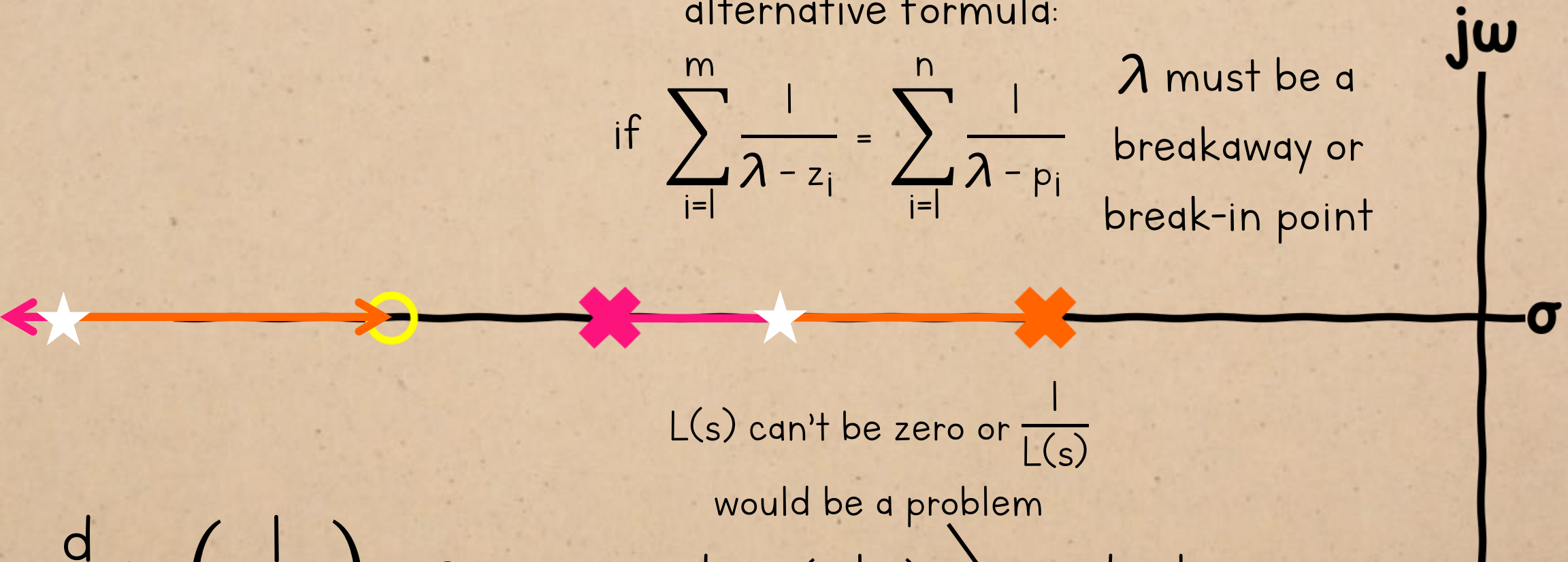


$$\frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = 0$$

BREAKAWAY/IN POINTS

alternative formula:

if
$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$
 λ must be a breakaway or break-in point



$L(s)$ can't be zero or $\frac{1}{L(s)}$

would be a problem

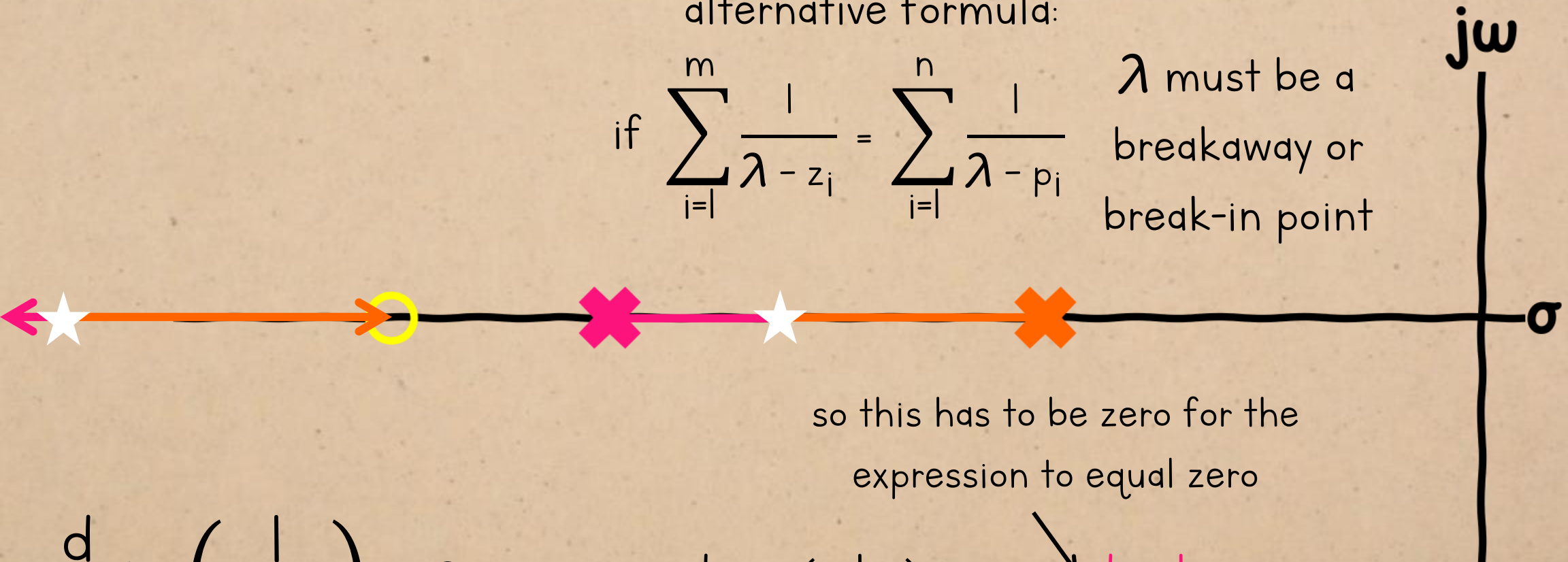
$$\frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = 0$$

$$\therefore \frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = L(s) \frac{d}{ds} \frac{1}{L(s)} = 0$$

BREAKAWAY/IN POINTS

alternative formula:

if
$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$
 λ must be a breakaway or break-in point



so this has to be zero for the expression to equal zero

$$\frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = 0$$

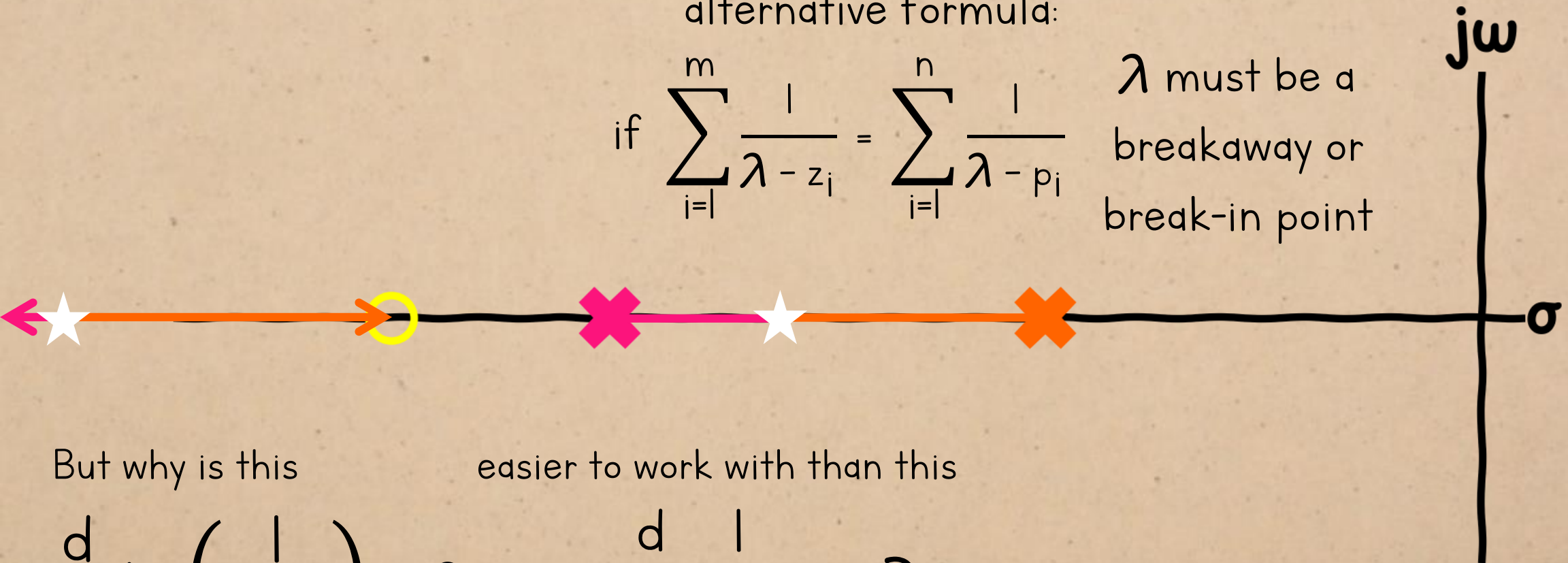
$$\therefore \frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = L(s) \frac{d}{ds} \frac{1}{L(s)} = 0$$

BREAKAWAY/IN POINTS

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



But why is this

$$\frac{d}{ds} \ln \left(\frac{1}{L(s)} \right) = 0$$

easier to work with than this

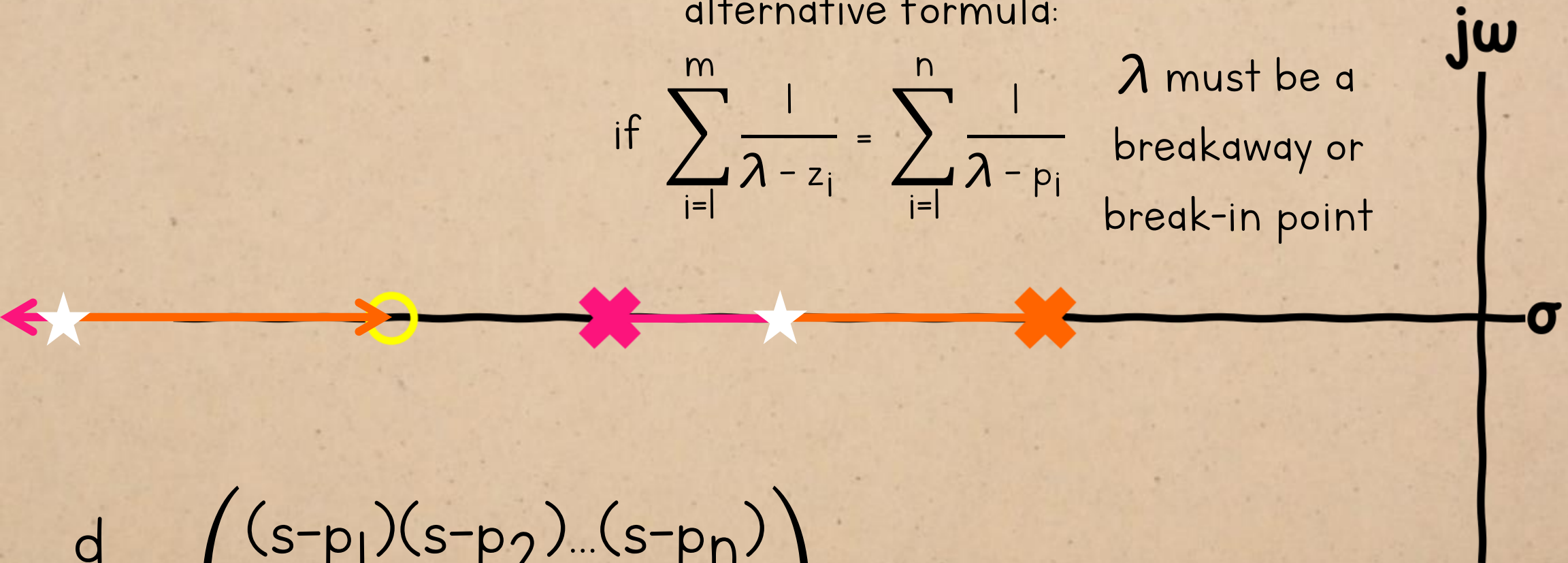
$$\frac{d}{ds} \frac{1}{L(s)} = 0?$$

BREAKAWAY/IN POINTS

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point

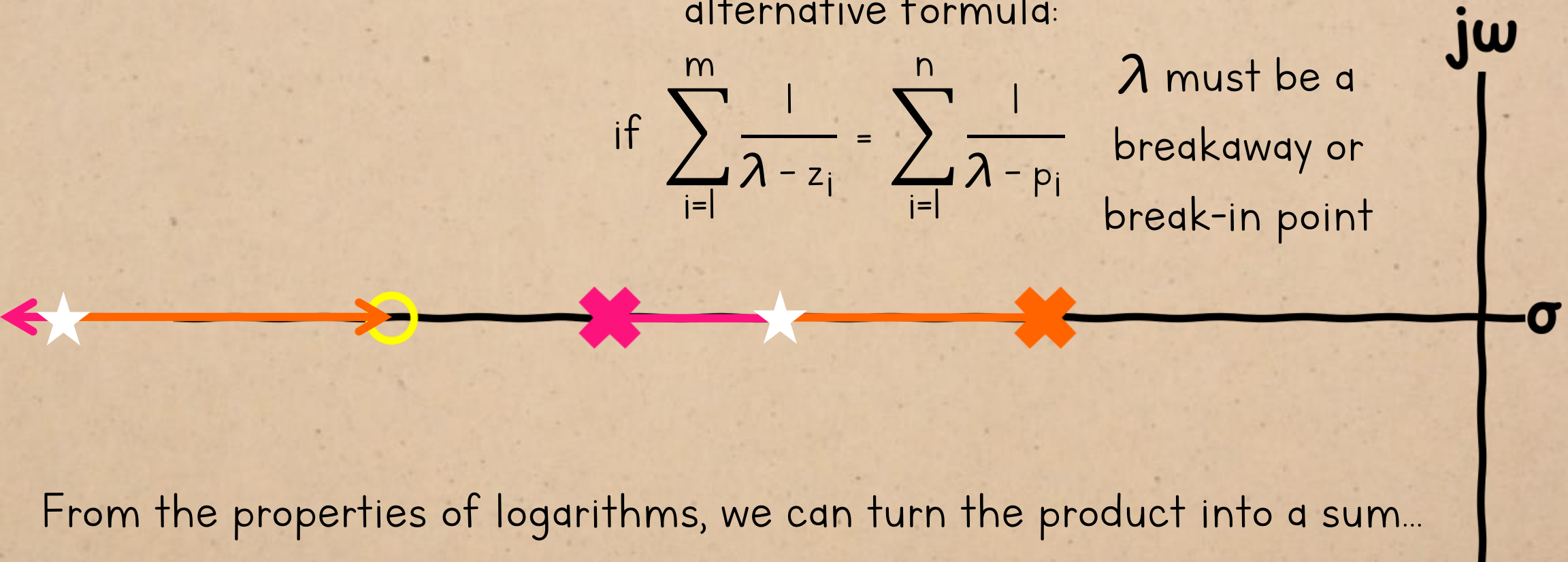


$$\frac{d}{ds} \ln \left(\frac{(s-p_1)(s-p_2)\dots(s-p_n)}{(s-z_1)(s-z_2)\dots(s-z_m)} \right) = 0$$

BREAKAWAY/IN POINTS

alternative formula:

if
$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$
 λ must be a breakaway or break-in point



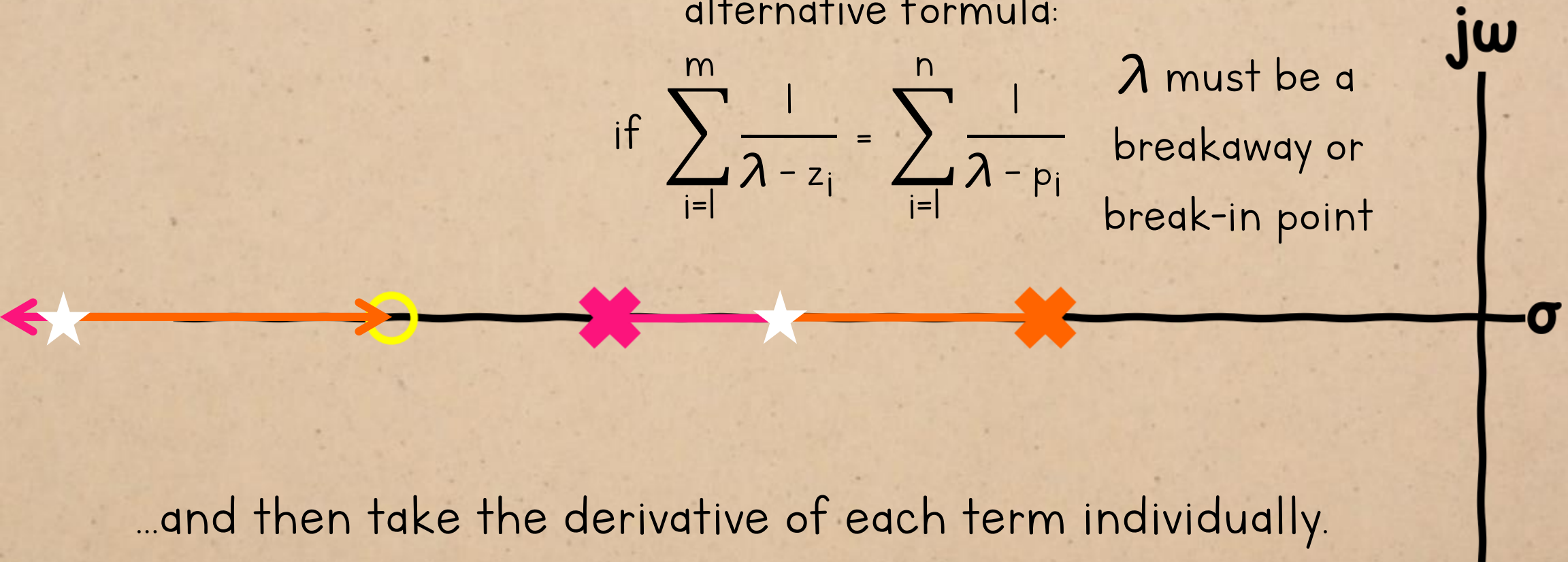
From the properties of logarithms, we can turn the product into a sum...

$$\frac{d}{ds} (\ln(s-p_1) + \ln(s-p_2) \dots + \ln(s-p_n) - \ln(s-z_1) - \ln(s-z_2) \dots - \ln(s-z_m)) = 0$$

BREAKAWAY/IN POINTS

alternative formula:

if
$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$
 λ must be a breakaway or break-in point



...and then take the derivative of each term individually.

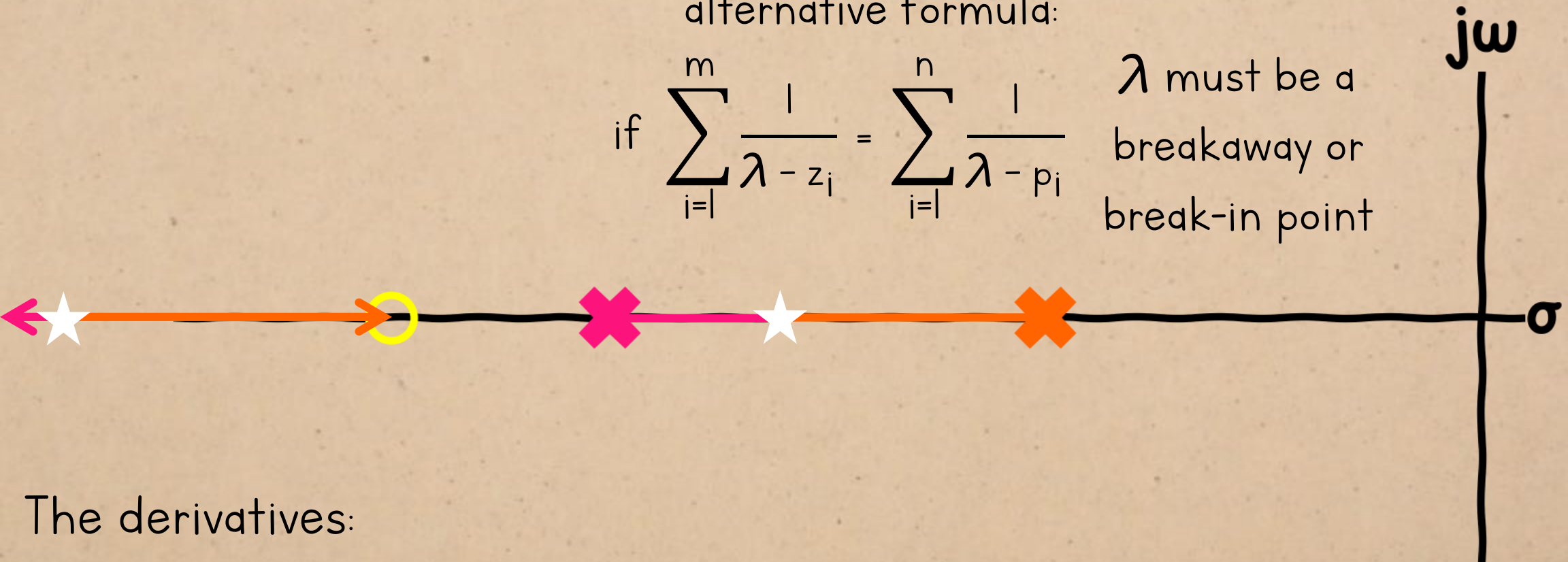
$$\frac{d}{ds} \ln(s-p_1) + \frac{d}{ds} \ln(s-p_2) \dots + \frac{d}{ds} \ln(s-p_n) - \frac{d}{ds} \ln(s-z_1) - \frac{d}{ds} \ln(s-z_2) \dots - \frac{d}{ds} \ln(s-z_m) = 0$$

BREAKAWAY/IN POINTS

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



The derivatives:

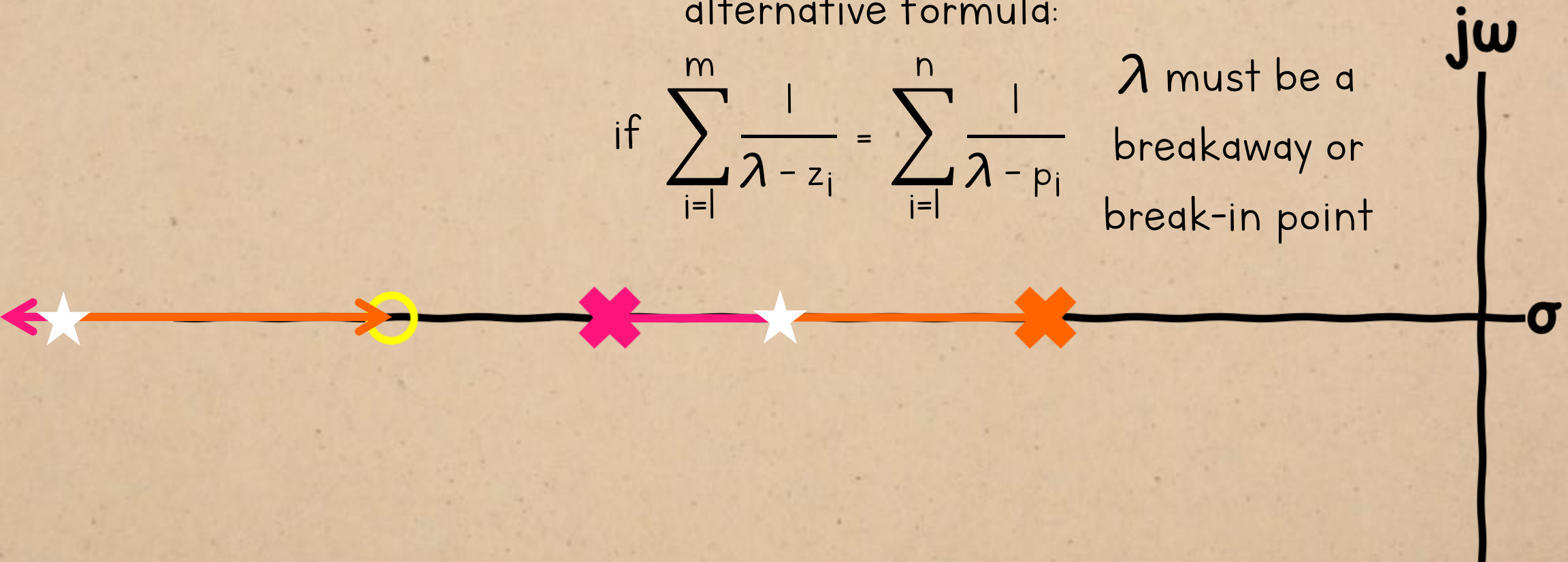
$$\frac{1}{s-p_1} + \frac{1}{s-p_2} + \dots - \frac{1}{s-p_n} - \frac{1}{s-z_1} - \frac{1}{s-z_2} \dots - \frac{1}{s-z_m} = 0$$

BREAKAWAY/IN POINTS

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



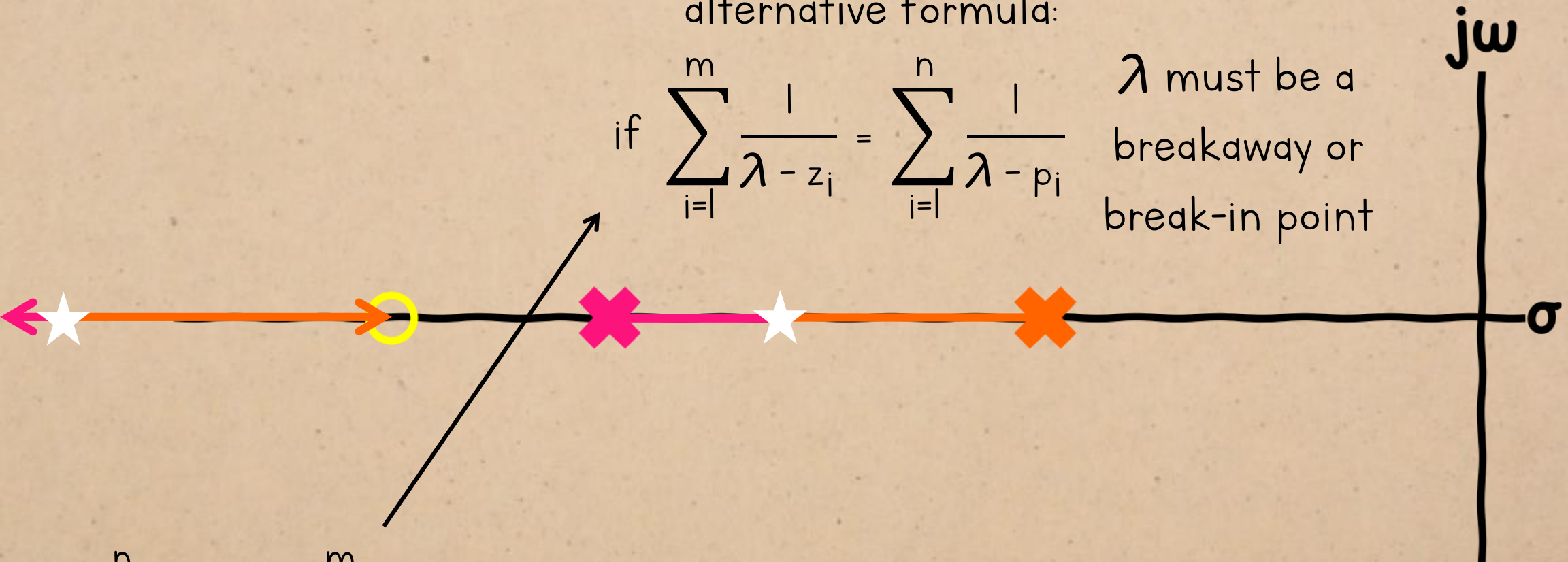
$$\frac{1}{s-p_1} + \frac{1}{s-p_2} + \dots + \frac{1}{s-p_n} = \frac{1}{s-z_1} + \frac{1}{s-z_2} + \frac{1}{s-z_m}$$

BREAKAWAY/IN POINTS

alternative formula:

if
$$\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



$$\sum_{i=1}^n \frac{1}{s - p_i} = \sum_{i=1}^m \frac{1}{s - z_i}$$

BREAKAWAY/IN POINTS

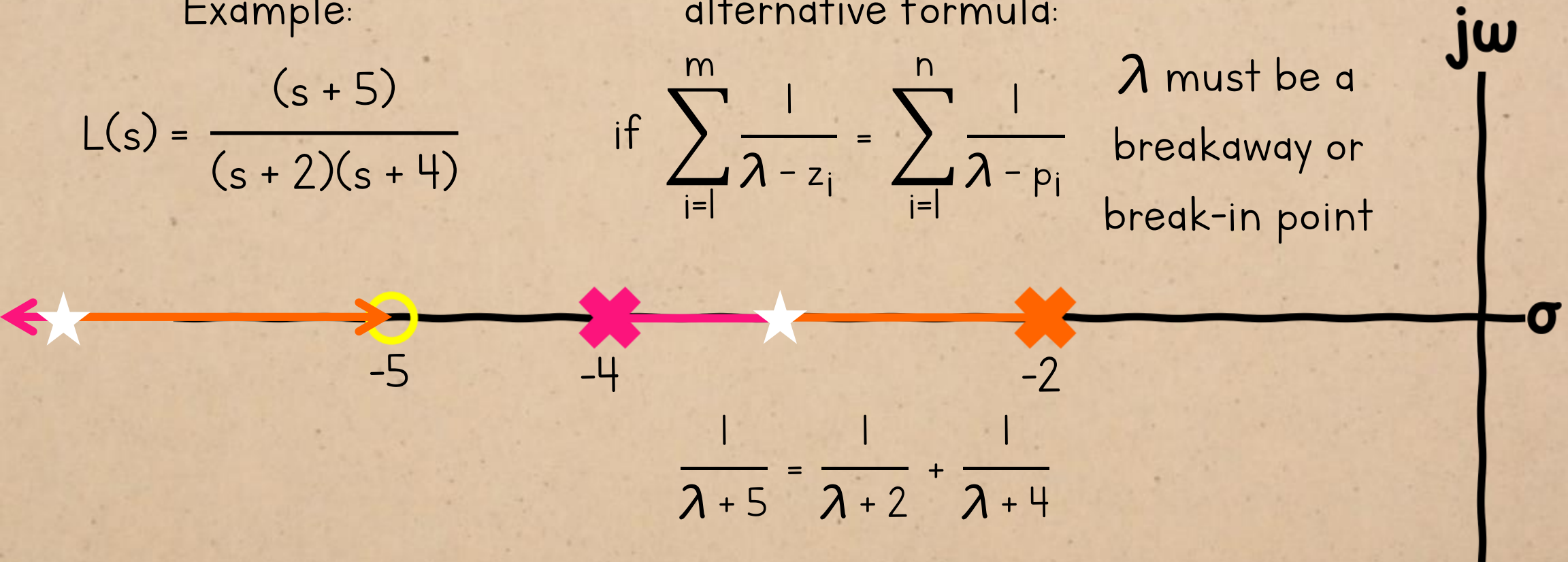
Example:

$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)}$$

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



BREAKAWAY/IN POINTS

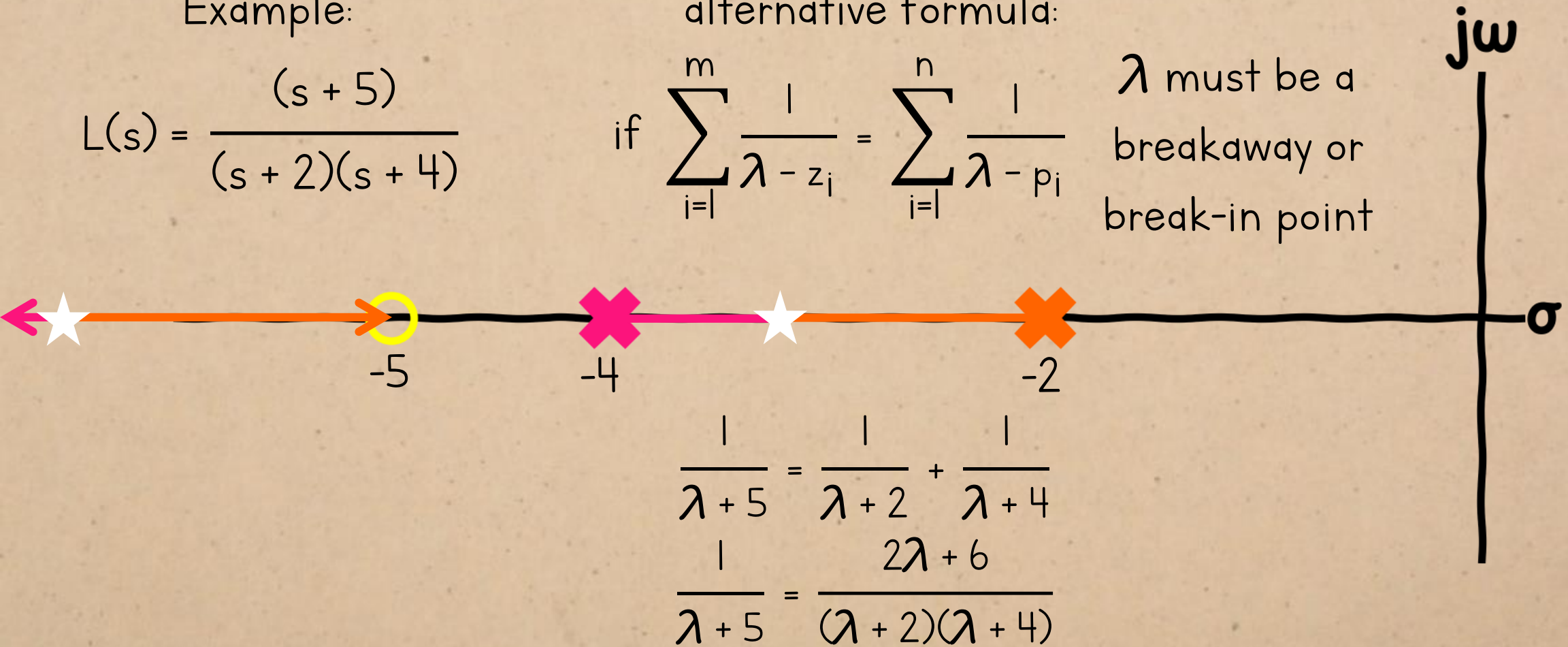
Example:

$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)}$$

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

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breakaway or
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BREAKAWAY/IN POINTS

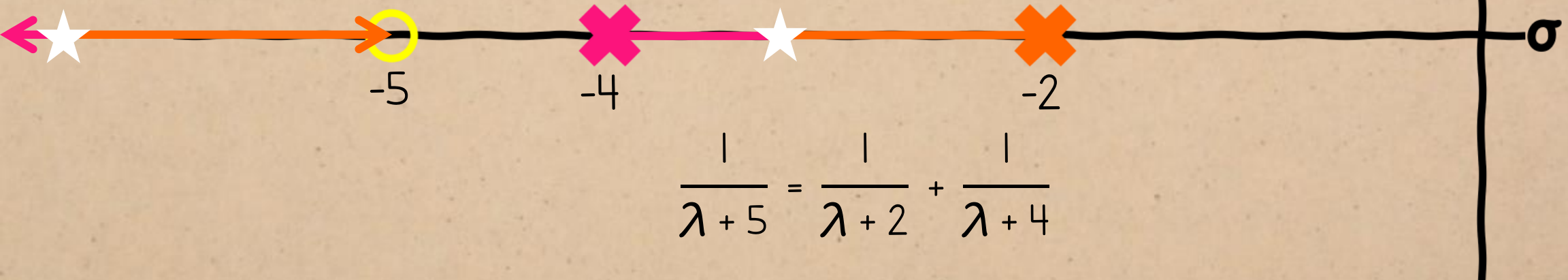
Example:

$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)}$$

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



$$\frac{1}{\lambda + 5} = \frac{1}{\lambda + 2} + \frac{1}{\lambda + 4}$$

$$(\lambda + 2)(\lambda + 4) = (2\lambda + 6)(\lambda + 5)$$

BREAKAWAY/IN POINTS

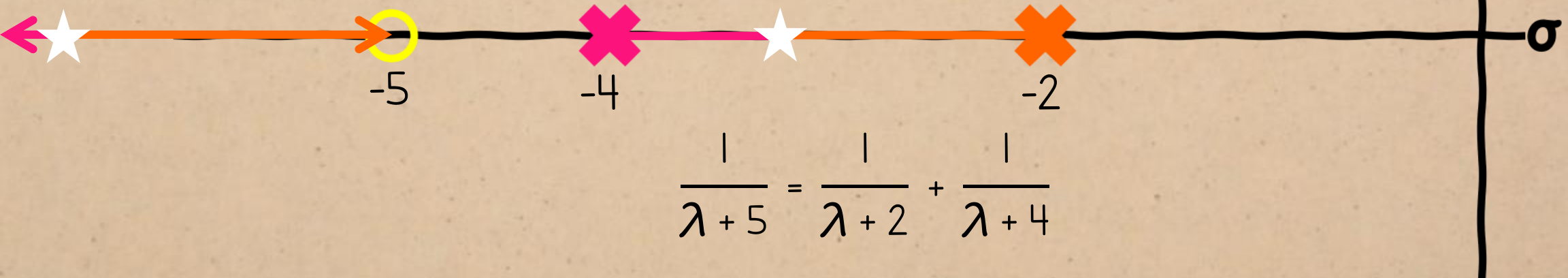
Example:

$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)}$$

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



$$\frac{1}{\lambda + 5} = \frac{1}{\lambda + 2} + \frac{1}{\lambda + 4}$$

$$\lambda^2 + 6\lambda + 8 = 2\lambda^2 + 16\lambda + 30$$

BREAKAWAY/IN POINTS

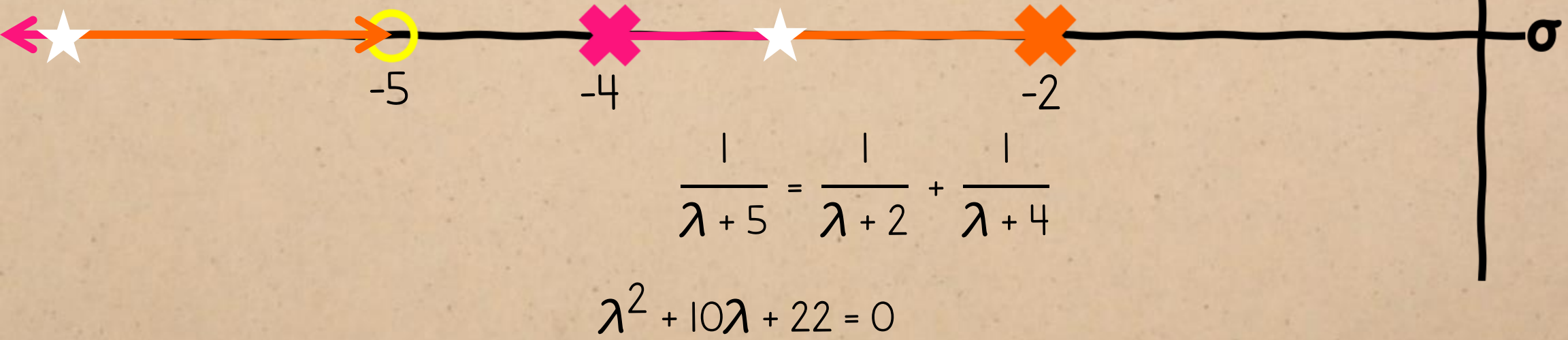
Example:

$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)}$$

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

λ must be a
breakaway or
break-in point



BREAKAWAY/IN POINTS

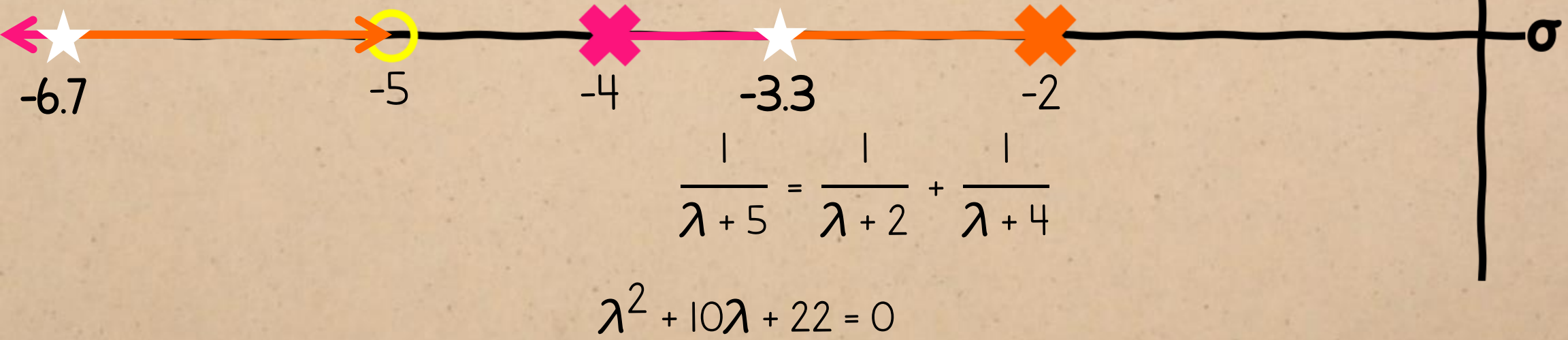
Example:

$$L(s) = \frac{(s + 5)}{(s + 2)(s + 4)}$$

alternative formula:

$$\text{if } \sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$$

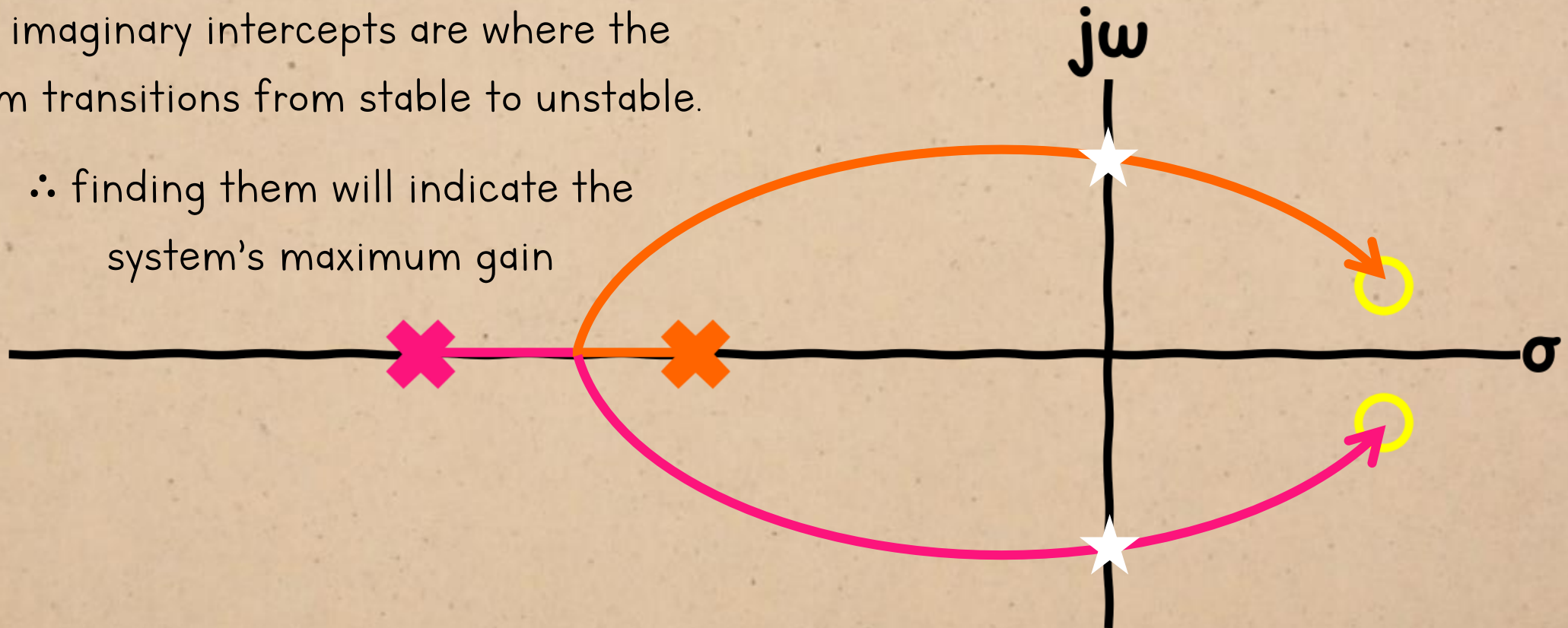
λ must be a
breakaway or
break-in point



IMAGINARY INTERCEPTS

The imaginary intercepts are where the system transitions from stable to unstable.

∴ finding them will indicate the system's maximum gain

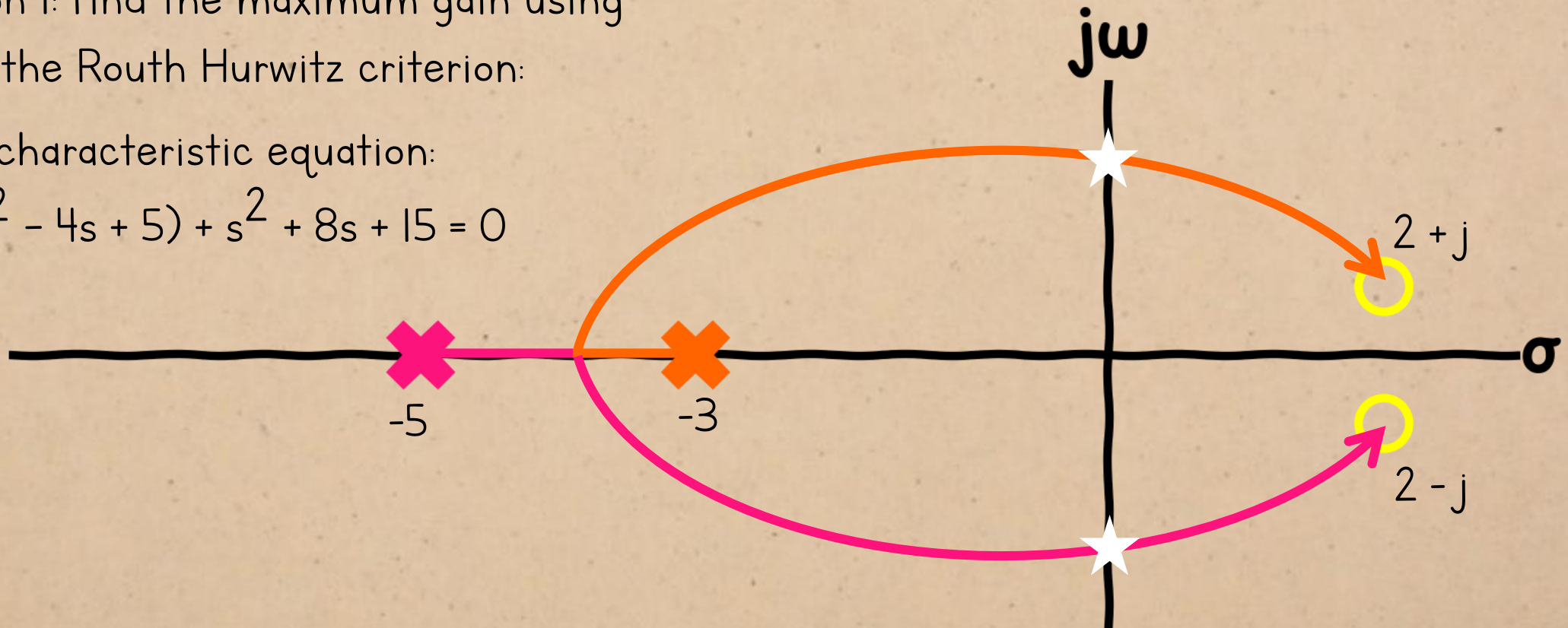


IMAGINARY INTERCEPTS

Option I: find the maximum gain using
the Routh Hurwitz criterion:

characteristic equation:

$$K(s^2 - 4s + 5) + s^2 + 8s + 15 = 0$$



IMAGINARY INTERCEPTS

Option 1: find the maximum gain using
the Routh Hurwitz criterion:

characteristic equation:

$$(1 + K)s^2 + (8 - 4K)s + (15 + 5K) = 0$$

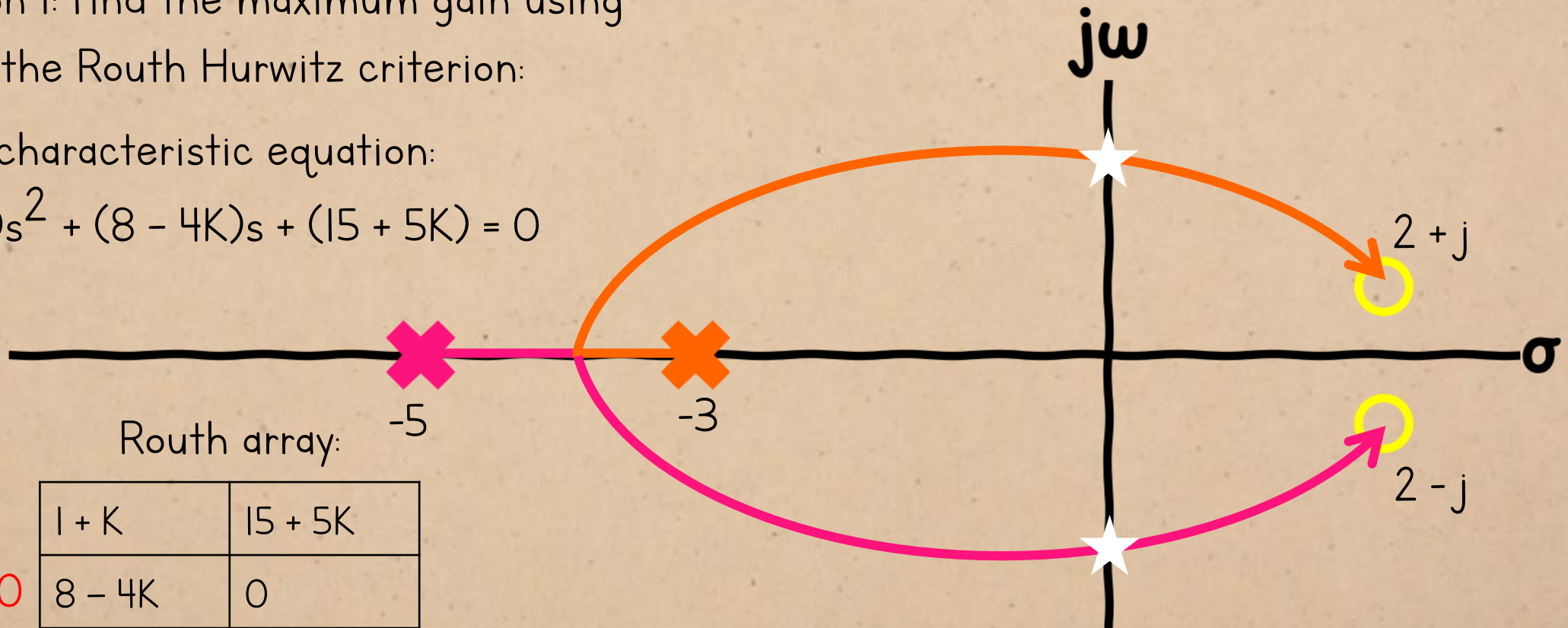
Routh array:

$1 + K$	$15 + 5K$
$8 - 4K$	0
$15 + 5K$	0

$$8 - 4K \geq 0$$

$$15 + 5K \geq 0$$

$$\therefore K \leq 2$$

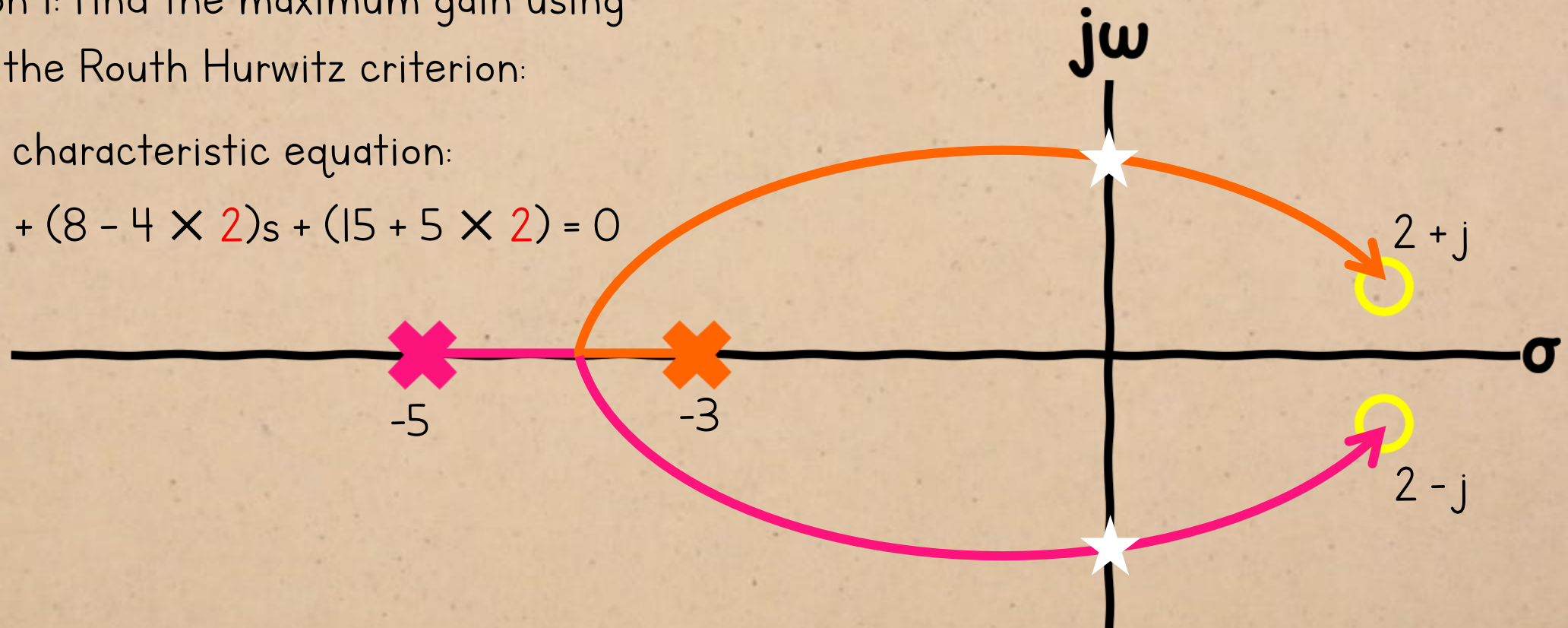


IMAGINARY INTERCEPTS

Option I: find the maximum gain using
the Routh Hurwitz criterion:

characteristic equation:

$$(1 + 2)s^2 + (8 - 4 \times 2)s + (15 + 5 \times 2) = 0$$

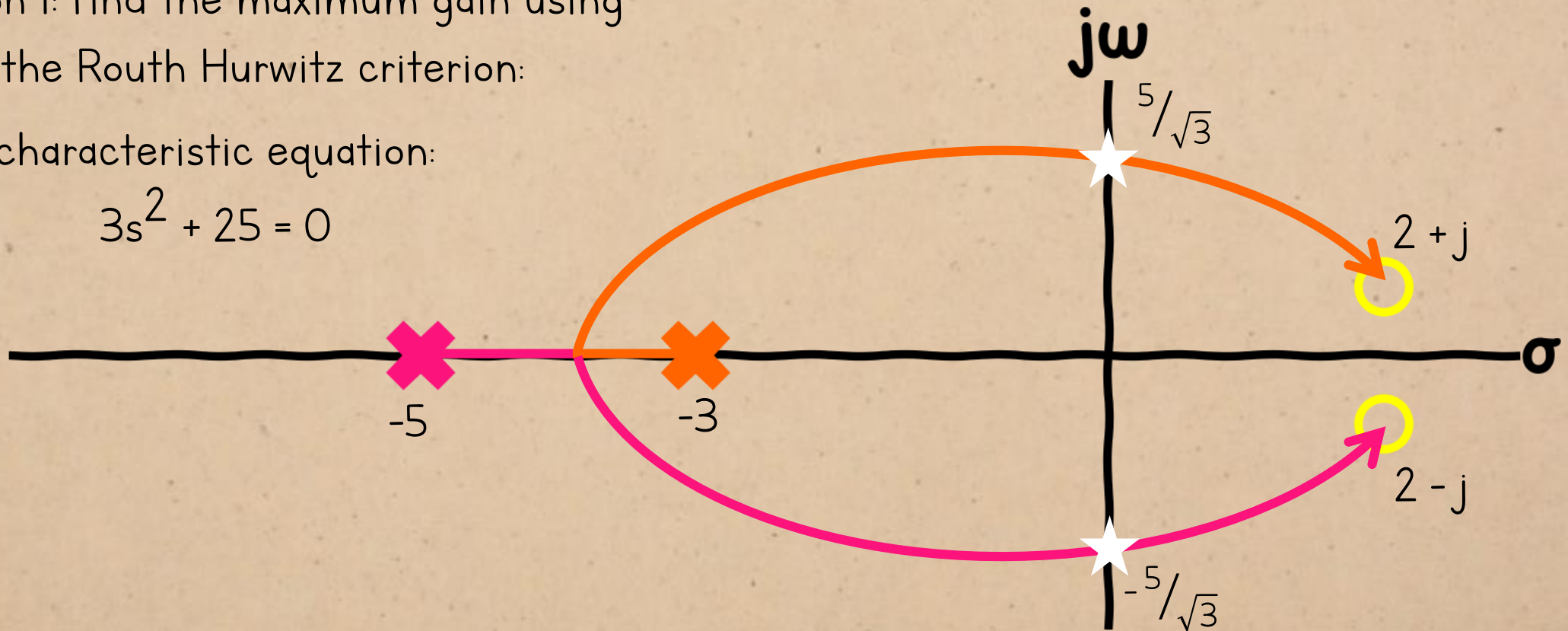


IMAGINARY INTERCEPTS

Option I: find the maximum gain using
the Routh Hurwitz criterion:

characteristic equation:

$$3s^2 + 25 = 0$$



IMAGINARY INTERCEPTS

Option 2: substitute $j\omega$ into the characteristic equation and solve for the values of K and ω for which the real and imaginary parts are zero:

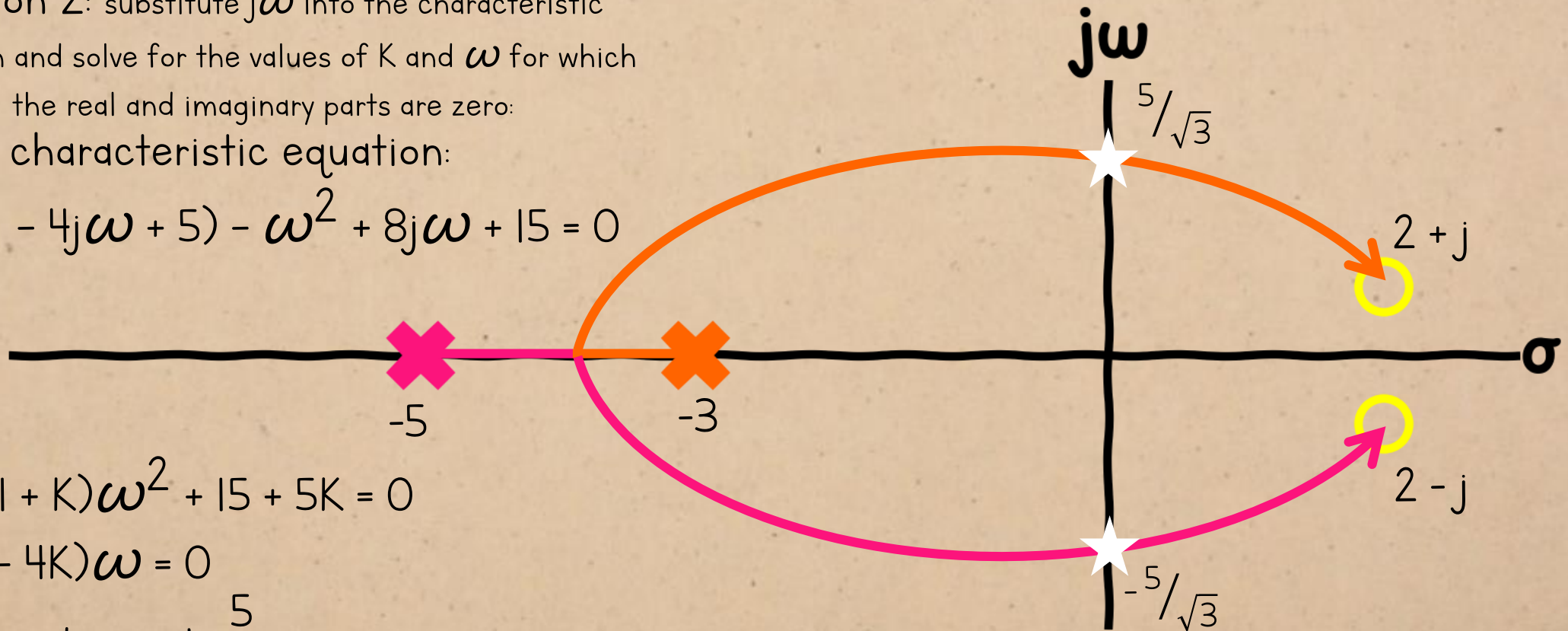
characteristic equation:

$$K(-\omega^2 - 4j\omega + 5) - \omega^2 + 8j\omega + 15 = 0$$

$$\text{real: } -(1 + K)\omega^2 + 15 + 5K = 0$$

$$\text{im: } (8 - 4K)\omega = 0$$

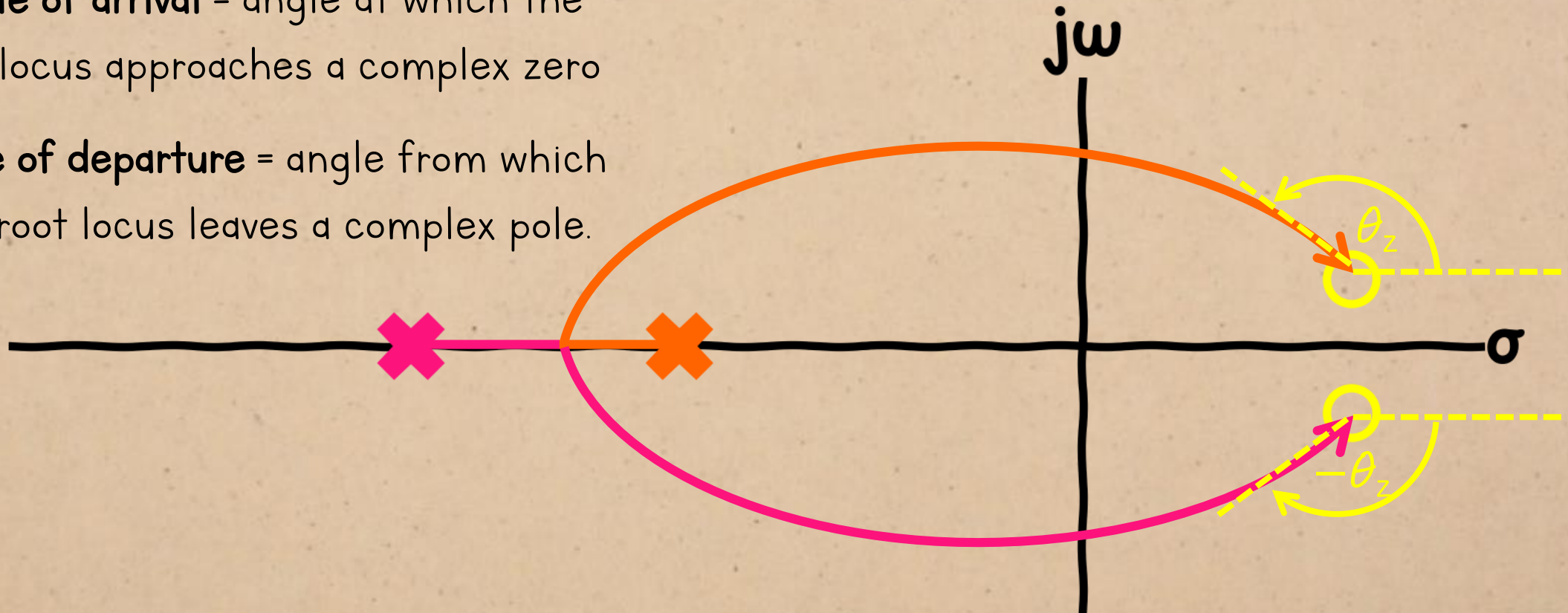
$$\therefore K = 2 \text{ and } \omega = \pm \frac{5}{\sqrt{3}}$$



ANGLES OF ARRIVAL/DEP.

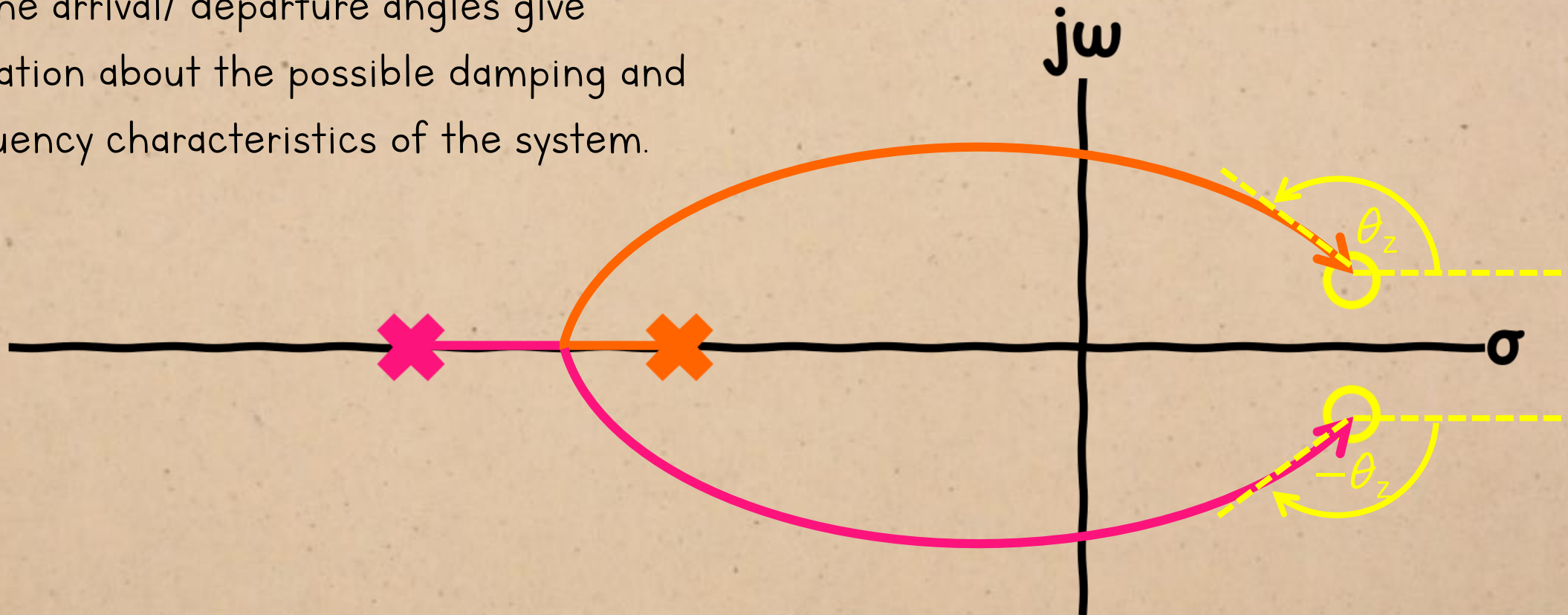
angle of arrival = angle at which the root locus approaches a complex zero

angle of departure = angle from which the root locus leaves a complex pole.

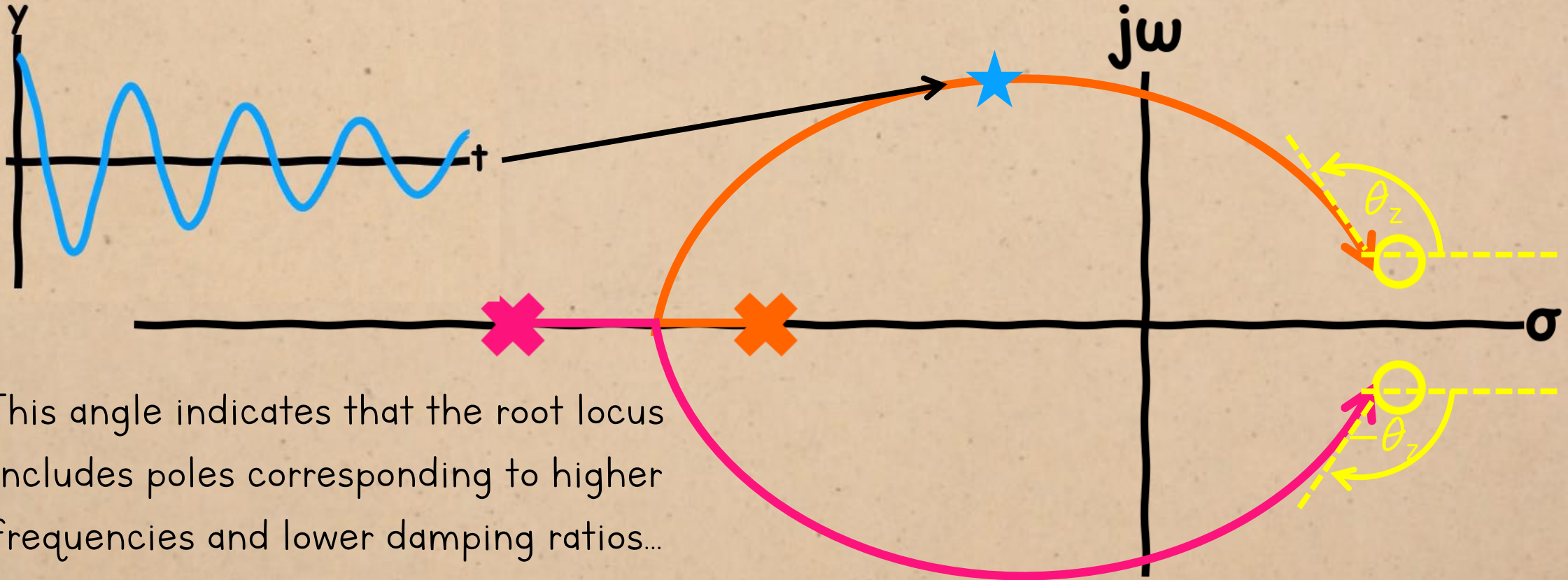


ANGLES OF ARRIVAL/DEP.

The arrival/ departure angles give information about the possible damping and frequency characteristics of the system.

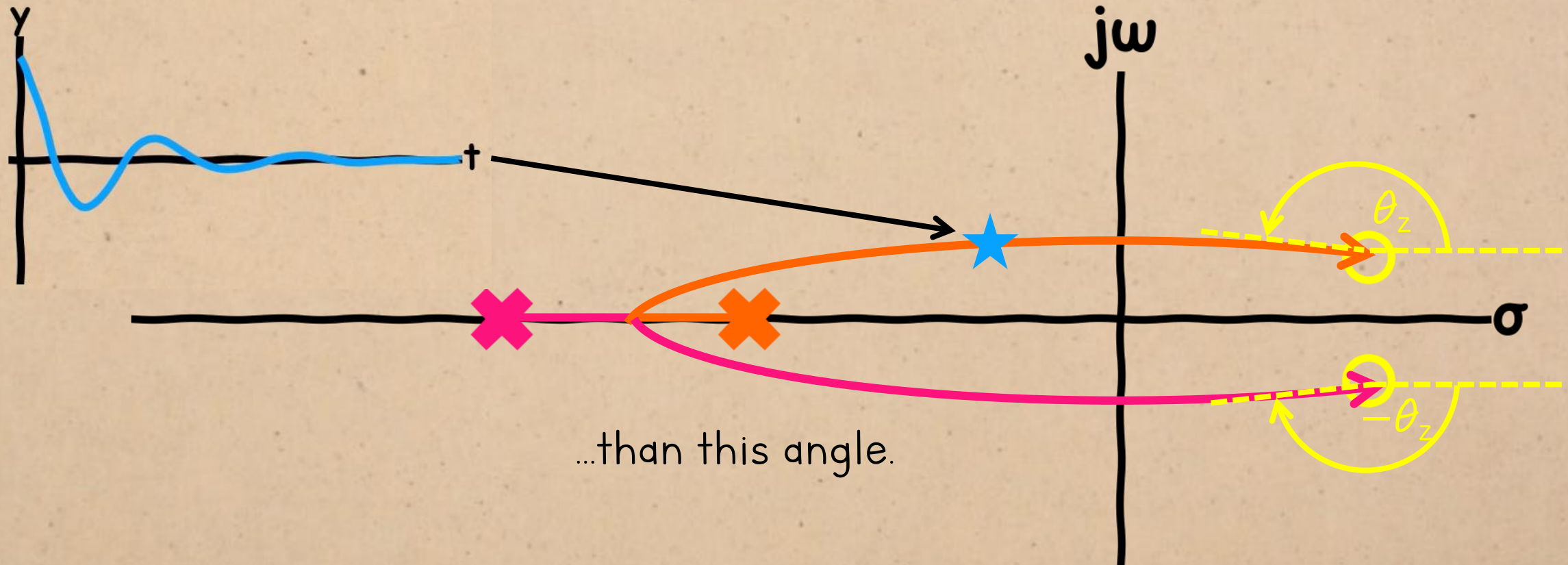


ANGLES OF ARRIVAL/DEP.

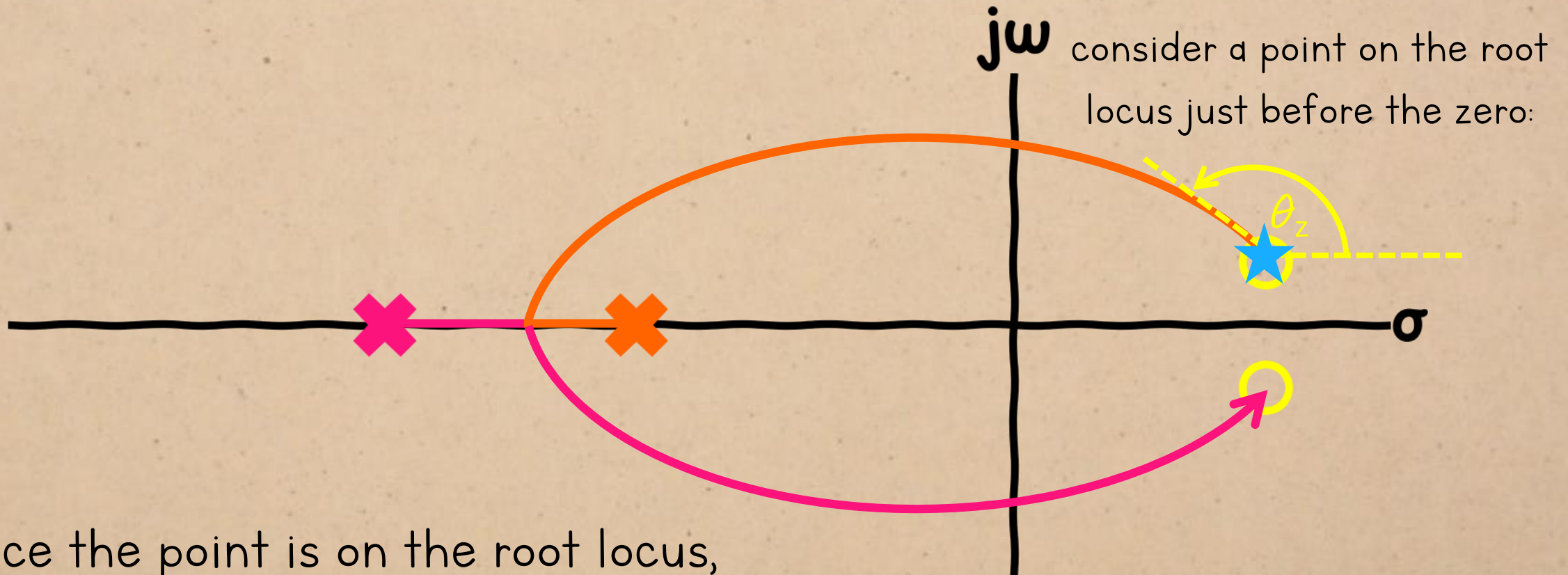


This angle indicates that the root locus includes poles corresponding to higher frequencies and lower damping ratios...

ANGLES OF ARRIVAL/DEP.

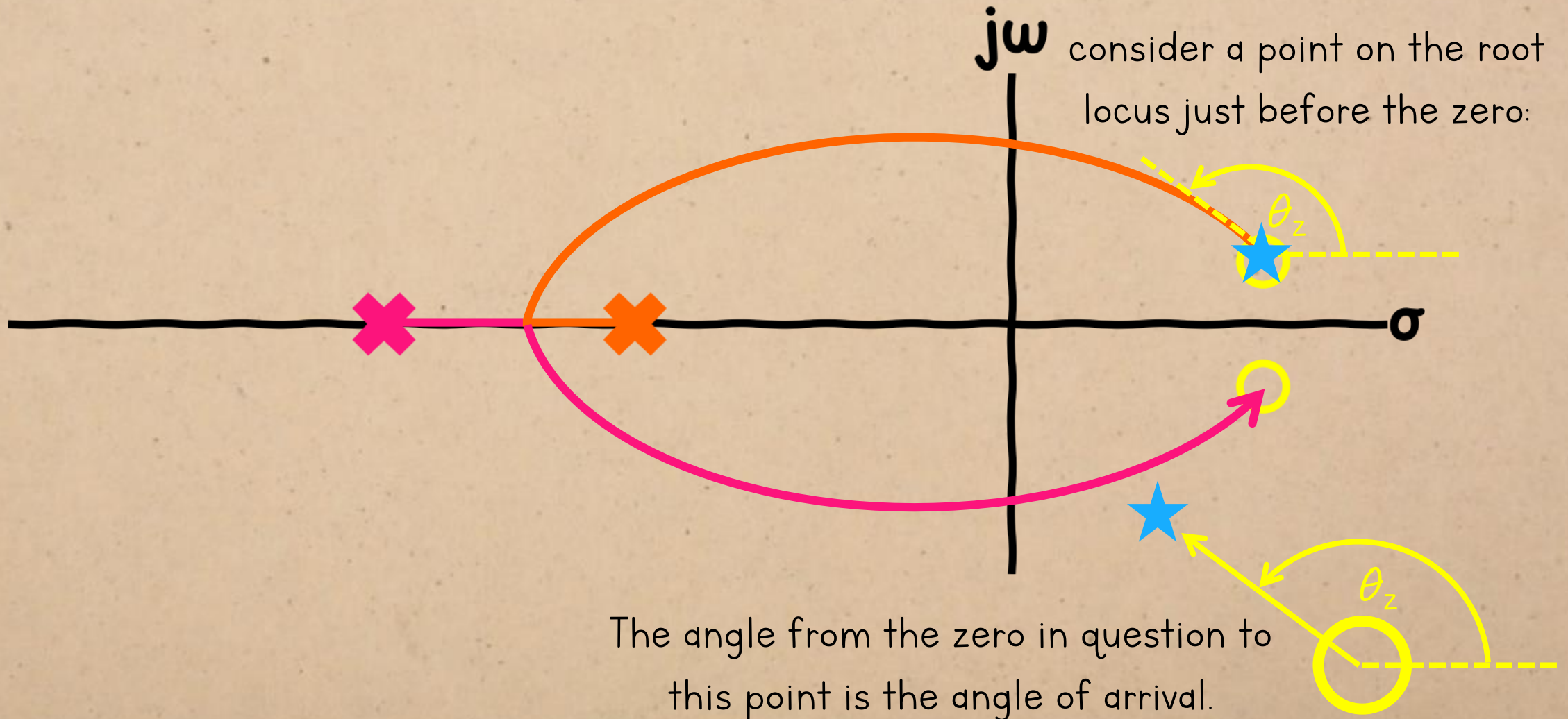


ANGLES OF ARRIVAL/DEP.



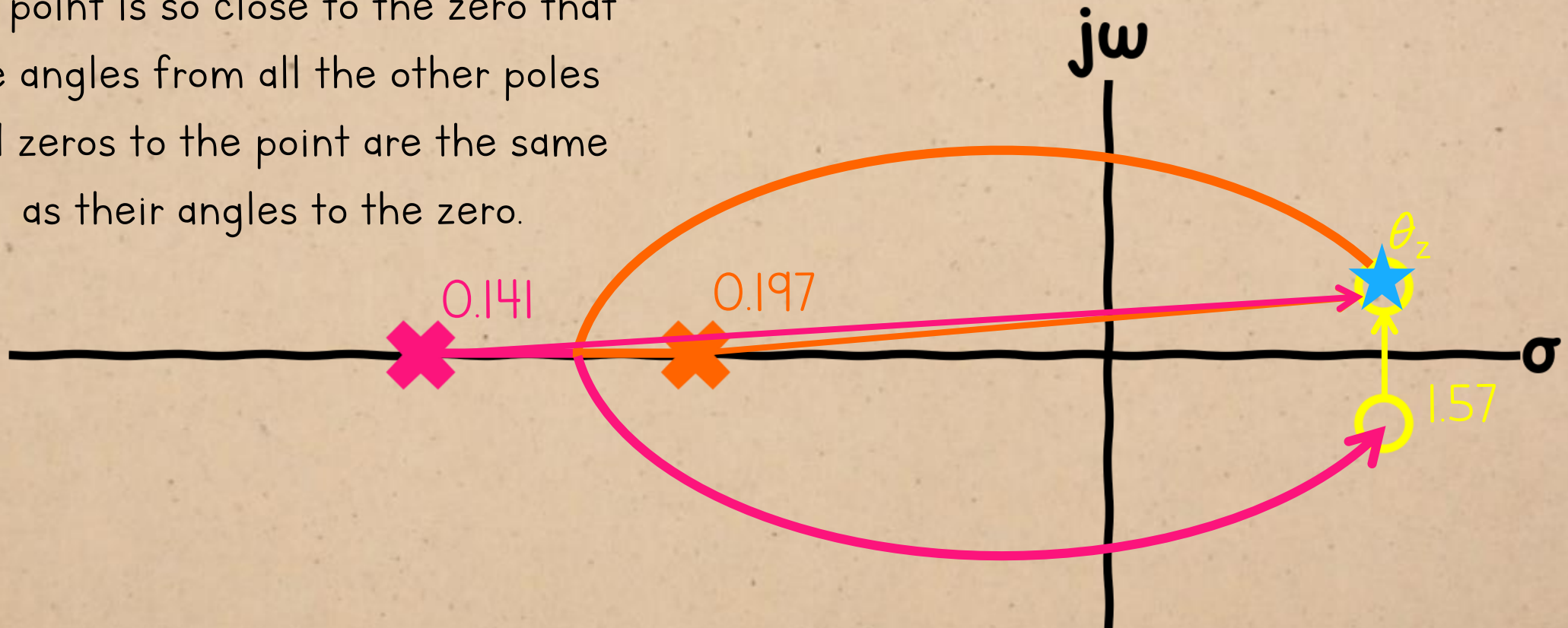
Since the point is on the root locus,
it must satisfy the angle criterion.

ANGLES OF ARRIVAL/DEP.

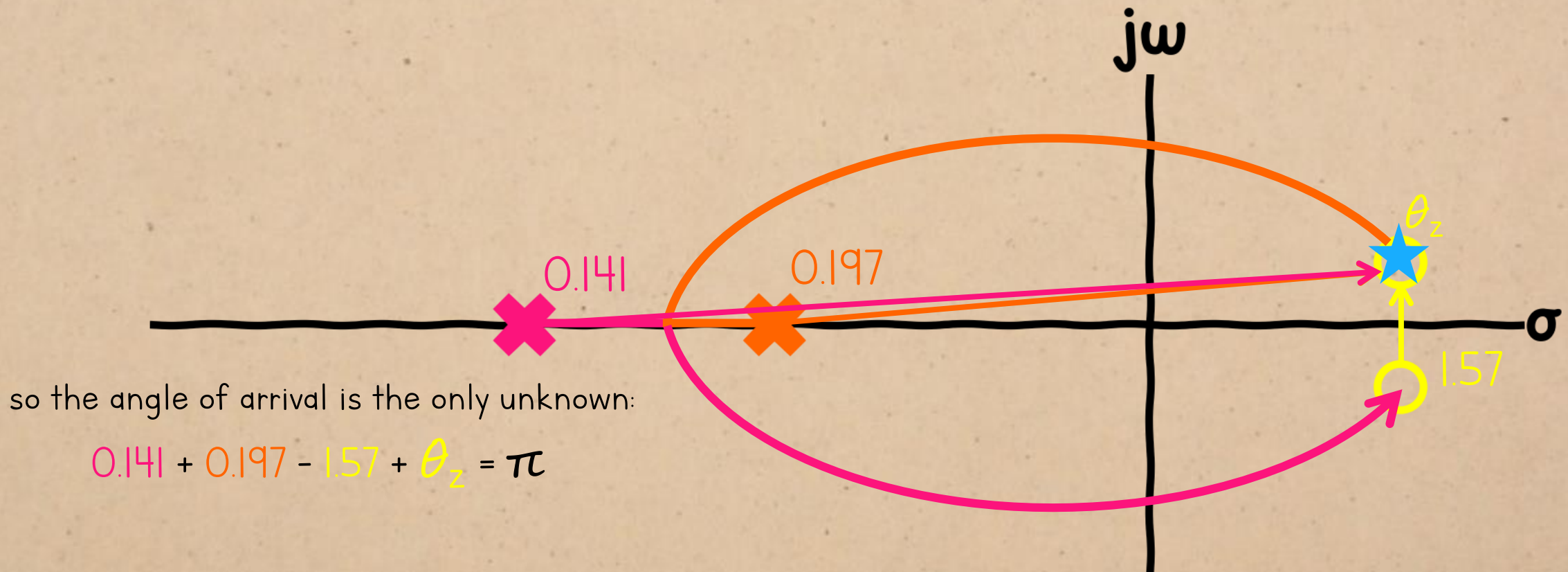


ANGLES OF ARRIVAL/DEP.

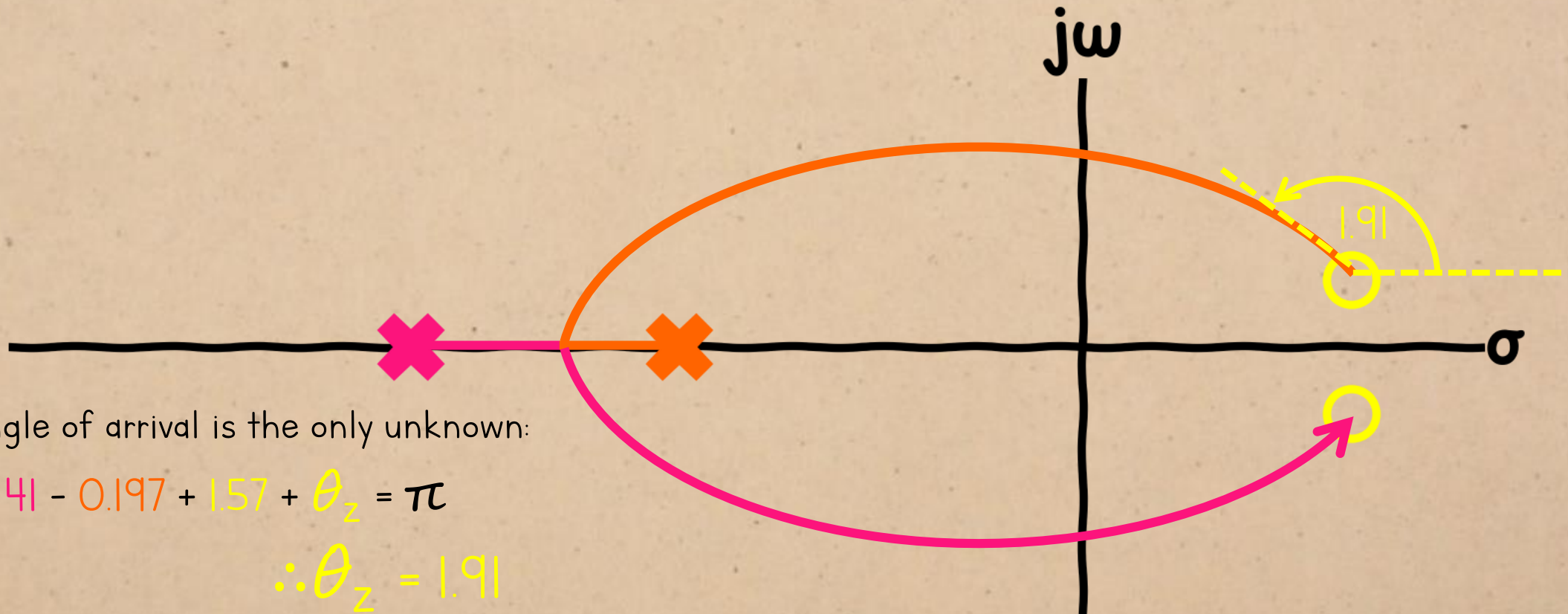
The point is so close to the zero that the angles from all the other poles and zeros to the point are the same as their angles to the zero.



ANGLES OF ARRIVAL/DEP.



ANGLES OF ARRIVAL/DEP.

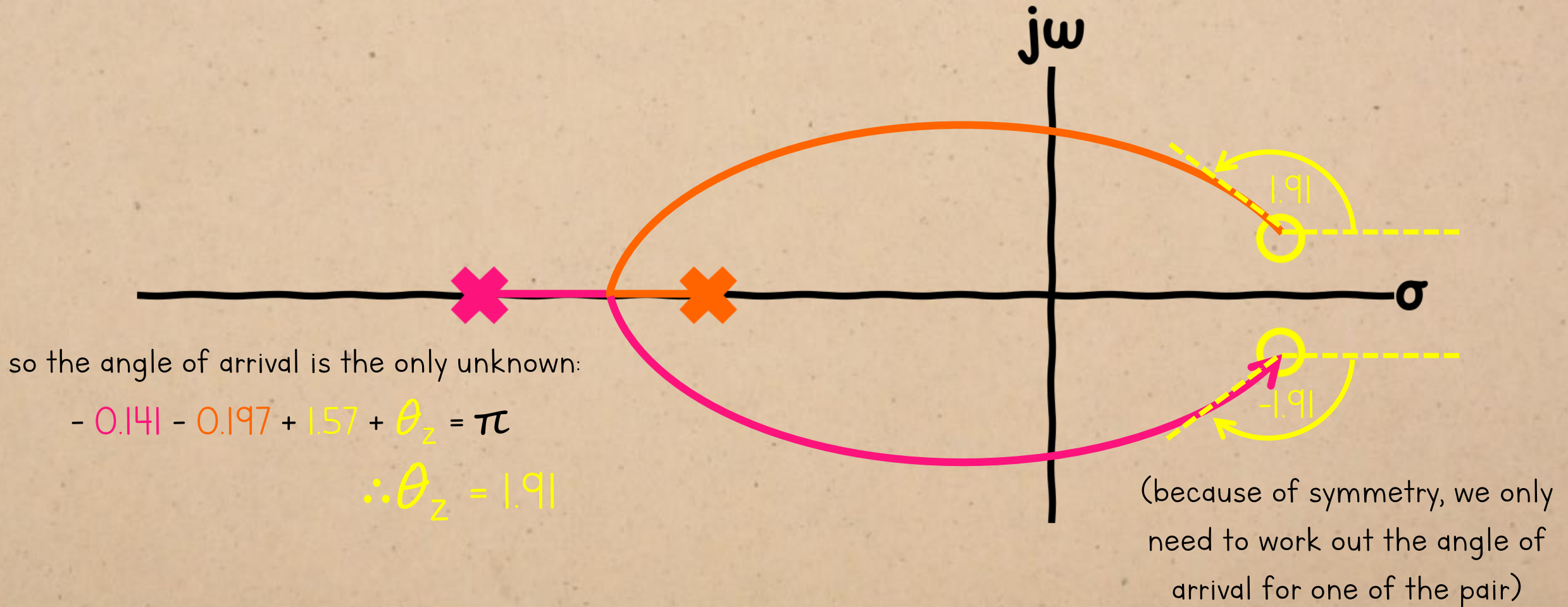


so the angle of arrival is the only unknown:

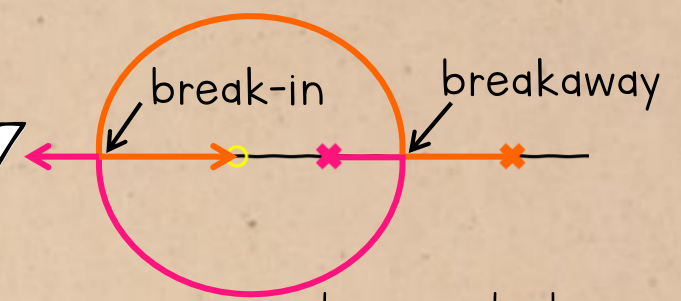
$$-0.141 - 0.197 + 1.57 + \theta_z = \pi$$

$$\therefore \theta_z = 1.91$$

ANGLES OF ARRIVAL/DEP.



SUMMARY



1

real axis segments: any section of the real axis with an odd number of real-valued critical frequencies on its right is on the root locus.

3

breakaway/-in points: where $\frac{d}{ds} K(s) = \frac{d}{ds} \frac{1}{L(s)} = 0$
OR solve for λ where $\sum_{i=1}^m \frac{1}{\lambda - z_i} = \sum_{i=1}^n \frac{1}{\lambda - p_i}$

2

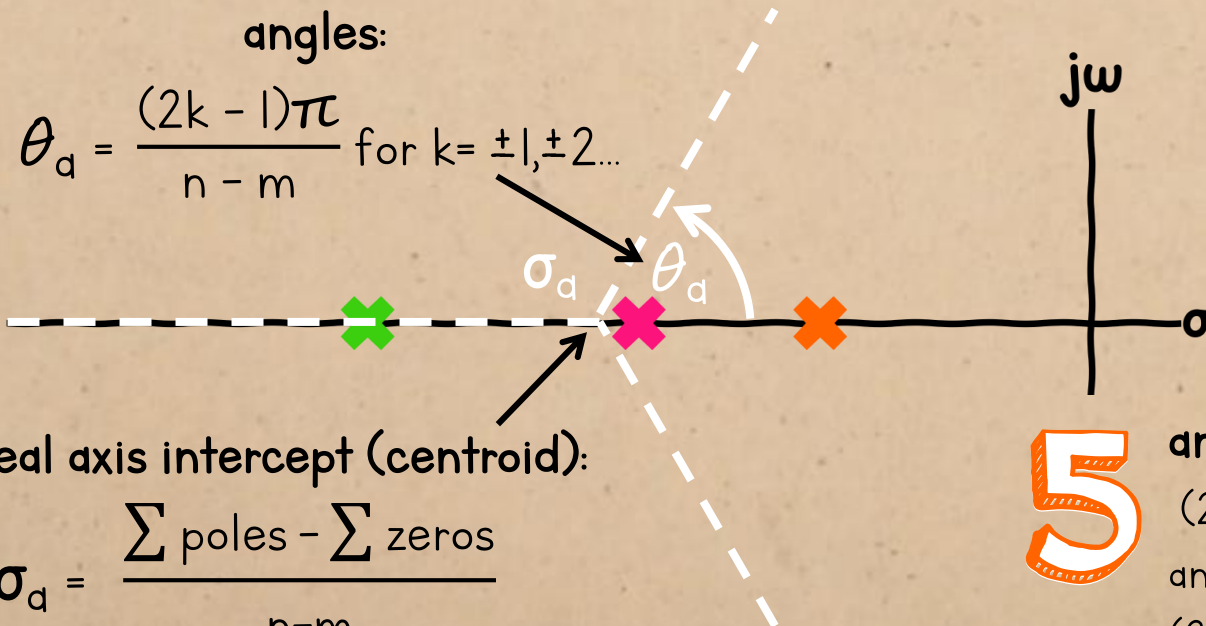
asymptotes: there will be $n-m$ asymptotes, where n = number of poles, m = number of zeros.

4

imaginary axis intercepts: find the maximum gain using the Routh array and then substitute this into the characteristic equation OR substitute $j\omega$ into the characteristic equation and solve for K and ω such that the real and imaginary parts = 0

angles:

$$\theta_d = \frac{(2k-1)\pi}{n-m} \text{ for } k = \pm 1, \pm 2, \dots$$



real axis intercept (centroid):

$$\sigma_d = \frac{\sum \text{poles} - \sum \text{zeros}}{n-m}$$

5

angles of arrival/ departure: the angle of arrival at a complex zero:
 $(2k-1)\pi - \sum \text{angles from poles} + \sum \text{angles from other zeros}$
angle of departure from a complex pole:
 $(2k-1)\pi - \sum \text{angles from other poles} + \sum \text{angles from zeros}$

